Lecture Notes

Vibrations and Waves

MIT OpenCourseWare

based on the lectures of

Prof. Yen-Jie Lee

MIT 8.03SC — Physics III (Fall 2016)

Notes by 9DA62C42, adapted from MIT OCW 8.03SC

October 21, 2026

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How to Use These Notes

These notes follow the 23 lectures of MIT 8.03SC one section at a time. They are written to be read on their own, but they work best next to the lecture videos and the problem sets. This page explains the layout and suggests a way of working through the material.

What the boxes mean

Box	Contains
Definition (red)	A new term, a law, or the equation that defines a system. Learn these exactly.
Theorem (navy)	A result you should be able to state and use.
Proof (grey)	The derivation of the result above it. Read it once carefully; reproduce it later from memory.
Example (brass)	A worked problem or a demonstration from the lecture.
Remark, Recipe, Summary (green)	Comments, step-by-step methods, the summary of each lecture, and the check-yourself questions.
! Warning (thick red)	A common mistake or a limit of validity.

Short asides sit in the outer margin: *Reading* gives the matching chapter of Georgi's *The Physics of Waves*, *Tip* a hint, and *Watch Out* a pitfall.

A way to study each lecture

(1) Read the opening paragraph and the summary box first, so you know where the lecture is going.

(2) Watch the lecture video, or read the section straight through.

(3) Close the notes and rederive each proof on paper. Open the notes only when stuck.

(4) Answer the three check-yourself questions, then compare with Appendix B.

(5) Do the problem set listed in the table below. The problem sets have no published solutions, so use the practice exams, which do.

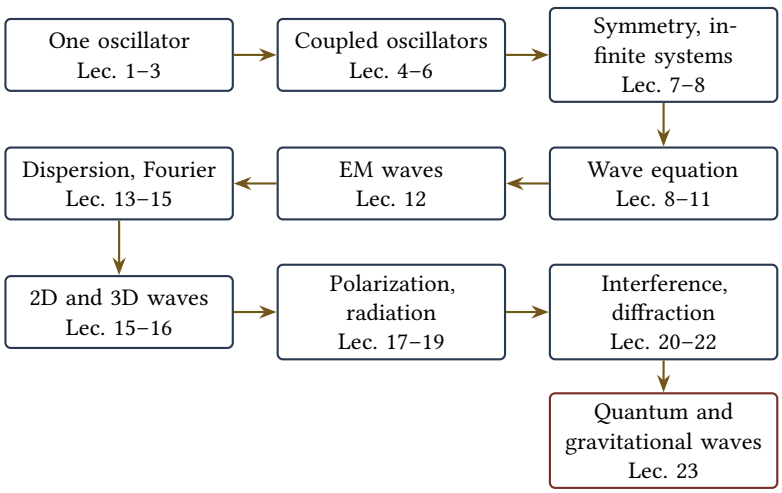
(6) Before an exam, work from Appendix A (the formula sheet) and make sure you can say where every line comes from.

Study map

Lec.	Topic	Georgi	Problem set	Exam
1	Simple harmonic motion	Ch. 1	1	1
2	Damped oscillators	Ch. 2		1
3	Driven oscillators, resonance	Ch. 2	2	1
4	Coupled oscillators, normal modes	Ch. 3		1
5	Beats	Ch. 3	3	1
6	Driven coupled oscillators	Ch. 3–5.2		1
7	Symmetry, infinite chain	Ch. 4, 5.1–5.2	4	1
8	Boundary conditions, wave equation	Ch. 5.3–6.1		1
9	Standing waves, Fourier series	Ch. 5.3–6.1	5	2
10	Traveling waves	Ch. 6.2–8.2		2
11	Sound	Ch. 7	6	2
12	Maxwell's equations, EM waves	Ch. 8.3–9.2		2
13	Dispersion, phase and group velocity	Ch. 8.3–9.2	7	2
14	Fourier transform, AM radio	Ch. 10		2
15	Uncertainty principle, 2D waves	Ch. 10	8	2
16	2D and 3D waves, Snell's law	Ch. 11–12.3		2
17	Polarization	Ch. 11–12.3		Final
18	Wave plates, radiation	Ch. 12.4–		Final
19	Waves in media, Brewster's angle	Ch. 12.4–	9	Final
20	Interference, thin films	Ch. 13	10	Final
21	Many sources, phased radar	Ch. 13		Final
22	Diffraction, resolution	Ch. 13		Final
23	Quantum and gravitational waves	—		Final

Exam 1 covers Lectures 1–8, Exam 2 covers Lectures 9–16, and the final covers everything. Each has a formula sheet and practice exams with solutions in the course materials.

How the ideas build on each other



Three ideas recur throughout: *linearity* (solutions add), *normal modes* (every motion is a sum of simple ones), and the *dispersion relation* $\omega(k)$ (the medium's signature).

Notation

Symbol	Meaning	Symbol	Meaning
ω_0	natural angular frequency	k	wave number, $2\pi/\lambda$
Γ	damping constant	$\vec{k}$	wave vector
ω_d	driving frequency	v_p, v_g	phase and group velocity
Q	quality factor, ω_0/Γ	ψ	wave displacement or wave function
A, ϕ	amplitude and phase	ρ_L	mass per unit length
M, K	mass and stiffness matrices	n	refractive index, c/v
S	symmetry matrix	δ	phase difference
κ	spring constant of the chain	Z	impedance; polarization vector
a	spacing of masses or beads	R, $\mathcal{T}$	reflection and transmission coefficients

A dot means a time derivative, $\dot{x} = \mathrm{d}x/\mathrm{d}t$, and Re means "the real part of". A few letters are reused, as in the lectures: T is a period, a tension or a transmission coefficient, and Q is a charge or a quality factor. The context always decides.

Mathematics to have at your fingertips

(1) Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$, hence $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$.

(2) $\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2}$ and $\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$.

(3) $\cos a\cos b = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$ and $\sin a\sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$.

(4) Small quantities: $\sin\theta \approx \theta$, $\cos\theta \approx 1$, $(1+\epsilon)^p \approx 1 + p\epsilon$.

(5) A 2×2 determinant is $ad - bc$; a system $EB = D$ is solved by Cramer's rule, $B_j = \det E_j/\det E$.

1 Lecture 1: Periodic Oscillations and Harmonic Oscillators

Part I — Mechanical Vibrations and Waves. The course opens with the simplest vibrating system there is: one mass on one ideal spring. Everything that follows — damping, driving, coupled systems, and eventually waves — is built by adding one complication at a time to this model, so it pays to solve it slowly and to notice *why* each step works.

1.1 From a physical situation to an equation of motion

A block of mass m rests on a frictionless table and is attached to a wall by a spring of spring constant k and natural length ℓ_0 (Figure 1). We choose the coordinate x along the spring, with $x = 0$ at the position where the spring is neither stretched nor compressed.

Reading

Georgi, The Physics of Waves, *Chapter 1: Harmonic Oscillation.*

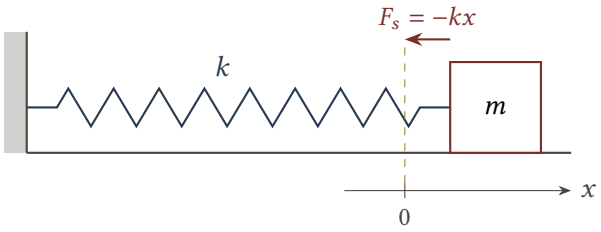


Figure 1 A mass on an ideal spring, displaced by x from equilibrium.

Three forces act on the block: the spring force $\vec{F}_s$, the normal force $\vec{F}_N$ and gravity $\vec{F}_g$. The block moves only horizontally, so the vertical forces cancel, $\vec{F}_N = -\vec{F}_g$, and the total force is the spring force alone.

Definition — Hooke's law and the simple harmonic oscillator

An *ideal spring* exerts a restoring force proportional to the displacement from equilibrium,

$$F = -kx.$$

Newton's second law $F = m\ddot{x}$ then gives the *equation of motion*

$$\ddot{x} + \omega^2 x = 0, \qquad \omega \equiv \sqrt{\frac{k}{m}}.$$

A system obeying this equation is a *simple harmonic oscillator*; ω is its *angular frequency*, measured in radians per second.

The equation of motion alone does not fix the motion. We also need the *initial conditions*: where the block starts, $x(0)$, and how fast it is moving, $\dot{x}(0)$. A second-order equation needs exactly two such numbers.

1.2 Solving the equation

Theorem — General solution

Every solution of $\ddot{x} + \omega^2 x = 0$ has the form

$$x(t) = a\cos\omega t + b\sin\omega t,$$

where the two constants a and b are fixed by the initial conditions:

$$a = x(0), \qquad b = \frac{\dot{x}(0)}{\omega}.$$

Uniqueness

For a linear second-order equation, position and velocity at one instant determine the whole motion.

Proof (verification)

Differentiate twice:

$$\ddot{x} = -\omega^2 a\cos\omega t - \omega^2 b\sin\omega t = -\omega^2 x,$$

so the form satisfies the equation for any a and b. Setting $t = 0$ gives $x(0) = a$, and differentiating once gives $\dot{x}(0) = \omega b$. The uniqueness theorem for ordinary differential equations says a solution with given $x(0)$ and $\dot{x}(0)$ is the only one, so nothing has been missed. ■

Example — Released from rest

The block is pulled out to $x(0) = x_{\text{initial}}$ and released with $\dot{x}(0) = 0$. Then $a = x_{\text{initial}}$ and $b = 0$, so

$$x(t) = x_{\text{initial}}\cos\omega t.$$

The block oscillates forever between $\pm x_{\text{initial}}$ with period

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}.$$

Notice that the period does not depend on the amplitude: pulling the block out twice as far does not change how long one cycle takes.

Remark — The three-step method of the course

(1) Translate the physical situation into a mathematical description (an equation of motion plus initial conditions).

(2) Solve the equation.

(3) Compare the solution with what the system really does.

That step (3) succeeds at all is a remarkable fact about nature. The same procedure is used for a block on a spring and for gravitational waves.

1.3 Why Hooke's law is everywhere

No real spring obeys Hooke's law for arbitrarily large stretches. The law is nevertheless useful, because it is what *any* smooth force looks like close to a stable

equilibrium.

Theorem — Small oscillations about a stable equilibrium

Let a particle move in a smooth potential $V(x)$ with a minimum at $x = 0$ and $V''(0) > 0$. For small enough x the force is

$$F(x) \approx -V''(0)\,x,$$

so the motion is simple harmonic with effective spring constant $k = V''(0)$ and $\omega = \sqrt{V''(0)/m}$.

Proof

Expand the potential in a Taylor series about the minimum:

$$V(x) = V(0) + V'(0)\,x + \frac{V''(0)}{2!}x^2 + \frac{V'''(0)}{3!}x^3 + \cdots$$

The force is $F = -\mathrm{d}V/\mathrm{d}x$, so

$$F(x) = -V'(0) - V''(0)\,x - \tfrac{1}{2}V'''(0)\,x^2 - \cdots$$

At a minimum $V'(0) = 0$, which removes the constant term. The quadratic term is negligible compared with the linear one as long as

$$\left|x\,V'''(0)\right| \ll V''(0),$$

leaving $F \approx -V''(0)\,x$. ∎

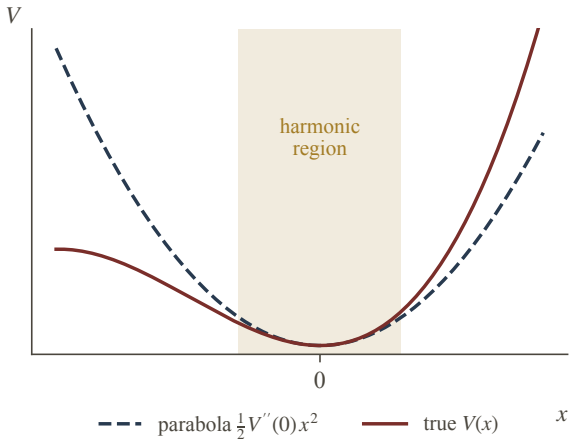


Figure 2 Near its minimum any smooth potential (solid) is indistinguishable from a parabola (dashed). Farther out the two separate, and the motion is no longer simple harmonic.

! Linear versus non-linear

When the amplitude grows, the $V'''(0)$ and higher terms can no longer be dropped. The equation of motion is then non-linear, and the two properties in the next subsection fail. This course deals with linear systems only.

Watch Out

The condition $|xV'''(0)| \ll V''(0)$ is what "small" means. It depends on the potential, not on a universal length.

1.4 Two properties of the linear equation of motion

Theorem — Linearity and time-translation invariance

For $\ddot{x} + \omega^2 x = 0$ with constant ω:

(1) **Superposition.** If $x_1(t)$ and $x_2(t)$ are solutions, then so is $x_1(t) + x_2(t)$, and so is any constant multiple of a solution.

(2) **Time-translation invariance.** If $x(t)$ is a solution, then so is $x(t + a)$ for any constant a.

Proof

(1) The equation contains x and its derivatives only to the first power, so

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}(x_1 + x_2) + \omega^2(x_1 + x_2) = (\ddot{x}_1 + \omega^2 x_1) + (\ddot{x}_2 + \omega^2 x_2) = 0.$$

(2) Put $t' = t + a$. By the chain rule $\mathrm{d}/\mathrm{d}t = \mathrm{d}/\mathrm{d}t'$, because $\mathrm{d}t'/\mathrm{d}t = 1$. The coefficients of the equation do not depend on t, so the equation in terms of t' is identical to the original one. ∎

Time-translation invariance says that the choice of the instant called $t = 0$ has no physical meaning. Most systems have this symmetry when no external force acts on them. To break it, a parameter such as k (and hence ω) would have to change with time.

1.5 Three ways to write the same solution

Definition — Amplitude and phase

The general solution can be written in three equivalent forms:

(1) $x(t) = a\cos\omega t + b\sin\omega t$, with constants a, b;

(2) $x(t) = A\cos(\omega t + \phi)$, with *amplitude* $A \geq 0$ and *phase* ϕ;

(3) $x(t) = \mathrm{Re}\left[A\, e^{i(\omega t + \phi)}\right]$.

Expanding (2) with the angle-addition formula gives the dictionary

$$a = A\cos\phi, \qquad b = -A\sin\phi,$$

$$A = \sqrt{a^2 + b^2}, \qquad \tan\phi = -\frac{b}{a}.$$

Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$. *Its modulus is 1 for every real θ.*

Form (3) looks like a detour, but it is the one used for the rest of the course. Writing $x(t)$ as the real part of something leaves the imaginary part free: $x(t) = \mathrm{Re}[A\cos(\omega t + \phi) + i f(t)]$ for any real $f(t)$. The useful choice is $f(t) = A\sin(\omega t + \phi)$, because Euler's formula then collapses the bracket into a single exponential.

Theorem — Why the complex exponential is convenient

The function $z(t) = A\,e^{i(\omega t+\phi)}$ has two properties:

(1) **It survives differentiation.** $\dot{z} = i\omega z$ and $\ddot{z} = -\omega^2 z$: each derivative only multiplies by a constant.

(2) **Exponents add.** $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$, so a time translation $t \longrightarrow t + a$ multiplies z by the constant phase factor $e^{i\omega a}$.

Property (1) turns a differential equation into algebra. Property (2) gives the picture in Figure 3: $z(t)$ is a point moving at constant angular speed ω around a circle of radius A in the complex plane, and the physical displacement $x(t)$ is its shadow on the real axis. A time translation is simply a rotation of that plane.

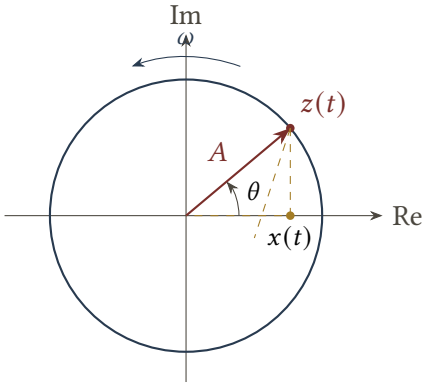


Figure 3 The complex solution $z(t) = Ae^{i(\omega t+\phi)}$ moves uniformly around a circle, at angle $\theta = \omega t + \phi$; the physical displacement is its projection on the real axis.

The meaning of the constants is easiest to see on a graph of the motion (Figure 4).

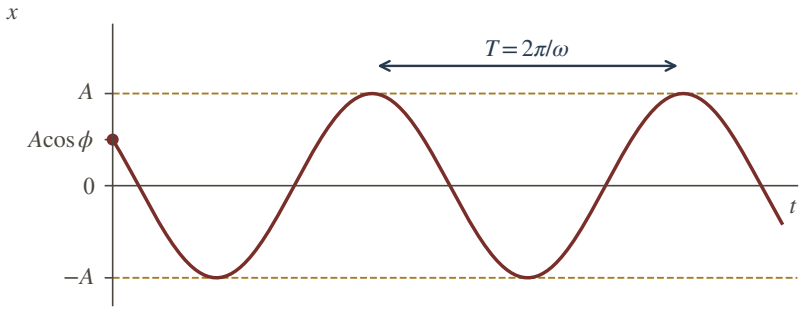


Figure 4 The motion $x = A\cos(\omega t + \phi)$. The amplitude A sets the height, the period T the spacing of the peaks, and the phase ϕ where the curve starts at $t = 0$.

Example — Initial conditions in amplitude–phase form

A block has $x(0) = x_0$ and $\dot{x}(0) = v_0$. From $x = A\cos(\omega t + \phi)$ and $\dot{x} = -A\omega\sin(\omega t + \phi)$,

$$x_0 = A\cos\phi, \qquad v_0 = -A\omega\sin\phi.$$

Squaring and adding, then dividing, gives

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}, \qquad \tan\phi = -\frac{v_0}{\omega x_0}.$$

Released from rest ($v_0 = 0$) this reduces to $A = x_0$, $\phi = 0$, as before.

> **Tip**
> *Choose the quadrant of ϕ from the signs of $\cos\phi$ and $\sin\phi$ separately, not from $\tan\phi$ alone.*

Summary of Lecture 1

(1) A mass on an ideal spring obeys $\ddot{x} + \omega^2 x = 0$ with $\omega = \sqrt{k/m}$.

(2) The solution is $x = A\cos(\omega t + \phi)$; two initial conditions fix A and ϕ.

(3) Any smooth potential near a stable minimum behaves like a spring with $k = V''(0)$.

(4) The equation is linear and time-translation invariant, which is what makes $e^{i\omega t}$ the natural tool.

Check yourself — Lecture 1

(1) A block of mass 0.5 kg hangs on a spring with $k = 8$ N/m. Find ω and the period T.

(2) An oscillator starts at $x(0) = 0$ with velocity $v_0 > 0$. Find A and ϕ in $x = A\cos(\omega t + \phi)$.

(3) A particle of mass m moves in $V(x) = V_0\,[1 - \cos(x/a)]$. What is the frequency of small oscillations about $x = 0$?

Answers are in Appendix B.

2 Lecture 2: Damped Free Oscillators

Lecture 1 solved one system. This lecture first shows that the same equation describes systems that look nothing like a spring, then looks at the energy of a harmonic oscillator, and finally adds the first realistic complication: a drag force that removes energy.

2.1 The same equation in other systems

Example — The LC circuit

An inductor L and a capacitor C are connected in a loop. Let $Q(t)$ be the charge on the capacitor and $I = \mathrm{d}Q/\mathrm{d}t$ the current, with clockwise current taken as positive. The voltage across the inductor is $L\,\mathrm{d}I/\mathrm{d}t$ and across the capacitor Q/C. Going once round the loop,

$$L\frac{\mathrm{d}I}{\mathrm{d}t} + \frac{Q}{C} = 0,$$

$$\ddot{Q} + \omega_0^2 Q = 0, \qquad \omega_0 = \frac{1}{\sqrt{LC}}.$$

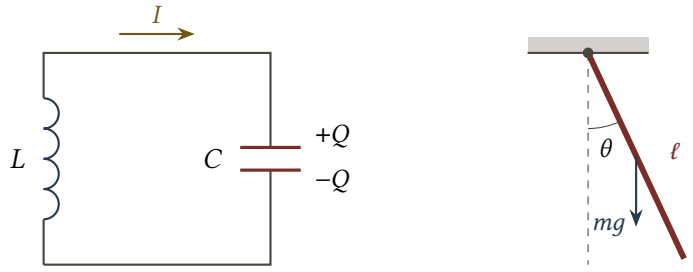
With $Q(0) = 0$ and $I(0) = I_{\text{initial}}$ the solution is

$$Q(t) = \frac{I_{\text{initial}}}{\omega_0}\sin\omega_0 t.$$

In the form $Q = A\cos(\omega_0 t + \phi)$ this is $A = I_{\text{initial}}/\omega_0$, $\phi = -\pi/2$.

Reading

Georgi, Chapters 1 and 2 (the damped oscillator is in §2.1).



LC circuit hanging rod

Figure 5 Two systems with the same mathematics as the mass on a spring: an LC circuit, and (in the next subsection) a rod swinging about one end.

The charge sloshes back and forth exactly as the block does. Comparing the two equations term by term gives a dictionary between the systems:

Quantity	Mass on spring	LC circuit
Generalised coordinate	x	Q
Generalised mass μ	m	L
Generalised spring constant k	k	$1/C$
Natural frequency ω_0	$\sqrt{k/m}$	$1/\sqrt{LC}$

Definition — Generalised harmonic oscillator

Any system whose equation of motion can be brought to the form

$$\mu\,\ddot{x} = -k\,x$$

is a harmonic oscillator with *natural angular frequency* $\omega_0 = \sqrt{k/\mu}$. Here x is a *generalised coordinate*, μ a *generalised mass* and k a *generalised spring constant*.

2.2 Energy of a harmonic oscillator

Theorem — Energy conservation

For $\mu\ddot{x} = -kx$ with solution $x(t) = A\cos(\omega_0 t + \phi)$, the total energy

$$E = \tfrac{1}{2}\mu\dot{x}^2 + \tfrac{1}{2}kx^2$$

is constant in time and equal to $\tfrac{1}{2}kA^2$.

Proof

From $\dot{x} = -A\omega_0 \sin(\omega_0 t + \phi)$ and $\mu\omega_0^2 = k$,

$$\begin{aligned} E &= \tfrac{1}{2}\mu A^2\omega_0^2 \sin^2(\omega_0 t + \phi) + \tfrac{1}{2}kA^2\cos^2(\omega_0 t + \phi) \\ &= \tfrac{1}{2}kA^2\left[\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)\right] \\ &= \tfrac{1}{2}kA^2. \end{aligned}$$

■

The energy is proportional to the *square* of the amplitude. It is never lost; it moves between kinetic form (largest as the oscillator passes through equilibrium) and potential form (largest at the turning points), as in Figure 6.

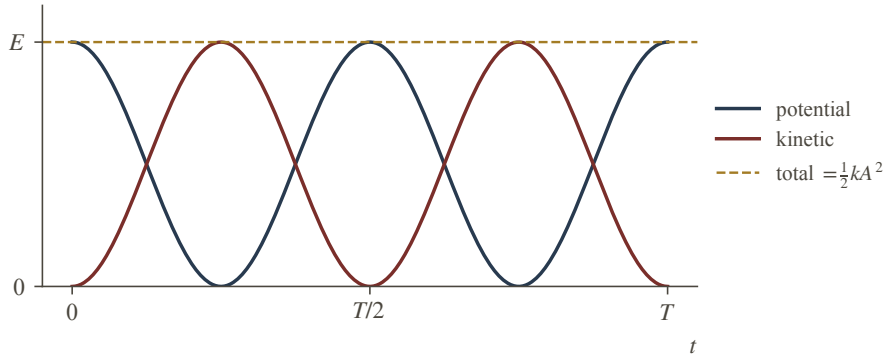


Figure 6 Potential and kinetic energy of an undamped oscillator released from rest. Their sum is constant.

Moment of inertia

A uniform rod about one end has $I = m\ell^2/3$. Gravity acts at the centre, a distance $\ell/2$ from the pivot.

2.3 A physical pendulum and the small-angle approximation

A uniform rod of mass m and length ℓ hangs from a pivot at one end. Let θ be its angle from the vertical, counter-clockwise positive, with $\theta(0) = \theta_{\text{initial}}$ and $\dot{\theta}(0) = 0$.

Example — Equation of motion of the rod

Gravity acts at the centre of mass and produces the torque

$$\tau_g = -mg\,\frac{\ell}{2}\,\sin\theta.$$

The rotational form of Newton's law, $\tau = I\ddot{\theta}$, gives

$$\ddot{\theta} = -\frac{mg\ell}{2I}\sin\theta = -\frac{3g}{2\ell}\sin\theta,$$

$$\ddot{\theta} = -\omega_0^2 \sin\theta, \qquad \omega_0 = \sqrt{\frac{3g}{2\ell}}.$$

This equation is exact, and it is *not* the harmonic-oscillator equation: $\sin\theta$ is not proportional to θ. It has no solution in elementary functions. We can either integrate it numerically or restrict ourselves to small swings.

Definition — Small-angle approximation

For $|\theta| \ll 1$ (in radians),

$$\sin\theta \approx \theta, \qquad \cos\theta \approx 1.$$

The pendulum equation then becomes $\ddot{\theta} = -\omega_0^2\,\theta$, with solution $\theta(t) = \theta_{\text{initial}}\cos\omega_0 t$ for a release from rest.

θ	$\sin\theta/\theta$
1°	99.99%
5°	99.9%
10°	99.5%

! The period depends on amplitude for large swings

The amplitude-independent period $T = 2\pi/\omega_0$ is a property of the linearised equation only. For large angles the true period is longer, and the motion is no longer a pure cosine.

Watch Out

The approximation needs θ in radians. It is the same idea as keeping only $V''(0)$ in ***§1****.*

2.4 Adding a drag force

A real pendulum slows down. We model the loss by a torque proportional to the angular velocity and opposing it,

$$\tau_{\text{drag}} = -b\,\dot{\theta}.$$

This form is chosen because it is reasonably realistic for slow motion through a fluid and because it keeps the equation linear, so it can be solved exactly. Other drag laws generally need a computer.

Definition — The damped oscillator equation

With $\tau = \tau_g + \tau_{\text{drag}}$ and the small-angle approximation, the rod obeys

$$\ddot{\theta} + \Gamma\,\dot{\theta} + \omega_0^2\,\theta = 0,$$

$$\omega_0^2 = \frac{3g}{2\ell}, \qquad \Gamma = \frac{b}{I} = \frac{3b}{m\ell^2}.$$

Γ is the *damping constant*; it has the dimensions of a frequency. Every linearly damped oscillator has this form, with its own ω_0 and Γ.

Theorem — Characteristic equation

The trial solution $z(t) = e^{i\alpha t}$, with $\theta = \operatorname{Re} z$, solves the damped equation exactly when

$$\alpha^2 - i\Gamma\alpha - \omega_0^2 = 0,$$

that is, when

$$\alpha = \frac{i\Gamma}{2} \pm \sqrt{\omega_0^2 - \frac{\Gamma^2}{4}}.$$

Proof

Each time derivative of $e^{i\alpha t}$ multiplies it by $i\alpha$, so substitution gives

$$\left(-\alpha^2 + i\Gamma\alpha + \omega_0^2\right)e^{i\alpha t} = 0.$$

The exponential is never zero, so the bracket must vanish. The quadratic formula gives the two roots. ∎

The sign of the quantity under the square root decides the character of the motion. There are three cases.

2.5 *Case 1: underdamped, $\omega_0^2 > \Gamma^2/4$*

Theorem — Underdamped solution

If the drag is weak, $\omega_0^2 > \Gamma^2/4$, define

$$\omega \equiv \sqrt{\omega_0^2 - \frac{\Gamma^2}{4}}.$$

The general solution is

$$\theta(t) = A\,e^{-\Gamma t/2}\cos(\omega t + \phi),$$

or equivalently $\theta(t) = e^{-\Gamma t/2}(a\cos\omega t + b\sin\omega t)$.

Proof

The two roots are $\alpha_\pm = i\Gamma/2 \pm \omega$, giving the complex solutions

$$z_\pm(t) = e^{-\Gamma t/2}\,e^{\pm i\omega t}.$$

The equation is linear with real coefficients, so sums and differences of these

are also solutions. Two real, independent combinations are

$$\theta_1 = \tfrac{1}{2}(z_+ + z_-) = e^{-\Gamma t/2}\cos\omega t,$$

$$\theta_2 = \tfrac{1}{2i}(z_+ - z_-) = e^{-\Gamma t/2}\sin\omega t.$$

A second-order equation has exactly two independent solutions, so the general solution is $a\,\theta_1 + b\,\theta_2$. ■

The motion is an oscillation at the *reduced* frequency $\omega < \omega_0$ inside an exponentially shrinking envelope $\pm Ae^{-\Gamma t/2}$ (Figure 7).

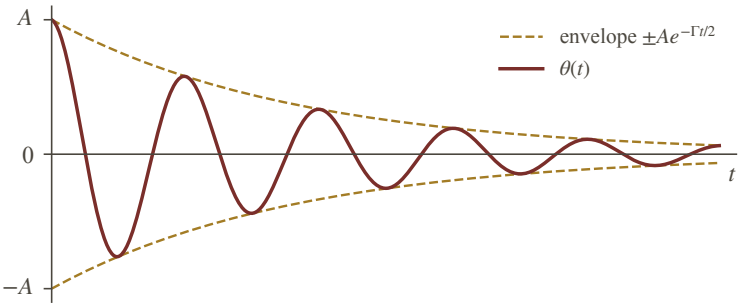


Figure 7 Underdamped motion: oscillation at frequency ω inside the envelope $\pm Ae^{-\Gamma t/2}$.

Example — Underdamped rod released from rest

Impose $\theta(0) = \theta_{\text{initial}}$ and $\dot\theta(0) = 0$ on $\theta = Ae^{-\Gamma t/2}\cos(\omega t + \phi)$:

$$\theta_{\text{initial}} = A\cos\phi,$$

$$0 = -\frac{\Gamma}{2}A\cos\phi - \omega A\sin\phi.$$

The second equation gives the phase and the first then gives the amplitude:

$$\tan\phi = -\frac{\Gamma}{2\omega}, \qquad A = \frac{\theta_{\text{initial}}}{\cos\phi}.$$

Because of the drag, the phase is not zero even though the rod starts at rest.

2.6 Case 2: critically damped, $\omega_0^2 = \Gamma^2/4$

Theorem — Critically damped solution

If $\omega_0^2 = \Gamma^2/4$ the general solution is

$$\theta(t) = (A + Bt)\,e^{-\Gamma t/2}.$$

There is no oscillation.

Proof (as a limit of Case 1)

Now $\omega = 0$ and the two roots coincide, so the method above yields only one

solution. Take the limit $\omega \to 0$ of the underdamped pair instead:

$$\theta_1 = e^{-\Gamma t/2}\cos\omega t \;\longrightarrow\; e^{-\Gamma t/2},$$

while $\theta_2 = e^{-\Gamma t/2}\sin\omega t \to 0$, which is useless. But θ_2/ω is also a solution for every $\omega \neq 0$, and

$$\frac{\theta_2}{\omega} = e^{-\Gamma t/2}\,\frac{\sin\omega t}{\omega} \;\longrightarrow\; t\,e^{-\Gamma t/2}.$$

The general solution is a linear combination of the two limits. ■

Critical damping returns the system to equilibrium as fast as possible without overshooting, which is why it is the design target for door closers and vehicle suspensions.

Tip

Check directly: substituting $te^{-\Gamma t/2}$ into the equation gives zero only when $\omega_0^2 = \Gamma^2/4$.

2.7 Case 3: overdamped, $\omega_0^2 < \Gamma^2/4$

Theorem — Overdamped solution

If the drag is strong, $\omega_0^2 < \Gamma^2/4$, define

$$\Gamma_\pm \equiv \frac{\Gamma}{2} \pm \sqrt{\frac{\Gamma^2}{4} - \omega_0^2}.$$

Both are real and positive, and the general solution is

$$\theta(t) = A_+\,e^{-\Gamma_+ t} + A_-\,e^{-\Gamma_- t}.$$

Proof

The square root in the characteristic equation is now imaginary:

$$\alpha = i\left(\frac{\Gamma}{2} \pm \sqrt{\frac{\Gamma^2}{4} - \omega_0^2}\right) = i\,\Gamma_\pm,$$

so $e^{i\alpha t} = e^{-\Gamma_\pm t}$, two real decaying exponentials. ■

There is again no oscillation. At late times the slower exponential, $e^{-\Gamma_- t}$, dominates. Since $\Gamma_- < \Gamma/2$, an overdamped system returns to equilibrium *more slowly* than a critically damped one with the same ω_0 (Figure 8).

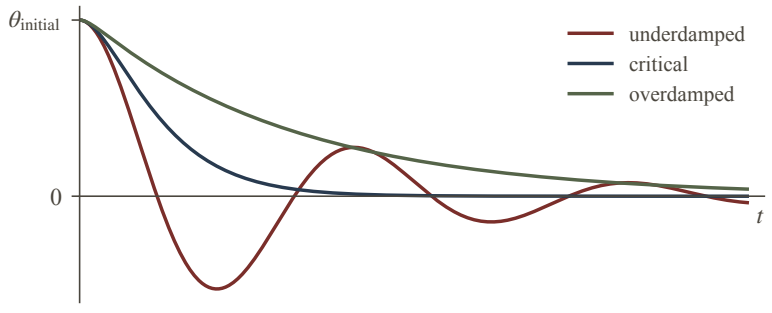


Figure 8 The three regimes for the same ω_0 and the same release from rest. Critical damping reaches equilibrium fastest.

Summary of Lecture 2

(1) $\mu\ddot{x} = -kx$ describes springs, LC circuits and small-angle pendula alike; the energy $\frac{1}{2}kA^2$ is conserved.

(2) Linear drag gives $\ddot{\theta} + \Gamma\dot{\theta} + \omega_0^2\theta = 0$.

(3) Underdamped: $Ae^{-\Gamma t/2}\cos(\omega t + \phi)$ with $\omega = \sqrt{\omega_0^2 - \Gamma^2/4}$.

(4) Critical: $(A + Bt)e^{-\Gamma t/2}$. Overdamped: two decaying exponentials.

Check yourself — Lecture 2

(1) An oscillator has $\omega_0 = 10\ \mathrm{s}^{-1}$. Classify the motion for $\Gamma = 4$, 20 and $30\ \mathrm{s}^{-1}$.

(2) How long does the amplitude of an underdamped oscillator take to fall to $1/e$ of its starting value?

(3) Find ω_0 for an LC circuit with $L = 1$ mH and $C = 1\ \mu$F.

Answers are in Appendix B.

3 Lecture 3: Driven Oscillators, Transients and Resonance

A damped oscillator left alone always comes to rest. To keep it moving, something must push it. This lecture adds a periodic driving force and finds that, after an initial messy period, the oscillator settles into motion at the *driver's* frequency, with an amplitude that can become very large when that frequency is close to the oscillator's own.

3.1 Recap: the free damped oscillator

Reading
Georgi, Chapter 2: Forced Oscillation and Resonance.

The solutions of $\ddot{\theta}+\Gamma\dot{\theta}+\omega_0^2\theta=0$ found in **§2** are collected here, because they reappear below as the transient part of the driven motion.

Regime	Condition	Solution $\theta(t)$
Undamped	$\Gamma = 0$	$A\cos(\omega_0 t+\alpha)$
Underdamped	$\omega_0^2 > \Gamma^2/4$	$Ae^{-\Gamma t/2}\cos(\omega t+\alpha)$
Critical	$\omega_0^2 = \Gamma^2/4$	$(A+Bt)\,e^{-\Gamma t/2}$
Overdamped	$\omega_0^2 < \Gamma^2/4$	$Ae^{-(\Gamma/2+\beta)t}+Be^{-(\Gamma/2-\beta)t}$

Here $\omega=\sqrt{\omega_0^2-\Gamma^2/4}$ and $\beta=\sqrt{\Gamma^2/4-\omega_0^2}$.

3.2 The driven equation and its steady-state solution

Return to the hanging rod and apply an external torque $\tau_{\text{drive}} = d_0\cos\omega_d t$. The total torque is $\tau_g+\tau_{\text{drag}}+\tau_{\text{drive}}$.

Definition — The driven damped oscillator

$$\ddot{\theta}+\Gamma\dot{\theta}+\omega_0^2\,\theta = f_0\cos\omega_d t,$$

where $f_0 = d_0/I$ measures the strength of the drive and ω_d is the *driving angular frequency*. For the rod, $\omega_0^2 = 3g/2\ell$ and $\Gamma = 3b/m\ell^2$ as before.

Watch Out
Three frequencies appear: natural ω_0, damped ω, and driving ω_d. Only ω_d is chosen by the experimenter.

! Keep the three frequencies apart

(1) ω_0 — the natural frequency, with no drag and no drive.

(2) $\omega=\sqrt{\omega_0^2-\Gamma^2/4}$ — the frequency of free oscillation when drag is present; slightly below ω_0.

(3) ω_d — the frequency of the external drive; it has nothing to do with the oscillator's own properties.

Theorem — Steady-state solution

A particular solution of the driven equation is

$$\theta(t) = A(\omega_d)\cos\big(\omega_d t-\delta(\omega_d)\big),$$

with amplitude and phase lag

$$A(\omega_d) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + \omega_d^2\Gamma^2}},$$

$$\tan\delta = \frac{\Gamma\,\omega_d}{\omega_0^2 - \omega_d^2}, \qquad 0 \le \delta \le \pi.$$

It contains no free constants.

Proof

Replace the real equation by the complex one,

$$\ddot{z} + \Gamma\dot{z} + \omega_0^2 z = f_0\, e^{i\omega_d t}, \qquad \theta = \operatorname{Re} z,$$

and try $z(t) = A\, e^{i(\omega_d t - \delta)}$ with real A and δ. The phase δ allows the response to lag behind the drive. Since $\dot{z} = i\omega_d z$ and $\ddot{z} = -\omega_d^2 z$,

$$\left(-\omega_d^2 + i\omega_d\Gamma + \omega_0^2\right) A\, e^{i(\omega_d t - \delta)} = f_0\, e^{i\omega_d t}.$$

Cancel $e^{i\omega_d t}$ and multiply by $e^{i\delta}$:

$$\left(\omega_0^2 - \omega_d^2 + i\omega_d\Gamma\right) A = f_0(\cos\delta + i\sin\delta).$$

The real and imaginary parts must hold separately:

$$(\omega_0^2 - \omega_d^2)A = f_0\cos\delta, \qquad \omega_d\Gamma A = f_0\sin\delta.$$

Squaring and adding gives A; dividing the second by the first gives $\tan\delta$. Since $\sin\delta \ge 0$, the phase lies between 0 and π. ■

3.3 The full solution and transients

The steady-state solution has no adjustable constants, so on its own it cannot match arbitrary initial conditions. The missing freedom comes from the free equation.

Theorem — General solution of the driven oscillator

The general solution is the steady-state solution plus the general solution of the undriven equation. In the underdamped case,

$$\theta(t) = A(\omega_d)\cos(\omega_d t - \delta) + B\, e^{-\Gamma t/2}\cos(\omega t + \alpha),$$

with B and α fixed by the initial conditions.

Proof

Let θ_p solve the driven equation and θ_h the undriven one. By linearity, $\theta_p + \theta_h$ gives $f_0\cos\omega_d t + 0$ on the right-hand side, so it is a solution. It contains two free constants, which is the number a second-order equation requires, so it is the general solution. ■

The second term is the *transient*. While it lasts, the motion is a superposition of two oscillations at different frequencies, ω_d and ω, and can look quite irregular. It

decays like $e^{-\Gamma t/2}$, so after a time of several $2/\Gamma$ only the steady state is left, and the oscillator has forgotten its initial conditions (Figure 9).

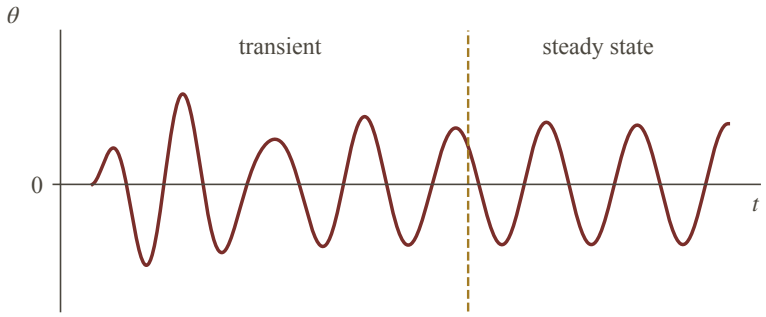


Figure 9 A driven oscillator started from rest. The irregular early motion is the beating of the transient against the steady state; later only the steady oscillation at ω_d remains.

3.4 Worked example: a pendulum driven at its support

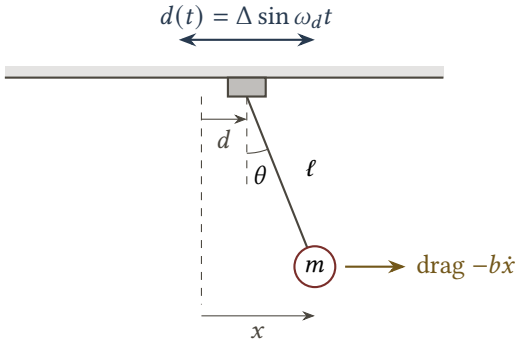


Figure 10 A pendulum driven by moving its support. Both x and d are measured from the same fixed vertical line.

Example — Deriving the equation of motion

A bob of mass m hangs on a string of length ℓ. The support is moved horizontally, $d(t) = \Delta \sin \omega_d t$, and the bob at horizontal position x feels a drag $-b\dot{x}$. The string makes the small angle $\theta \approx (x - d)/\ell$ with the vertical, so its tension has components

$$T_x \approx -T\,\frac{x-d}{\ell}, \qquad T_y \approx T.$$

There is no vertical motion for small angles, so $T = mg$. Horizontally,

$$m\ddot{x} = -b\dot{x} - \frac{mg}{\ell}(x - d),$$

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{g}{\ell}x = \frac{g\Delta}{\ell}\sin\omega_d t.$$

This is the driven equation with

$$\Gamma = \frac{b}{m}, \qquad \omega_0^2 = \frac{g}{\ell}, \qquad f_0 = \frac{g\Delta}{\ell} = \omega_0^2\Delta.$$

The drive is a sine instead of a cosine, which only shifts the origin of time; the amplitude and phase formulas are unchanged.

3.5 Three regimes of the response

Theorem — Low, high and resonant driving

For the driven pendulum, with $f_0 = \omega_0^2 \Delta$:

(1) **Slow drive**, $\omega_d \to 0$: $A \to f_0/\omega_0^2 = \Delta$ and $\delta \to 0$. The bob follows the support.

(2) **Fast drive**, $\omega_d \to \infty$: $A \to 0$ and $\delta \to \pi$. The bob barely moves, and what motion there is opposes the support.

(3) **On resonance**, $\omega_d = \omega_0$: $\delta = \pi/2$ and

$$A(\omega_0) = \frac{f_0}{\omega_0 \Gamma} = \frac{\omega_0}{\Gamma}\,\Delta = Q\,\Delta, \qquad Q \equiv \frac{\omega_0}{\Gamma}.$$

Proof

Each follows from the amplitude and phase formulas. (1) Set $\omega_d = 0$: the square root becomes ω_0^2 and $\tan\delta = 0$. (2) For large ω_d the square root grows like ω_d^2, so $A \to 0$; $\tan\delta \to 0$ from the negative side, which with $\sin\delta \geq 0$ means $\delta \to \pi$. (3) At $\omega_d = \omega_0$ the first term under the root vanishes, leaving $\omega_0\Gamma$, and $\tan\delta \to \infty$ gives $\delta = \pi/2$. ∎

Definition — Quality factor and resonance

The dimensionless number

$$Q \equiv \frac{\omega_0}{\Gamma}$$

is the *quality factor* of the oscillator. Driving near ω_0 is called driving *on resonance*; the response is then Q times larger than the response to a very slow drive of the same strength.

A lightly damped oscillator has $Q \gg 1$, so even a tiny periodic push at the right frequency builds up a large motion. This is how a singer can shatter a wine glass, and why a child on a swing must be pushed in time with the swing. Figure 11 shows the amplitude and phase across the whole range of ω_d. The amplitude peak sits slightly below ω_0, at $\omega_d^2 = \omega_0^2 - \Gamma^2/2$; for large Q the difference is negligible.

Remark — Reading the phase physically

Well below resonance the restoring force dominates and the oscillator moves with the drive. Well above resonance its inertia dominates and it moves against the drive. Exactly on resonance those two effects cancel, the drive works only against the drag, and the displacement lags the force by a quarter of a cycle. The velocity is then in phase with the force, so the drive delivers power at every instant.

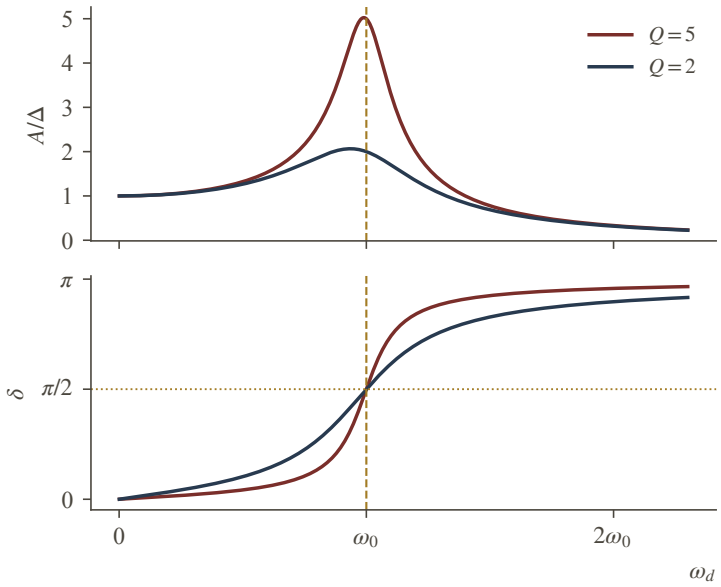


Figure 11 Steady-state amplitude (top) and phase lag (bottom) against driving frequency for two values of Q. Less damping gives a taller, narrower peak and a sharper jump in phase.

Summary of Lecture 3

(1) The driven equation is $\ddot{\theta} + \Gamma\dot{\theta} + \omega_0^2\theta = f_0 \cos \omega_d t$.

(2) Steady state: $A(\omega_d)\cos(\omega_d t - \delta)$, at the driving frequency, with no free constants.

(3) Full solution = steady state + a transient that decays like $e^{-\Gamma t/2}$.

(4) On resonance the amplitude is enhanced by $Q = \omega_0/\Gamma$ and the phase lag is $\pi/2$.

Check yourself — Lecture 3

(1) An oscillator has $\omega_0 = 100\ \mathrm{s}^{-1}$ and $\Gamma = 2\ \mathrm{s}^{-1}$. Its support is shaken with amplitude 1 mm on resonance. What is the steady-state amplitude?

(2) State the phase lag δ for $\omega_d \ll \omega_0$, $\omega_d = \omega_0$ and $\omega_d \gg \omega_0$.

(3) Roughly how long after switching on the drive does the motion become steady?

Answers are in Appendix B.

4 Lecture 4: Coupled Oscillators and Normal Modes

So far there has been one moving object. When several oscillators are connected, each one pushes on its neighbours and the motion can look hopelessly complicated: energy wanders from one part to another and nothing repeats. The key idea of this lecture is that such motion is always a sum of a few very simple motions, the *normal modes*, and that there is a mechanical procedure for finding them.

Reading
Georgi, Chapter 3: Normal Modes.

4.1 What a normal mode is

Definition — Normal mode

A *normal mode* of a coupled system is a motion in which every part of the system oscillates with the *same frequency* and the *same phase*. Only the amplitudes differ from part to part, and their ratios stay fixed in time. (A negative amplitude ratio means two parts move exactly opposite to each other.)

A system with n coordinates has n normal modes, and its most general motion is a superposition of them. Finding the modes therefore solves the problem completely.

4.2 Guessing the modes of a symmetric system

Consider three blocks on a frictionless line (Figure 12). The middle block has mass $2m$ and displacement x_1; the outer blocks have mass m and displacements x_2 (right) and x_3 (left). Two identical springs of constant k join the middle block to each outer block. All displacements are measured from equilibrium.

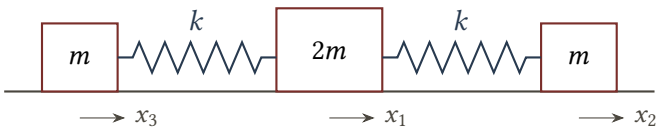


Figure 12 Three blocks joined by two identical springs. The heavy block sits in the middle.

The system is symmetric under exchanging left and right, which suggests looking for motions that respect that symmetry.

Example — Three modes found by intuition

Mode A. The outer blocks move equally and oppositely, so the two spring forces on the middle block cancel and it stays still. Each outer block is then a mass m on a spring k attached to a fixed point:

$$x_1 = 0, \quad x_2 = A\cos(\omega_A t + \phi_A), \quad x_3 = -x_2, \qquad \omega_A^2 = \frac{k}{m}.$$

Mode B. The outer blocks move together and the middle block moves the opposite way with the same amplitude, which keeps the centre of mass at rest ($2m \cdot B + m(-B) + m(-B) = 0$):

$$x_1 = B\cos(\omega_B t + \phi_B), \quad x_2 = x_3 = -x_1, \qquad \omega_B^2 = \frac{2k}{m}.$$

To see the frequency, look at an outer block: its spring is stretched by $x_2 - x_1 = 2x_2$, so $m\ddot{x}_2 = -2kx_2$.

Mode C. All three blocks move together, no spring is stretched and no force acts:

$$x_1 = x_2 = x_3 = c + vt, \qquad \omega_C = 0.$$

This is uniform translation of the whole system, not an oscillation.

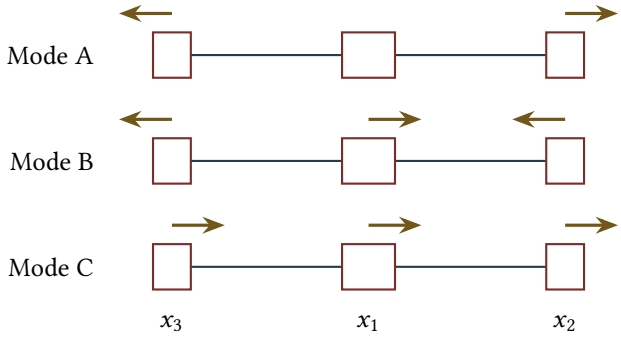


Figure 13 The three normal modes of the three-block system. Arrows show the relative displacements at one instant; half a period later all arrows in Modes A and B are reversed.

Theorem — General motion of the three blocks

$$\begin{aligned} x_1 &= B\cos(\omega_B t + \phi_B) + c + vt, \\ x_2 &= A\cos(\omega_A t + \phi_A) - B\cos(\omega_B t + \phi_B) + c + vt, \\ x_3 &= -A\cos(\omega_A t + \phi_A) - B\cos(\omega_B t + \phi_B) + c + vt. \end{aligned}$$

The six constants A, B, ϕ_A, ϕ_B, c, v are fixed by the six initial conditions (three positions and three velocities).

Three second-order equations need six constants, and there are six, so this is the complete solution. The rest of the lecture shows how to reach it without guessing.

4.3 *The matrix method*

Example — Equations of motion in matrix form

A spring between blocks i and j pulls block i with force $k(x_j - x_i)$. Hence

$$\begin{aligned} 2m\ddot{x}_1 &= k(x_2 - x_1) + k(x_3 - x_1), \\ m\ddot{x}_2 &= k(x_1 - x_2), \\ m\ddot{x}_3 &= k(x_1 - x_3). \end{aligned}$$

Collecting terms, this is $M\ddot{X} = -KX$ with

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad M = \begin{pmatrix} 2m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{pmatrix},$$

$$K = \begin{pmatrix} 2k & -k & -k \\ -k & k & 0 \\ -k & 0 & k \end{pmatrix}.$$

Theorem — Normal modes from the eigenvalue problem

For a system $M\ddot{X} = -KX$, a normal mode has the form

$$X(t) = \mathrm{Re}\left[A\, e^{i(\omega t+\phi)}\right],$$

with A a constant column vector. Such a motion is a solution exactly when

$$\left(M^{-1}K - \omega^2 I\right)A = 0.$$

A non-zero A exists only if

$$\det\left(M^{-1}K - \omega^2 I\right) = 0.$$

The roots ω^2 of this equation are the squared normal-mode frequencies, and for each root the corresponding A gives the amplitude ratios. In the language of linear algebra this is an eigenvalue problem: ω^2 is an eigenvalue of $M^{-1}K$ and A its eigenvector.

Proof

Write $Z = A\, e^{i(\omega t+\phi)}$, so that $\ddot{Z} = -\omega^2 Z$. Substituting into $M\ddot{Z} = -KZ$ gives $\omega^2 MZ = KZ$. Cancel the exponential and multiply by M^{-1}:

$$\omega^2 A = M^{-1}KA.$$

This is a set of homogeneous linear equations for the components of A. It has a solution other than $A = 0$ only if the matrix $M^{-1}K - \omega^2 I$ is singular, that is, has zero determinant. ■

Example — Frequencies of the three blocks

With $\omega_0^2 \equiv k/m$,

$$M^{-1}K - \omega^2 I = \begin{pmatrix} \omega_0^2 - \omega^2 & -\frac{1}{2}\omega_0^2 & -\frac{1}{2}\omega_0^2 \\ -\omega_0^2 & \omega_0^2 - \omega^2 & 0 \\ -\omega_0^2 & 0 & \omega_0^2 - \omega^2 \end{pmatrix}.$$

Expanding the determinant along the first row,

$$(\omega_0^2 - \omega^2)^3 - \omega_0^4(\omega_0^2 - \omega^2) = 0,$$

$$(\omega_0^2 - \omega^2)\,\omega^2\,(\omega^2 - 2\omega_0^2) = 0.$$

The three roots are

$$\omega^2 = \frac{k}{m}, \qquad \omega^2 = \frac{2k}{m}, \qquad \omega^2 = 0,$$

exactly the frequencies of Modes A, B and C.

Example — Amplitude ratios

Put each frequency back into $(K - \omega^2 M)A = 0$.
Mode B, $\omega^2 = 2k/m$. Then $\omega^2 M$ has diagonal entries $4k, 2k, 2k$, and the three

equations are

$$0 = -2kA_1 - kA_2 - kA_3,$$
$$0 = -kA_1 - kA_2,$$
$$0 = -kA_1 - kA_3.$$

The last two give $A_2 = A_3 = -A_1$; the first is then satisfied automatically. So $A \propto (1, -1, -1)$.
Mode A, $\omega^2 = k/m$. The second equation becomes $-kA_1 = 0$, so $A_1 = 0$, and the first gives $A_2 = -A_3$. So $A \propto (0, 1, -1)$.
Mode C, $\omega^2 = 0$. The equations give $A_1 = A_2 = A_3$, so $A \propto (1, 1, 1)$.

Watch Out

One equation is always redundant at a mode frequency. Only ratios of amplitudes are determined, never their overall size.

! The overall amplitude is not fixed by the mode

The eigenvalue equation fixes the *shape* of each mode, not how much of it is present. The amount and phase of each mode in a given motion come from the initial conditions.

4.4 Choice of origin does not matter

One may instead measure each block's position from a common fixed origin, so that the natural length ℓ_0 of the springs appears explicitly:

$$2m\ddot{x}_1 = k(x_2 - x_1 - \ell_0) + k(x_3 - x_1 + \ell_0),$$
$$m\ddot{x}_2 = k(x_1 - x_2 + \ell_0),$$
$$m\ddot{x}_3 = k(x_1 - x_3 - \ell_0).$$

Shifting the coordinates by constants, $x_2' = x_2 - \ell_0$ and $x_3' = x_3 + \ell_0$, removes ℓ_0 and returns exactly the equations above. A constant shift does not change any second derivative, so the frequencies and mode shapes are the same. Measuring from equilibrium simply makes this automatic.

Recipe — Finding normal modes

(1) Choose coordinates measured from equilibrium and write Newton's law for each one.

(2) Arrange the equations as $M\ddot{X} = -KX$.

(3) Solve $\det(M^{-1}K - \omega^2 I) = 0$ for the mode frequencies.

(4) For each frequency solve $(M^{-1}K - \omega^2 I)A = 0$ for the amplitude ratios.

(5) Write the general solution as a sum of all modes, each with its own amplitude and phase, and fix those from the initial conditions.

Summary of Lecture 4

(1) In a normal mode all parts oscillate with one frequency and one phase.

(2) Symmetry often lets the modes be guessed; the matrix method finds them in every case.

(3) A system with n coordinates has n modes; a mode with $\omega = 0$ is a uniform translation.

(4) The general motion is a superposition of the normal modes.

Check yourself — Lecture 4

(1) How many normal modes has a line of five masses joined by springs, moving in one dimension?

(2) Why does Mode C of the three blocks have zero frequency?

(3) In Mode B, why does the middle block have the same amplitude as the outer ones, with opposite sign?

Answers are in Appendix B.

5 Lecture 5: Beat Phenomena

This lecture applies the recipe of **§4** to two pendula joined by a spring. The result explains a striking demonstration: start one pendulum swinging, and after a while it stops while the other swings; then the motion comes back. The energy passes back and forth at a slow *beat* frequency.

5.1 Two pendula coupled by a spring

Lecturer
Lectures 5 and 6 were given by Prof. Bolek Wyslouch.

Two identical pendula, each a bob of mass m on a string of length ℓ, hang side by side and are joined by a spring of constant k (Figure 14). Let x_1 and x_2 be the horizontal displacements of the bobs from equilibrium. There is no drag. Initially neither bob moves; bob 2 is displaced by x_0 and bob 1 is at equilibrium.

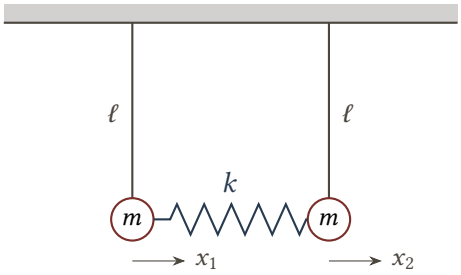


Figure 14 Two identical pendula coupled by a spring.

Example — Equations of motion

For bob 1, with string tension T_1 at angle θ_1 to the vertical:

$$m\ddot{y}_1 = T_1\cos\theta_1 - mg,$$

$$m\ddot{x}_1 = -T_1\sin\theta_1 + k(x_2 - x_1).$$

For small angles $\cos\theta_1 \approx 1$ and $\sin\theta_1 \approx x_1/\ell$. There is no vertical motion to this order, so $T_1 = mg$, and

$$m\ddot{x}_1 = -\left(k + \frac{mg}{\ell}\right)x_1 + k\,x_2.$$

By the same argument,

$$m\ddot{x}_2 = k\,x_1 - \left(k + \frac{mg}{\ell}\right)x_2.$$

In matrix form $M\ddot{X} = -KX$ with $M = mI$ and

$$M^{-1}K = \begin{pmatrix} \dfrac{g}{\ell} + \dfrac{k}{m} & -\dfrac{k}{m} \\[2mm] -\dfrac{k}{m} & \dfrac{g}{\ell} + \dfrac{k}{m} \end{pmatrix}.$$

Theorem — Normal modes of the coupled pendula

The two normal modes are

(1) the **symmetric mode**, $A^{(1)} = (1, 1)$, with $\omega_1^2 = \dfrac{g}{\ell}$;

(2) the **antisymmetric mode**, $A^{(2)} = (1, -1)$, with $\omega_2^2 = \dfrac{g}{\ell} + \dfrac{2k}{m}$.

Proof

The determinant condition is

$$\left(\frac{g}{\ell} + \frac{k}{m} - \omega^2\right)^2 - \left(\frac{k}{m}\right)^2 = 0,$$

so $\dfrac{g}{\ell} + \dfrac{k}{m} - \omega^2 = \pm\dfrac{k}{m}$, giving the two values of ω^2. For $\omega^2 = g/\ell$ the matrix $M^{-1}K - \omega^2 I$ is

$$\frac{k}{m}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

which annihilates $(1, 1)$. For $\omega^2 = g/\ell + 2k/m$ it is

$$-\frac{k}{m}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

which annihilates $(1, -1)$. ∎



symmetric: $\omega_1^2 = g/\ell$ antisymmetric: $\omega_2^2 = g/\ell + 2k/m$

Figure 15 The two normal modes of the coupled pendula. In the symmetric mode the spring (dashed) keeps its natural length; in the antisymmetric mode it is stretched and compressed.

Both results make physical sense. In the symmetric mode the bobs move together, the spring is never stretched, and each bob is simply a free pendulum with $\omega^2 = g/\ell$. In the antisymmetric mode the spring is stretched by twice each bob's displacement and adds its own restoring force, raising the frequency.

5.2 The general solution and the initial conditions

Theorem — General solution

$$x_1 = \alpha\cos(\omega_1 t + \phi_1) + \beta\cos(\omega_2 t + \phi_2),$$
$$x_2 = \alpha\cos(\omega_1 t + \phi_1) - \beta\cos(\omega_2 t + \phi_2),$$

with α, β, ϕ_1, ϕ_2 fixed by the initial conditions.

Example — One bob displaced, both at rest

The conditions are $x_1(0) = 0$, $x_2(0) = x_0$, $\dot{x}_1(0) = \dot{x}_2(0) = 0$. Zero initial

velocities give $\phi_1 = \phi_2 = 0$. The positions give

$$\alpha + \beta = 0, \qquad \alpha - \beta = x_0,$$

so $\alpha = x_0/2$ and $\beta = -x_0/2$. Therefore

$$x_1 = \frac{x_0}{2}\left(\cos\omega_1 t - \cos\omega_2 t\right),$$

$$x_2 = \frac{x_0}{2}\left(\cos\omega_1 t + \cos\omega_2 t\right).$$

Displacing one bob excites both modes equally.

5.3 Beats

Theorem — Product form of the motion

The solution of the example above can be written

$$x_1 = x_0 \sin\!\left(\frac{\omega_2 - \omega_1}{2}\, t\right) \sin\!\left(\frac{\omega_1 + \omega_2}{2}\, t\right),$$

$$x_2 = x_0 \cos\!\left(\frac{\omega_2 - \omega_1}{2}\, t\right) \cos\!\left(\frac{\omega_1 + \omega_2}{2}\, t\right).$$

Identities

$\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2}$, *and* $\cos a - \cos b = -2\sin\frac{a+b}{2}\sin\frac{a-b}{2}$.

Definition — Carrier and beat

When the two mode frequencies are close, $\omega_1 \approx \omega_2$, each displacement is a fast oscillation at the *carrier* frequency

$$\omega_{\text{carrier}} = \frac{\omega_1 + \omega_2}{2},$$

multiplied by a slowly varying amplitude at

$$\omega_{\text{mod}} = \frac{\omega_2 - \omega_1}{2}.$$

The slow rise and fall of the amplitude is called *beating*.

For example, if $\omega_1 = 0.9\,\omega_2$ the carrier is at $0.95\,\omega_2$ and the modulation at $0.05\,\omega_2$, nineteen times slower. The envelope of bob 2 is a cosine and that of bob 1 a sine, so when one bob swings with full amplitude the other is at rest, and they exchange roles every quarter of a modulation period (Figure 16).

Watch Out

The amplitude peaks twice per modulation period, so the audible beat frequency is $\omega_2 - \omega_1$, not half of it.

! Beat frequency versus modulation frequency

The envelope $\cos\!\left(\frac{1}{2}(\omega_2 - \omega_1)t\right)$ has period $4\pi/(\omega_2 - \omega_1)$, but the *magnitude* of the amplitude returns to its maximum twice in that time. Bursts of large motion therefore repeat with period $2\pi/(\omega_2 - \omega_1)$. For two tuning forks of nearly equal pitch, the loudness pulses at the difference of their frequencies.

Beats require weak coupling here: $\omega_2^2 - \omega_1^2 = 2k/m$, so a soft spring gives nearly equal frequencies and a slow, clearly visible exchange.

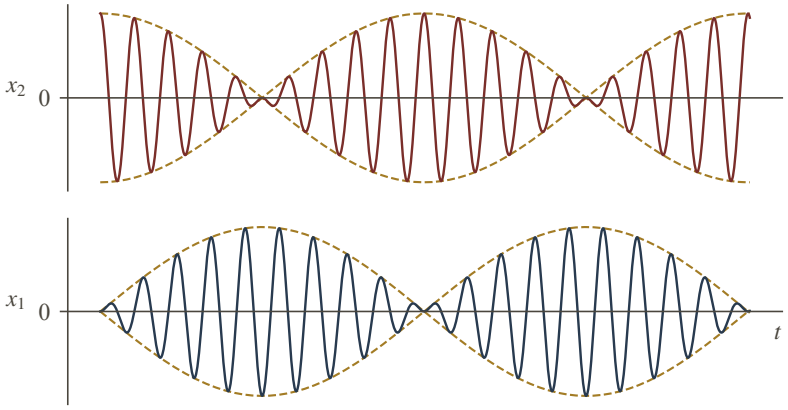


Figure 16 Beats in two weakly coupled pendula. When bob 2 (top) is swinging fully, bob 1 (bottom) is at rest, and vice versa.

5.4 Normal coordinates

The modes can also be found without matrices, by choosing better coordinates.

Definition — Normal coordinates

For the coupled pendula define

$$U_1 \equiv x_1 + x_2, \qquad U_2 \equiv x_1 - x_2.$$

These are the *normal coordinates* of the system.

Theorem — Normal coordinates decouple the equations

$$m\ddot{U}_1 = -\frac{mg}{\ell} U_1, \qquad m\ddot{U}_2 = -\left(\frac{mg}{\ell} + 2k\right) U_2.$$

Each normal coordinate is an independent simple harmonic oscillator, with frequency ω_1 and ω_2 respectively.

Proof

Add the two equations of motion: the spring terms cancel,

$$m(\ddot{x}_1 + \ddot{x}_2) = -\frac{mg}{\ell}(x_1 + x_2).$$

Subtract them: the spring terms add,

$$m(\ddot{x}_1 - \ddot{x}_2) = -\left(\frac{mg}{\ell} + 2k\right)(x_1 - x_2).$$

■

In the original coordinates the two equations were tangled together. In normal coordinates there is no coupling at all: U_1 never learns what U_2 is doing. Exciting a single mode means setting one normal coordinate in motion and leaving the other at zero.

Summary of Lecture 5

(1) Two coupled pendula have a symmetric mode with $\omega_1^2 = g/\ell$ and an antisymmetric mode with $\omega_2^2 = g/\ell + 2k/m$.

(2) Displacing one bob excites both modes equally.

(3) Two nearby frequencies superpose into a carrier at their average, modulated at half their difference: beats.

(4) In normal coordinates $x_1 \pm x_2$ the equations decouple.

Check yourself — Lecture 5

(1) Two coupled pendula have $\omega_1 = 10$ and $\omega_2 = 11\ \mathrm{s}^{-1}$. Find the carrier frequency, the modulation frequency, and the time between successive moments when a given bob is at rest.

(2) What initial conditions excite only the antisymmetric mode?

(3) What are the normal coordinates of this system, and why are they useful?

Answers are in Appendix B.

6 Lecture 6: Driven Coupled Oscillators and Resonance

Lecture 3 drove a single oscillator and found one resonance. A coupled system has several normal modes, and this lecture shows that it has one resonance for each of them. Driving at a mode frequency makes the whole system respond in the shape of that mode.

Reading

Georgi, Chapter 3 (forced oscillation is §3.5), then Chapter 4 and Chapter 5 through §5.2.

6.1 Arbitrary motion versus a single normal mode

It is worth restating what makes a normal mode special, by comparing it with the general motion of the same system.

	Arbitrary excitation	One normal mode
Motion of each part	not harmonic	harmonic
Amplitude ratios	change in time	constant
Energy	migrates between parts	stays where it is

6.2 Adding a driving force

Take the two coupled pendula of **§5** and push the left bob with a horizontal force $F_d = F_0 \cos \omega_d t$.

Definition — Driven coupled system

The equations of motion are

$$m\ddot{x}_1 = -\left(k + \frac{mg}{\ell}\right)x_1 + k\,x_2 + F_0 \cos \omega_d t,$$
$$m\ddot{x}_2 = k\,x_1 - \left(k + \frac{mg}{\ell}\right)x_2,$$

or, in matrix form,

$$\ddot{X} + M^{-1}KX = M^{-1}F \cos \omega_d t, \qquad M^{-1}F = \begin{pmatrix} F_0/m \\ 0 \end{pmatrix},$$

with $M^{-1}K$ as in **§5**.

Without the drive, the solution is the homogeneous one found last time:

$$X_h = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(\omega_1 t + \phi_1) + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(\omega_2 t + \phi_2),$$

with $\omega_1^2 = g/\ell$ and $\omega_2^2 = g/\ell + 2k/m$. A useful fact for what follows is that the determinant factorises in terms of these:

$$\det\left(M^{-1}K - \omega^2 I\right) = (\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2).$$

6.3 The steady-state response

Theorem — Particular solution

The driven system has the particular solution $X_p = B\cos\omega_d t$, where the constant vector $B = (B_1, B_2)$ satisfies

$$\left(M^{-1}K - \omega_d^2 I\right)B = M^{-1}F.$$

Proof

Go to complex notation, $X = \operatorname{Re} Z$, with

$$\ddot{Z} + M^{-1}KZ = M^{-1}F\,e^{i\omega_d t}.$$

Try $Z = B\,e^{i\omega_d t}$. Then $\ddot{Z} = -\omega_d^2 Z$ and the common factor $e^{i\omega_d t}$ cancels, leaving the stated equation. There is no damping, so the equation is real and B is real: the response is exactly in phase or exactly out of phase with the drive. ∎

Written out, these are two ordinary simultaneous equations:

$$\left(\frac{k}{m} + \frac{g}{\ell} - \omega_d^2\right)B_1 - \frac{k}{m}B_2 = \frac{F_0}{m},$$
$$-\frac{k}{m}B_1 + \left(\frac{k}{m} + \frac{g}{\ell} - \omega_d^2\right)B_2 = 0.$$

They can be solved by elimination, but a rule that scales to many coupled oscillators is more useful.

Linear algebra

Cramer's rule is practical for 2 or 3 unknowns. For larger systems use elimination.

Theorem — Cramer's rule

Let $E\,B = D$ be n linear equations in n unknowns with $\det E \neq 0$. Then

$$B_j = \frac{\det E_j}{\det E},$$

where E_j is the matrix E with its j-th column replaced by the column vector D.

Example — Amplitudes of the two bobs

Here $E = M^{-1}K - \omega_d^2 I$ and $D = (F_0/m,\, 0)$, with $\det E = (\omega_d^2 - \omega_1^2)(\omega_d^2 - \omega_2^2)$. Replacing the first column of E by D:

$$\det E_1 = \begin{vmatrix} \dfrac{F_0}{m} & -\dfrac{k}{m} \\ 0 & \dfrac{k}{m} + \dfrac{g}{\ell} - \omega_d^2 \end{vmatrix} = \frac{F_0}{m}\left(\frac{k}{m} + \frac{g}{\ell} - \omega_d^2\right).$$

Replacing the second column:

$$\det E_2 = \begin{vmatrix} \dfrac{k}{m} + \dfrac{g}{\ell} - \omega_d^2 & \dfrac{F_0}{m} \\ -\dfrac{k}{m} & 0 \end{vmatrix} = \frac{F_0}{m}\,\frac{k}{m}.$$

Therefore

$$B_1 = \frac{\frac{F_0}{m}\left(\frac{k}{m} + \frac{g}{\ell} - \omega_d^2\right)}{(\omega_d^2 - \omega_1^2)(\omega_d^2 - \omega_2^2)},$$

$$B_2 = \frac{\frac{F_0}{m}\frac{k}{m}}{(\omega_d^2 - \omega_1^2)(\omega_d^2 - \omega_2^2)}.$$

6.4 Resonance excites a normal mode

Theorem — Resonances of the coupled system

(1) Both amplitudes diverge as $\omega_d \to \omega_1$ and as $\omega_d \to \omega_2$: there is one resonance per normal mode.

(2) The ratio of the two amplitudes is

$$\frac{B_1}{B_2} = \frac{k/m + g/\ell - \omega_d^2}{k/m}.$$

(3) At $\omega_d = \omega_1$ this ratio is $+1$, and at $\omega_d = \omega_2$ it is -1: the response has the shape of the mode being driven.

Proof

(1) The common denominator vanishes at the two mode frequencies while the numerator of B_2 does not. (2) Divide B_1 by B_2. (3) Put $\omega_d^2 = g/\ell$ to get $(k/m)/(k/m) = 1$, and $\omega_d^2 = g/\ell + 2k/m$ to get $(-k/m)/(k/m) = -1$. ∎

This is the practical meaning of the lecture. Although only the left bob is pushed, driving at ω_1 makes both bobs swing together, and driving at ω_2 makes them swing against each other. The drive *selects* a normal mode by its frequency (Figure 17).

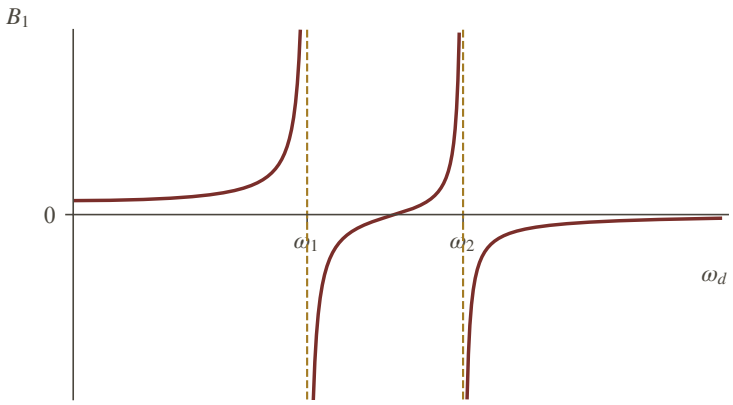


Figure 17 Amplitude B_1 of the driven bob against driving frequency, without damping. It diverges at both mode frequencies and changes sign across each one.

Watch Out

The infinite amplitudes come from neglecting damping. With drag the peaks are finite, as in §3.

! The divergence is not physical

An infinite amplitude signals that the model has left something out. Any real system has some damping, which keeps the peaks finite, and for large amplitudes the small-angle approximation fails as well. The sign changes of B_1 in the undamped model are what remains of the phase lag δ going from 0 to π across a resonance.

Between the resonances there is a frequency, $\omega_d^2 = g/\ell + k/m$, at which $B_1 = 0$: the driven bob stands still while the other one swings. The spring force from bob 2 then cancels the applied force exactly.

6.5 *The full solution*

Theorem — General solution of the driven coupled pendula

$$x_1 = \alpha\cos(\omega_1 t + \phi_1) + \beta\cos(\omega_2 t + \phi_2) + B_1\cos\omega_d t,$$
$$x_2 = \alpha\cos(\omega_1 t + \phi_1) - \beta\cos(\omega_2 t + \phi_2) + B_2\cos\omega_d t.$$

The B terms are the particular solution, fixed entirely by the drive. The α and β terms are the homogeneous solution, fixed by the initial conditions.

With no damping the homogeneous part never dies away. In a real system it is a transient, and after it has decayed only the steady-state B terms remain.

Summary of Lecture 6

(1) A driven coupled system obeys $\ddot{X} + M^{-1}KX = M^{-1}F\cos\omega_d t$.

(2) The steady-state amplitudes solve $(M^{-1}K - \omega_d^2 I)B = M^{-1}F$; Cramer's rule gives them directly.

(3) There is a resonance at every normal-mode frequency.

(4) Driving at a mode frequency excites that mode's shape, wherever the force is applied.

Check yourself — Lecture 6

(1) The left bob is driven exactly at ω_1. What is the ratio B_1/B_2?

(2) At what driving frequency does the driven bob stand still?

(3) Why are the infinite amplitudes at ω_1 and ω_2 not physical?

Answers are in Appendix B.

7 Lecture 7: Symmetry and Infinitely Many Coupled Oscillators

Solving $\det(M^{-1}K - \omega^2 I) = 0$ becomes hopeless as the number of oscillators grows, and impossible when it is infinite. This lecture shows that the *symmetry* of a system gives its normal modes with almost no calculation. Applied to an endless chain of masses and springs, the symmetry under shifting the chain by one step produces sinusoidal modes: this is where sine waves come from.

Reading

Georgi, Chapter 4: Symmetries, and Chapter 5: Waves (through §5.2).

7.1 Reflection symmetry of two coupled oscillators

Take again the two coupled pendula of **§5**. The system looks the same in a mirror. Reflection swaps the two bobs and reverses the direction of each displacement:

$$x_1 \longrightarrow -x_2, \qquad x_2 \longrightarrow -x_1.$$

Definition — Symmetry matrix and commutator

A *symmetry* of a system is a transformation that turns every solution $X(t)$ of the equation of motion into another solution $\tilde{X}(t) = SX(t)$. For the reflection above,

$$S = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

The *commutator* of two matrices is $[A, B] \equiv AB - BA$. If it is zero the matrices are said to *commute*.

Theorem — A symmetry commutes with $M^{-1}K$

S is a symmetry of $\ddot{X} = -M^{-1}KX$ if and only if

$$\left[S,\, M^{-1}K\right] = 0.$$

Proof

Let X be a solution. Multiplying the equation of motion by S gives

$$S\ddot{X} = -S\,M^{-1}K\,X.$$

For $\tilde{X} = SX$ to be a solution we need $S\ddot{X} = -M^{-1}K\,S\,X$. The two agree for every solution X exactly when $SM^{-1}K = M^{-1}KS$. ∎

Theorem — Normal modes are eigenvectors of the symmetry

Suppose $[S, M^{-1}K] = 0$.

(1) If the mode frequencies are all different, every normal mode vector $A^{(n)}$ is an eigenvector of S: $SA^{(n)} = \beta_n A^{(n)}$.

(2) Conversely, if the eigenvalues of S are all different, every eigenvector of S is a normal mode vector.

Linear algebra

Commuting matrices with distinct eigenvalues share their eigenvectors.

Proof

(1) $X = A^{(n)} \cos \omega_n t$ is a solution, so $SX = SA^{(n)} \cos \omega_n t$ is also a solution, oscillating at ω_n. Since no other mode has that frequency, $SA^{(n)}$ must be proportional to $A^{(n)}$.
(2) Let $SA = \beta A$. Then

$$S\,(M^{-1}KA) = M^{-1}K\,SA = \beta\,(M^{-1}KA),$$

so $M^{-1}KA$ is an eigenvector of S with the same eigenvalue β. Because that eigenvalue belongs to only one direction, $M^{-1}KA \propto A$, which says that A is an eigenvector of $M^{-1}K$. ■

Example — The two pendula, without a determinant

Reflecting twice does nothing, $S^2 = I$, so $\beta^2 = 1$ and $\beta = \pm 1$.

(1) $\beta = -1$: $SA = -A$ means $-A_2 = -A_1$, so $A^{(1)} = (1, 1)$.

(2) $\beta = +1$: $SA = A$ means $-A_2 = A_1$, so $A^{(2)} = (1, -1)$.

The frequencies follow by applying $M^{-1}K$ to each vector:

$$M^{-1}K \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{g}{\ell} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

$$M^{-1}K \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \left(\frac{g}{\ell} + \frac{2k}{m} \right) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These are the results of **§5**.

Remark — Solve one, solve them all

The mode shapes $(1, 1)$ and $(1, -1)$ came from the symmetry alone. Every two-coordinate system with this reflection symmetry therefore has the same two mode shapes, whatever its masses, springs or pendulum lengths. Only the frequencies depend on the details.

7.2 An infinite chain of masses and springs

Watch Out

From here on k means the wave number. The spring constant is written κ to keep the two apart.

Now consider identical masses m on a line, joined by identical springs of constant κ and natural length a, extending without end in both directions (Figure 18). Mass j has displacement x_j along the line from its equilibrium position.

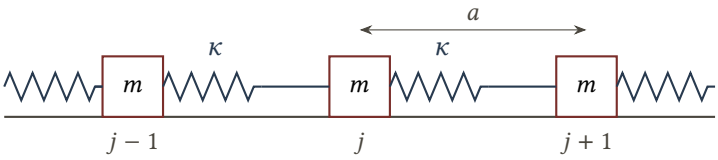


Figure 18 Part of an infinite chain of identical masses and springs.

Definition — Equation of motion of the chain

Each mass feels the springs on either side of it:

$$m\ddot{x}_j = -\kappa(x_j - x_{j-1}) - \kappa(x_j - x_{j+1}),$$

$$m\ddot{x}_j = \kappa\, x_{j-1} - 2\kappa\, x_j + \kappa\, x_{j+1}.$$

In a normal mode $x_j = \mathrm{Re}\left[A_j\, e^{i(\omega t+\phi)}\right]$, and with $\omega_0^2 \equiv \kappa/m$ this becomes

$$\omega^2 A_j = \omega_0^2\left(-A_{j-1} + 2A_j - A_{j+1}\right).$$

The matrix $M^{-1}K$ now has infinitely many rows, each containing $-\omega_0^2,\ 2\omega_0^2,\ -\omega_0^2$ centred on the diagonal. Its determinant cannot be written down. Symmetry does the work instead.

7.3 Space-translation symmetry

Shifting the whole chain by one spacing a leaves it looking exactly the same. In terms of amplitudes this is the map $A \longrightarrow A' = SA$ with

$$A'_j = A_{j+1},$$

so S is an infinite matrix with 1s just above the diagonal and zeros elsewhere.

Theorem — Eigenvectors of translation

The eigenvectors of the translation S are the geometric sequences

$$A_j = \beta^j,$$

one for every non-zero number β, with eigenvalue β.

Proof

$SA = \beta A$ reads $A_{j+1} = \beta A_j$ for all j. Choosing the normalisation $A_0 = 1$ gives $A_1 = \beta$, $A_2 = \beta^2$, and in general $A_j = \beta^j$, for negative j as well. ∎

There are infinitely many eigenvectors, which is as it should be: an infinite system has infinitely many degrees of freedom and so infinitely many normal modes. Since S commutes with $M^{-1}K$, these eigenvectors are the normal modes, and we only need their frequencies.

Theorem — Dispersion relation of the chain

The mode $A_j = \beta^j$ has

$$\omega^2 = \omega_0^2\left(2 - \beta - \beta^{-1}\right).$$

For the amplitude to stay finite along the whole chain we need $|\beta| = 1$. Writing $\beta = e^{ika}$ with real k,

$$\omega^2 = 2\omega_0^2(1 - \cos ka) = 4\omega_0^2 \sin^2\frac{ka}{2}.$$

Proof

Substitute $A_j = \beta^j$ in the mode equation and divide by β^j:

$$\omega^2 = \omega_0^2\left(-\beta^{-1} + 2 - \beta\right).$$

If $|\beta| > 1$ then $\beta^j \to \infty$ as $j \to +\infty$, and if $|\beta| < 1$ the same happens as $j \to -\infty$; only $|\beta| = 1$ is acceptable. With $\beta = e^{ika}$, $\beta + \beta^{-1} = 2\cos ka$, and $1 - \cos ka = 2\sin^2(ka/2)$. ■

Definition — Wave number and dispersion relation

In the mode $A_j \propto e^{ijka}$ the mass at equilibrium position $x = ja$ has amplitude e^{ikx}. The quantity k is the *wave number*; the pattern repeats after a *wavelength*

$$\lambda = \frac{2\pi}{k}.$$

A relation $\omega = \omega(k)$ between frequency and wave number is called a *dispersion relation*. It is fixed by the physics of the medium.

7.4 Reading the dispersion relation

(1) **One mode for each k.** Every real k gives a normal mode, so there are infinitely many.

(2) **Real mode shapes.** β and $1/\beta$, that is k and $-k$, give the same ω. Any combination of e^{ijka} and e^{-ijka} is therefore also a mode at that frequency, in particular the real shapes

$$A_j = \sin(jka) \quad \text{and} \quad A_j = \cos(jka).$$

(3) **Lowest frequency.** $ka = 0$ ($\beta = 1$) gives $\omega = 0$: all masses move together, as in Mode C of **§4**.

(4) **Highest frequency.** $ka = \pi$ ($\beta = -1$) gives $\omega = 2\omega_0$: neighbouring masses move exactly opposite to each other. The chain cannot oscillate faster than this.

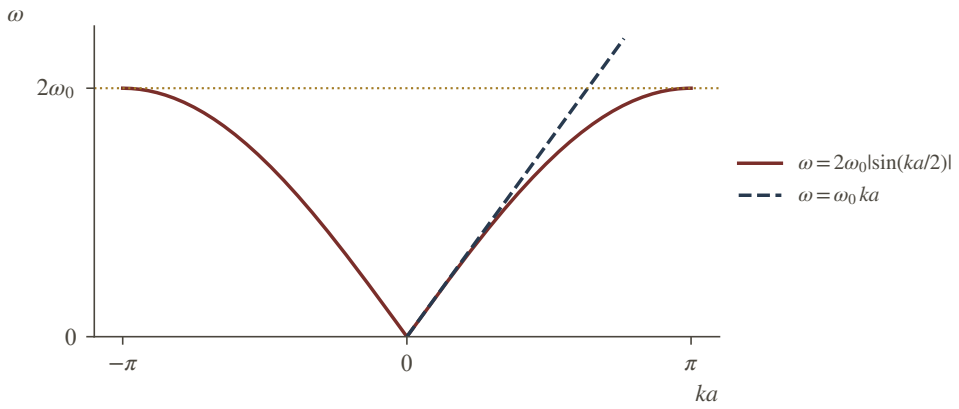


Figure 19 Dispersion relation $\omega = 2\omega_0|\sin(ka/2)|$ of the chain. For small ka it is a straight line.

Each mode, with a real shape such as $\sin(jka)$, is a *standing wave*: every mass oscillates in time at the same frequency while the amplitudes trace out a fixed sinusoid

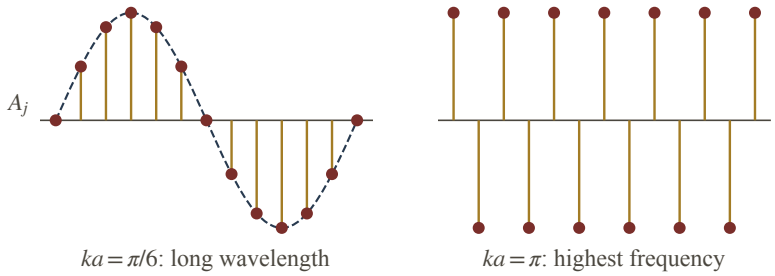


Figure 20 Amplitudes A_j of the masses in two modes of the chain. Left: neighbours move almost together. Right: neighbours move exactly opposite, the shortest wavelength the chain can carry.

along the chain. The most general motion of the chain is a superposition of these modes.

Summary of Lecture 7

(1) If $[S, M^{-1}K] = 0$, the normal modes are eigenvectors of the symmetry S.

(2) Reflection symmetry gives the modes $(1, 1)$ and $(1, -1)$ directly.

(3) Translation symmetry gives $A_j \propto e^{ijka}$: sinusoidal mode shapes.

(4) The chain's dispersion relation is $\omega^2 = 4\omega_0^2 \sin^2(ka/2)$, with $0 \le \omega \le 2\omega_0$.

Check yourself — Lecture 7

(1) Why are the eigenvalues of the reflection matrix ± 1?

(2) What is the largest frequency the infinite chain can carry, and what does the motion look like?

(3) Why must $|\beta| = 1$ for the infinite chain?

Answers are in Appendix B.

8 Lecture 8: Boundary Conditions and the Wave Equation

Reading

Georgi, Chapter 5: Waves (§5.3 to end), and Chapter 6 through §6.1.

The infinite chain of Lecture 7 does not exist. Real systems have ends. This lecture shows that a finite system can be solved using the modes of the infinite one: the ends simply select which wave numbers are allowed. At the end of the lecture the spacing between masses is shrunk to zero, and the infinitely many coupled equations become a single partial differential equation, the wave equation.

8.1 Recap, and a new system: beads on a string

For an infinite periodic system, translation symmetry gave $A_j \propto \beta^j$, and finiteness at both ends of the chain forced $\beta = e^{ika}$, so that $A_j \propto e^{ijka}$.

The same reasoning applies to a different system (Figure 21): point masses m threaded on a massless string under constant tension T, a horizontal distance a apart. Each bead moves only up and down, by a small amount $y_j \ll a$.

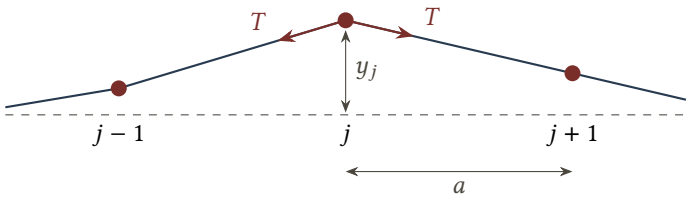


Figure 21 Beads on a stretched string. Each bead is pulled by the tension in the two string segments attached to it.

Example — Equation of motion of a bead

Let the string segments on the left and right of bead j make small angles θ_1 and θ_2 with the horizontal. Horizontally,

$$m\ddot{x}_j = -T\cos\theta_1 + T\cos\theta_2 \approx 0,$$

so there is no horizontal motion. Vertically,

$$m\ddot{y}_j = -T\sin\theta_1 - T\sin\theta_2,$$

with $\sin\theta_1 \approx (y_j - y_{j-1})/a$ and $\sin\theta_2 \approx (y_j - y_{j+1})/a$. Hence

$$m\ddot{y}_j = \frac{T}{a}\left(y_{j-1} - 2y_j + y_{j+1}\right).$$

This has exactly the form of the spring chain with the spring constant replaced by T/a. Nothing needs to be solved again.

Theorem — Dispersion relation of the beaded string

With $y_j = \mathrm{Re}\left[A_j e^{i(\omega t+\phi)}\right]$ and $A_j \propto e^{ijka}$,

$$\omega^2 = 4\omega_0^2 \sin^2\frac{ka}{2}, \qquad \omega_0^2 \equiv \frac{T}{ma}.$$

On an infinite string every wave number k is allowed, and each gives a standing-wave mode with frequency $\omega(k)$.

8.2 Finite systems: boundary conditions

Definition — Boundary condition

A *boundary condition* is a requirement on the motion at the ends of a finite system. A finite system of N beads, $j = 1, \ldots, N$, is treated as part of an infinite one, with fictitious beads $j = 0$ and $j = N + 1$ whose motion is prescribed by the boundary condition.

Because $\omega(k) = \omega(-k)$, the general mode shape at a given frequency is a combination of both signs of k:

$$y_j = \mathrm{Re}\left[e^{i(\omega t+\phi)}\left(\alpha\, e^{ijka} + \beta\, e^{-ijka}\right)\right].$$

The boundary conditions fix the ratio α/β and restrict k.

Theorem — Both ends fixed

With $y_0 = 0$ and $y_{N+1} = 0$, the normal modes are

$$A_j \propto \sin(jka), \qquad ka = \frac{n\pi}{N+1}, \quad n = 1, 2, \ldots, N.$$

Proof

$y_0 = 0$ gives $\alpha + \beta = 0$, so the shape is $\alpha(e^{ijka} - e^{-ijka}) = 2i\alpha \sin(jka)$. Then $y_{N+1} = 0$ requires

$$\sin\big((N+1)ka\big) = 0, \qquad (N+1)ka = n\pi.$$

The values $n = 1, \ldots, N$ give N different shapes. $n = 0$ and $n = N + 1$ give zero displacement everywhere, and larger n repeat shapes already found. ∎

Theorem — Both ends free

A free end exerts no vertical force, which requires the last string segment to be horizontal: $y_0 = y_1$ and $y_{N+1} = y_N$. The normal modes are

$$A_j \propto \cos\big((j - \tfrac{1}{2})ka\big), \qquad ka = \frac{n\pi}{N}, \quad n = 0, 1, \ldots, N-1.$$

Proof

The first condition, $\alpha + \beta = \alpha e^{ika} + \beta e^{-ika}$, gives

$$\alpha\big(1 - e^{ika}\big) = \beta\big(e^{-ika} - 1\big).$$

The second, written the same way, gives

$$\alpha\, e^{iNka}\big(1 - e^{ika}\big) = \beta\, e^{-iNka}\big(e^{-ika} - 1\big).$$

Dividing one by the other, $e^{2iNka} = 1$, so $ka = n\pi/N$. The first condition simplifies to $\beta = \alpha e^{ika}$, and then

$$\alpha e^{ijka} + \beta e^{-ijka} = \alpha e^{ika/2}\big(e^{i(j-1/2)ka} + e^{-i(j-1/2)ka}\big),$$

which is proportional to $\cos\big((j - \tfrac{1}{2})ka\big)$. ∎

> **Tip**
> *N beads, N modes: the count always matches the number of degrees of freedom.*

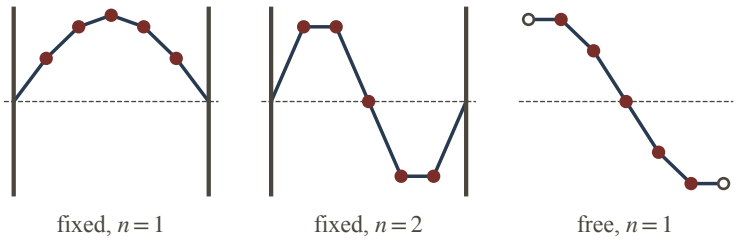


Figure 22 Mode shapes of five beads. With fixed ends the fictitious beads $j = 0$ and $j = 6$ sit at zero. With free ends they (open circles) copy their neighbours, so the end segments are horizontal.

Fixed ends select sines, with a node at each end. Free ends select cosines, with zero slope at each end; the $n = 0$ mode is a uniform shift of the whole string, with $\omega = 0$.

8.3 A driven end

Example — One end fixed, the other shaken

Take $y_0 = 0$ and $y_{N+1} = \Delta \cos \omega_d t$. We look for the steady state, in which every bead oscillates at ω_d. The dispersion relation gives the wave number that goes with this frequency:

$$\omega_d^2 = 2\omega_0^2(1 - \cos k_d a), \qquad k_d a = \cos^{-1}\Big(1 - \frac{\omega_d^2}{2\omega_0^2}\Big).$$

Try $y_j = \mathrm{Re}\big[e^{i\omega_d t}(\alpha e^{ijk_d a} + \beta e^{-ijk_d a})\big]$. The fixed end gives $\beta = -\alpha$, so

$$y_j = \mathrm{Re}\big[2i\alpha\, e^{i\omega_d t} \sin(jk_d a)\big].$$

The driven end requires $2i\alpha \sin\big((N+1)k_d a\big) = \Delta$. Therefore

$$y_j = \Delta \frac{\sin(jk_d a)}{\sin\big((N+1)k_d a\big)} \cos \omega_d t.$$

The amplitude diverges when $\sin\big((N+1)k_d a\big) = 0$, that is when $k_d a = n\pi/(N+1)$. These are exactly the wave numbers of the fixed-end modes: the string resonates whenever the driving frequency matches one of its normal-mode frequencies, as in **§6**.

Recipe — Finite periodic systems

(1) Translation symmetry, plus finiteness far away, gives $A_j \propto e^{\pm ijka}$.

(2) Physical laws give the equation of motion.

(3) Steps (1) and (2) together give the dispersion relation $\omega(k)$.

(4) Boundary conditions select the allowed values of k.

(5) The full solution is a superposition of the allowed modes; initial conditions fix their amplitudes and phases.

8.4 The continuum limit

Now let the beads become so closely spaced that the string looks continuous. Replace the label j by the position $x = ja$, and the displacement $y_j(t)$ by a function $\psi(x,t)$ with mode shape $A(x)$.

Definition — The wave equation

The equation

$$\frac{\partial^2 \psi}{\partial t^2} = v_p^2 \frac{\partial^2 \psi}{\partial x^2}$$

is the (one-dimensional) *wave equation*. The constant v_p has the dimensions of a speed and is called the *phase velocity*.

Theorem — Continuum limit of the beaded string

When the bead spacing a is much smaller than the wavelength, the beaded string obeys the wave equation with

$$v_p = \sqrt{\frac{T}{\rho_L}}, \qquad \rho_L \equiv \frac{m}{a},$$

where ρ_L is the mass per unit length. Its dispersion relation becomes $\omega = v_p k$.

Proof

The mode equation is

$$\omega^2 A(x) = \frac{T}{ma}\big[-A(x-a) + 2A(x) - A(x+a)\big].$$

Expand in Taylor series:

$$A(x \pm a) = A(x) \pm aA'(x) + \tfrac{1}{2}a^2 A''(x) \pm \cdots$$

The constant and first-derivative terms cancel in the bracket, leaving $-a^2 A''(x)$ plus terms of order a^4. So

$$\omega^2 A(x) = -\frac{T}{ma}\, a^2\, A''(x) = -\frac{T}{\rho_L} A''(x).$$

In other words, the matrix $M^{-1}K$ has become the differential operator $-(T/\rho_L)\,\partial^2/\partial x^2$. Since $\ddot{\psi} = -M^{-1}K\psi$, this is the wave equation. Substituting the mode $e^{ikx}e^{i\omega t}$ gives $\omega^2 = (T/\rho_L)k^2$. ∎

The same result follows from the discrete dispersion relation: for $ka \ll 1$, $\sin(ka/2) \approx ka/2$, so

$$\omega^2 \approx \frac{4T}{ma}\left(\frac{ka}{2}\right)^2 = \frac{T}{\rho_L}k^2.$$

This is the straight-line part of the dispersion curve near the origin. A medium with $\omega \propto k$ is called *non-dispersive*; the departure from a straight line at large k is *dispersion*, taken up in Lecture 13.

> **Watch Out**
> *The continuous string is an approximation. It fails when the wavelength becomes comparable to the spacing a.*

Summary of Lecture 8

(1) Beads on a string obey the same equations as the spring chain, with $\omega_0^2 = T/ma$.

(2) Fixed ends: $A_j \propto \sin(jka)$, $ka = n\pi/(N+1)$. Free ends: $A_j \propto \cos((j - \frac{1}{2})ka)$, $ka = n\pi/N$.

(3) A driven end resonates at each normal-mode frequency.

(4) As $a \to 0$ the system obeys the wave equation with $v_p = \sqrt{T/\rho_L}$.

Check yourself — Lecture 8

(1) List the allowed values of ka for three beads on a string with both ends fixed.

(2) How is a free end expressed as a condition on the beads?

(3) A string has $T = 100$ N and $\rho_L = 0.01$ kg/m. Find the wave speed.

Answers are in Appendix B.

9 Lecture 9: Wave Equation, Standing Waves and Fourier Series

The wave equation replaces infinitely many coupled equations by one. This lecture finds its normal modes, the standing waves of a continuous string, and then shows how *any* initial shape of the string can be written as a sum of those modes. That decomposition is the Fourier series.

Reading

Georgi, Chapter 5 (§5.3 to end) and Chapter 6: Continuum Limit and Fourier Series, through §6.1.

9.1 From the discrete string to the wave equation

In Lecture 8 the mode equation $M^{-1}KA = \omega^2 A$ of the beaded string became, for spacing a much smaller than the wavelength,

$$M^{-1}K \longrightarrow -\frac{T}{\rho_L}\frac{\partial^2}{\partial x^2}, \qquad \psi_j(t) \longrightarrow \psi(x,t),$$

so that $\ddot{\psi} = -M^{-1}K\psi$ turned into

$$\frac{\partial^2 \psi}{\partial t^2} = v_p^2 \frac{\partial^2 \psi}{\partial x^2}, \qquad v_p = \frac{\omega}{k} = \sqrt{\frac{T}{\rho_L}}.$$

This is still an infinite set of coupled oscillators: the second x-derivative couples each point of the string to its neighbours.

9.2 Normal modes by separation of variables

A normal mode is a motion in which every point oscillates with the same frequency and phase, with an amplitude that depends on position. That is a product of a function of x and a function of t.

Theorem — Normal modes of the wave equation

The solutions of the wave equation of the form $\psi(x,t) = A(x)\,B(t)$ are

$$\psi_m(x,t) = A_m \sin(\omega_m t + \beta_m)\,\sin(k_m x + \alpha_m), \qquad \omega_m = v_p\, k_m.$$

Proof

Substitute $\psi = A(x)B(t)$ and divide by $v_p^2 A(x)B(t)$:

$$\frac{1}{v_p^2\,B(t)}\frac{\mathrm{d}^2 B}{\mathrm{d}t^2} = \frac{1}{A(x)}\frac{\mathrm{d}^2 A}{\mathrm{d}x^2}.$$

The left side depends only on t and the right side only on x. Changing x with t held fixed cannot change the left side, so the right side does not depend on x either: both sides equal a constant. Call it $-k_m^2$. Then

$$\frac{\mathrm{d}^2 B}{\mathrm{d}t^2} = -k_m^2 v_p^2\, B, \qquad \frac{\mathrm{d}^2 A}{\mathrm{d}x^2} = -k_m^2\, A.$$

Both are harmonic-oscillator equations, one in time with frequency $\omega_m = v_p k_m$ and one in space with "frequency" k_m. Their solutions are $B = B_m \sin(\omega_m t + \beta_m)$ and $A = C_m \sin(k_m x + \alpha_m)$. ∎

Tip

A positive separation constant gives exponentials in x and t, which cannot satisfy the boundary conditions of a string.

Remark — Who decides what

(1) The relation $\omega_m = v_p k_m$ is fixed by the string itself (its tension and mass density).

(2) k_m and α_m are fixed by the *boundary conditions*.

(3) A_m and β_m are fixed by the *initial conditions*.

Each mode is a *standing wave*: every point of the string moves up and down harmonically, in step with every other point, and the amplitudes trace a sine curve, just as for the discrete system.

9.3 A string with one end fixed and one end free

Example — Allowed modes

A string of length L is tied down at $x = 0$. At $x = L$ it ends on a massless ring that slides without friction on a vertical rod. The boundary conditions are

$$\psi(0, t) = 0, \qquad \frac{\partial \psi}{\partial x}(L, t) = 0.$$

The second holds because a non-zero slope at $x = L$ would give the tension a vertical component acting on a massless ring, producing infinite acceleration. The first condition gives $\sin \alpha_m = 0$, so $\alpha_m = 0$. The second gives $\cos(k_m L) = 0$:

$$k_m L = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots \qquad k_m = \frac{(2m-1)\pi}{2L}, \quad m = 1, 2, 3, \ldots$$

Mode m	k_m	$\lambda_m = 2\pi/k_m$	$\omega_m = v_p k_m$
1	$\pi/2L$	$4L$	$\pi v_p/2L$
2	$3\pi/2L$	$4L/3$	$3\pi v_p/2L$
3	$5\pi/2L$	$4L/5$	$5\pi v_p/2L$

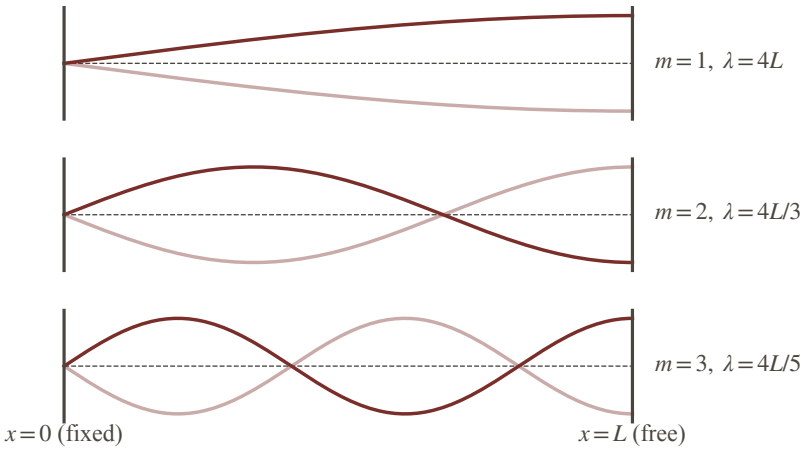


Figure 23 The first three standing waves of a string fixed at $x = 0$ and free at $x = L$, shown at the two extremes of the motion. The fixed end is a node, the free end an antinode.

Only odd multiples of the lowest frequency appear. The lowest mode fits one quarter of a wavelength into the string.

9.4 The general solution and the initial conditions

Theorem — General motion of the fixed–free string

$$\psi(x,t) = \sum_{m=1}^{\infty} A_m \, \sin(\omega_m t + \beta_m) \, \sin(k_m x),$$

$$k_m = \frac{(2m-1)\pi}{2L}, \qquad \omega_m = v_p k_m.$$

The constants A_m and β_m come from the shape and velocity of the string at $t = 0$. Suppose the string starts at rest with a known shape:

$$\text{(a)} \ \frac{\partial \psi}{\partial t}(x,0) = 0, \qquad \text{(b)} \ \psi(x,0) \text{ given.}$$

Condition (a) requires $\cos\beta_m = 0$ for every m, so $\beta_m = \pi/2$ and $\sin(\omega_m t + \beta_m) = \cos\omega_m t$. Then

$$\psi(x,0) = \sum_{m=1}^{\infty} A_m \sin(k_m x).$$

The question is how to read off the numbers A_m from the function $\psi(x,0)$.

Theorem — Orthogonality of the modes

For $k_m = (2m-1)\pi/2L$,

$$\int_0^L \sin(k_m x) \, \sin(k_n x) \, dx = \begin{cases} L/2, & m = n, \\ 0, & m \neq n. \end{cases}$$

Analogy

This is like finding a vector's components by taking dot products with orthogonal basis vectors.

Proof

Use $\sin a \sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$. For $m \neq n$, both $(k_m - k_n)L = (m-n)\pi$ and $(k_m + k_n)L = (m+n-1)\pi$ are integer multiples of π, so the integrals of both cosines are proportional to the sine of a multiple of π and vanish. For $m = n$ the first cosine is 1 and contributes $L/2$, while the second still integrates to zero. ∎

Theorem — Fourier coefficients

If $\psi(x,0) = \sum_m A_m \sin(k_m x)$, then

$$A_m = \frac{2}{L} \int_0^L \psi(x,0) \, \sin(k_m x) \, dx.$$

Proof

Multiply the series by $\sin(k_n x)$ and integrate from 0 to L. By orthogonality every term on the right vanishes except $m = n$, which gives $A_n \, L/2$. ∎

Definition — Fourier decomposition

Writing a given function as a sum of the normal-mode shapes of a system, with coefficients found by orthogonality, is its *Fourier decomposition*. The sum

is a *Fourier series.*

Example — A step-shaped initial displacement

Let the string start at rest with

$$\psi(x,0) = \begin{cases} 0, & 0 \le x < L/2, \\ h, & L/2 < x \le L. \end{cases}$$

Then

$$\begin{aligned} A_m &= \frac{2}{L}\int_{L/2}^{L} h\,\sin(k_m x)\,\mathrm{d}x \\ &= -\frac{2h}{k_m L}\left[\cos(k_m L) - \cos\frac{k_m L}{2}\right]. \end{aligned}$$

Since $\cos(k_m L) = 0$ for every allowed k_m,

$$A_m = \frac{2h}{k_m L}\cos\frac{k_m L}{2} = \frac{4h}{(2m-1)\pi}\cos\frac{(2m-1)\pi}{4}.$$

The later motion is

$$\psi(x,t) = \sum_{m=1}^{\infty} A_m \cos(\omega_m t)\sin(k_m x).$$

Watch Out

The modes have frequencies in the ratio 1:3:5, so the shape repeats only after the period of the lowest mode, $4L/v_p$.

! A superposition of modes is not a mode

Each term of the sum oscillates at its own frequency ω_m. Their relative phases change in time, so the shape of the string changes: it does not simply scale up and down. Only a single mode keeps its shape.

The coefficients fall off like $1/m$, so the low modes dominate, but the sharp step needs many terms to be reproduced well (Figure 24).

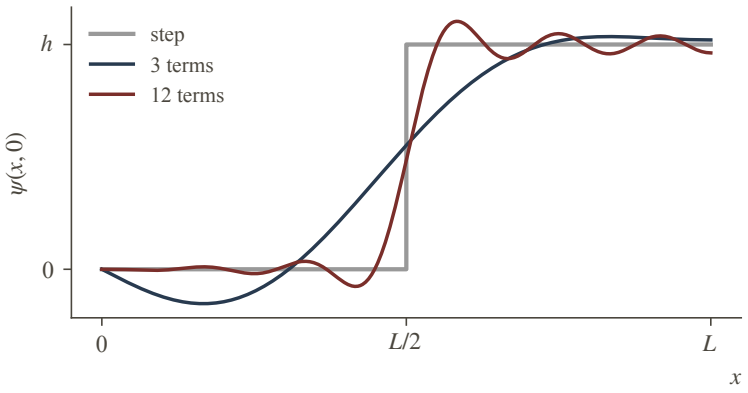


Figure 24 The step (grey) and its Fourier series truncated after 3 terms (navy) and 12 terms (red).

Summary of Lecture 9

(1) Separation of variables gives the standing-wave modes $\sin(\omega_m t + \beta_m)\sin(k_m x + \alpha_m)$ with $\omega_m = v_p k_m$.

(2) Boundary conditions quantise k_m; for a fixed–free string $k_m = (2m - 1)\pi/2L$.

(3) The general motion is a sum over all modes.

(4) Orthogonality of the mode shapes gives the coefficients: $A_m = \frac{2}{L}\int_0^L \psi(x,0)\sin(k_m x)\,\mathrm{d}x$.

Check yourself — Lecture 9

(1) What is the longest wavelength of a standing wave on a string of length L that is fixed at one end and free at the other? With both ends fixed?

(2) Which constants of a normal mode are fixed by the boundary conditions and which by the initial conditions?

(3) A fixed–free string is released from rest in the shape $\psi(x,0) = \sin(k_3 x)$. Find all the A_m.

Answers are in Appendix B.

10 Lecture 10: Traveling Waves

Reading

Georgi, Chapter 6 (§6.2 to end), Chapter 7, and Chapter 8: Traveling Waves (through §8.2).

Standing waves are one way to describe the motion of a string. This lecture introduces a second: shapes that slide along the string without changing. The two descriptions are equivalent, but traveling waves are far easier to use for pulses, and they show clearly what happens when a wave meets a boundary between two different strings.

10.1 Progressing-wave solutions

Theorem — d'Alembert's solutions

For *any* twice-differentiable function f, both

$$\psi(x,t) = f(x - v_p t) \qquad \text{and} \qquad \psi(x,t) = f(x + v_p t)$$

satisfy the wave equation $\partial^2\psi/\partial t^2 = v_p^2\,\partial^2\psi/\partial x^2$.

Proof

Let $\tau \equiv x - v_p t$, so that $\partial\tau/\partial x = 1$ and $\partial\tau/\partial t = -v_p$. By the chain rule,

$$\frac{\partial f}{\partial x} = f'(\tau), \qquad \frac{\partial^2 f}{\partial x^2} = f''(\tau),$$

$$\frac{\partial f}{\partial t} = -v_p f'(\tau), \qquad \frac{\partial^2 f}{\partial t^2} = v_p^2 f''(\tau).$$

Hence $\partial^2 f/\partial t^2 = v_p^2\,\partial^2 f/\partial x^2$. The same steps with $\tau = x + v_p t$ give $(+v_p)^2 = v_p^2$ and the same conclusion. ∎

The same holds for any function of $kx \pm \omega t$ provided $\omega = v_p k$, since $kx \pm \omega t = k(x \pm v_p t)$.

Definition — Traveling wave

$f(\tau)$ is the *shape* of the wave. The solution $f(x - v_p t)$ is that shape moving in the $+x$ direction at speed v_p: at a later time t the same value of f is found at a position larger by $v_p t$. Likewise $f(x + v_p t)$ moves in the $-x$ direction. The string itself does not travel: each bit of string only moves up and down, and it is the shape that moves along.

Example — Reading off the velocity

For

$$\psi(x,t) = \frac{1}{1 + (x - 3t)^4}$$

we recognise $\psi = f(x - 3t)$ with $f(\tau) = 1/(1 + \tau^4)$. The pulse moves to the right with speed 3 (in the units of x and t). Without spotting the pattern, the speed can still be found from the wave equation itself:

$$v_p = \sqrt{\frac{\partial^2\psi}{\partial t^2} \Big/ \frac{\partial^2\psi}{\partial x^2}}.$$

10.2 Superposition of pulses

The wave equation is linear, so the sum of two solutions is a solution. Two pulses sent towards each other therefore simply add where they overlap and then continue unchanged: they pass through each other.

A striking case is a pulse meeting its mirror image with opposite sign. At the instant of complete overlap the string is exactly flat (Figure 25). Yet the pulses reappear afterwards. How does the string "remember" them?

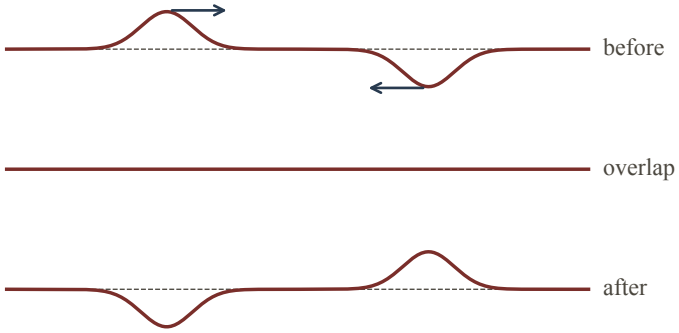


Figure 25 Two opposite pulses pass through each other. At the moment of overlap the displacement vanishes everywhere, but the string is moving.

The answer is that the flat string is *not* at rest. Its displacement is zero but its transverse velocity is not, and that velocity carries exactly the information needed to launch the two outgoing pulses. The state of a string is its displacement *and* its velocity.

10.3 Energy stored in a string

Theorem — Kinetic and potential energy

For small transverse vibrations of a string with tension T and mass per unit length ρ_L,

$$E_{\text{kin}} = \int \frac{\rho_L}{2}\left(\frac{\partial \psi}{\partial t}\right)^2 \mathrm{d}x, \qquad E_{\text{pot}} = \int \frac{T}{2}\left(\frac{\partial \psi}{\partial x}\right)^2 \mathrm{d}x.$$

Proof

A piece of string of length $\mathrm{d}x$ has mass $\mathrm{d}m = \rho_L\,\mathrm{d}x$ and transverse speed $\partial\psi/\partial t$, giving the kinetic energy.
When the string is displaced, that piece is stretched from $\mathrm{d}x$ to $\sqrt{\mathrm{d}x^2 + \mathrm{d}\psi^2}$. The work done against the tension is the tension times the extension:

$$\mathrm{d}W = T\left(\sqrt{\mathrm{d}x^2 + \mathrm{d}\psi^2} - \mathrm{d}x\right) = T\,\mathrm{d}x\left(\sqrt{1 + \left(\frac{\partial \psi}{\partial x}\right)^2} - 1\right).$$

For small slopes, $\sqrt{1+s^2} \approx 1 + \tfrac{1}{2}s^2$, so $\mathrm{d}W \approx \tfrac{1}{2}T(\partial\psi/\partial x)^2\,\mathrm{d}x$. ∎

In the cancelling-pulse example the potential energy is momentarily zero (the string is flat) and all the energy is kinetic. For a single traveling wave $f(x - v_p t)$, by contrast, $\partial\psi/\partial t = -v_p\,\partial\psi/\partial x$, and since $\rho_L v_p^2 = T$ the kinetic and potential energy densities are equal at every point.

10.4 Releasing a stationary shape

Suppose a string is held in some shape $g(x)$ and released from rest. There are two ways to predict what happens.

(1) **Brute force.** Decompose $g(x)$ into infinitely many standing waves, as in **§9**, and evolve each one.

(2) **Traveling waves.** Write the motion as two half-height copies moving in opposite directions.

Theorem — A shape released from rest splits in two

The solution with $\psi(x,0) = g(x)$ and $\partial\psi/\partial t\,(x,0) = 0$ on a long string is

$$\psi(x,t) = \tfrac{1}{2}\,g(x - v_p t) + \tfrac{1}{2}\,g(x + v_p t).$$

Proof

Each term solves the wave equation, so the sum does. At $t = 0$ the sum is $g(x)$. Its time derivative is $-\frac{1}{2}v_p g'(x - v_p t) + \frac{1}{2}v_p g'(x + v_p t)$, which vanishes at $t = 0$. ■

The same idea applies to a normal mode:

$$\sin kx\,\cos\omega t = \tfrac{1}{2}\sin(kx - \omega t) + \tfrac{1}{2}\sin(kx + \omega t).$$

> **Tip**
> *The converse also holds: a standing wave is two equal sine waves traveling in opposite directions.*

10.5 Reflection and transmission at a junction

Two strings are joined at $x = 0$. The left one has mass density ρ_L and the right one $4\rho_L$; the tension T is the same in both. The wave speeds are

$$v_1 = \sqrt{\frac{T}{\rho_L}}, \qquad v_2 = \sqrt{\frac{T}{4\rho_L}} = \frac{v_1}{2}.$$

A wave arriving from the left produces a reflected wave going back and a transmitted wave going on.

Definition — Boundary conditions at a junction

(1) **The string is continuous:** $y_L(0,t) = y_R(0,t)$. Both sides must therefore oscillate with the same frequency ω.

(2) **The slope is continuous:** $\partial y_L/\partial x = \partial y_R/\partial x$ at $x = 0$. Otherwise the tension would exert a finite net force on a point of zero mass, giving infinite acceleration.

Theorem — Reflection and transmission coefficients

An incident wave f_i produces a reflected wave $f_r = R\,f_i$ and a transmitted wave $f_t = \mathcal{T}\,f_i$, with

$$R = \frac{v_2 - v_1}{v_2 + v_1}, \qquad \mathcal{T} = \frac{2v_2}{v_1 + v_2}.$$

Proof

Write, with $k_1 = \omega/v_1$ and $k_2 = \omega/v_2$,

$$y_L = f_i(-k_1 x + \omega t) + f_r(k_1 x + \omega t), \qquad y_R = f_t(-k_2 x + \omega t).$$

Condition (1) at $x = 0$ gives $f_i + f_r = f_t$. Condition (2) gives $-k_1 f_i' + k_1 f_r' = -k_2 f_t'$, all evaluated at ωt. Integrate over the argument and use $k \propto 1/v$:

$$-v_2 f_i + v_2 f_r = -v_1 f_t.$$

Substituting $f_t = f_i + f_r$ gives $(v_2 + v_1) f_r = (v_2 - v_1) f_i$, and then $f_t = f_i + f_r = 2v_2 f_i/(v_1 + v_2)$. ∎

Example — Light string into heavy string

With $v_2 = v_1/2$,

$$R = \frac{\frac{1}{2} - 1}{\frac{1}{2} + 1} = -\frac{1}{3}, \qquad \mathcal{T} = \frac{1}{\frac{3}{2}} = \frac{2}{3}.$$

One third of the amplitude comes back, upside down, and two thirds goes on. The frequency is unchanged, so the wavelength in the heavy string is halved: $\lambda = 2\pi v/\omega \propto v$.

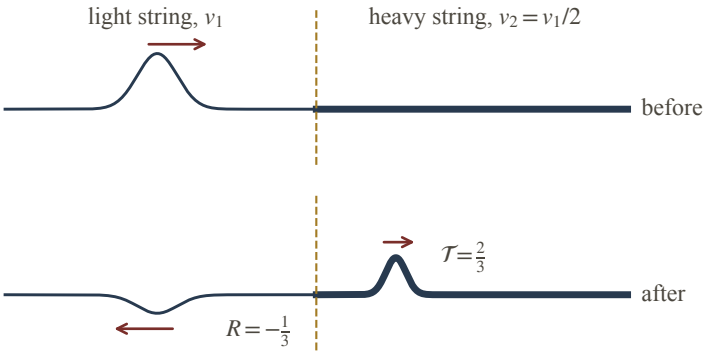


Figure 26 A pulse meeting a junction from a light string to one four times heavier. The reflected pulse is inverted; the transmitted pulse is slower and therefore half as long.

Definition — Impedance

For a string the *impedance* is

$$Z = \frac{T}{v} = \sqrt{T\rho_L}.$$

In terms of it, $R = (Z_1 - Z_2)/(Z_1 + Z_2)$ and $\mathcal{T} = 2Z_1/(Z_1 + Z_2)$. Reflection happens only where the impedance changes.

Case	Limit	R, $\mathcal{T}$	Result
Fixed end (wall)	$\rho_L \to \infty$, $v_2 \to 0$	-1, 0	Pulse returns inverted; nothing transmitted.
Free end	$\rho_L \to 0$, $v_2 \to \infty$	$+1$, 2	Pulse returns upright; the end moves with twice the amplitude.
Matched	$v_2 = v_1$	0, 1	No reflection at all.

10.6 Other boundary conditions

Why it matters

A matched damper absorbs a wave completely: this is how cables are terminated to stop echoes.

Two further ends appear in the lecture, both with the string starting at a driven point $x = 0$, $y(0, t) = P(t)$:

(1) **Massless ring on a frictionless rod.** The vertical component of the tension must vanish, so $-T\,\partial y/\partial x = 0$ at the ring.

(2) **Massless ring with a damper** of drag coefficient b. The tension's vertical component balances the drag:

$$-T\frac{\partial y}{\partial x} - b\frac{\partial y}{\partial t} = 0.$$

Remark — The perfectly absorbing end

For an incoming wave $y = f(x-vt)$, $\partial y/\partial t = -v\,\partial y/\partial x$, so the damper condition reads $(-T + bv)\,\partial y/\partial x = 0$. It is satisfied with no reflected wave exactly when $b = T/v = Z$. A damper matched to the impedance of the string looks, to the wave, like more string.

Summary of Lecture 10

(1) $f(x \mp v_p t)$ solves the wave equation for any shape f.

(2) Pulses superpose and pass through each other; the string's state is displacement plus velocity.

(3) Energy densities: $\frac{1}{2}\rho_L \dot{\psi}^2$ and $\frac{1}{2}T(\partial_x \psi)^2$.

(4) At a junction, $R = (v_2 - v_1)/(v_2 + v_1)$ and $\mathcal{T} = 2v_2/(v_1 + v_2)$; frequency is preserved and wavelength changes.

Check yourself — Lecture 10

(1) In which direction, and how fast, does $\psi = e^{-(x+2t)^2}$ move?

(2) A pulse goes from a string with speed v_1 to one with $v_2 = 2v_1$. Find R and $\mathcal{T}$.

(3) Two opposite pulses overlap and the string is momentarily flat. Where is the energy?

Answers are in Appendix B.

11 Lecture 11: Sound Waves

The waves so far were transverse: the string moved at right angles to the direction of travel. Sound is different. The air moves back and forth *along* the direction of travel. This lecture shows that sound nevertheless obeys the same wave equation, computes its speed from the properties of air, and applies the standing-wave machinery to an organ pipe.

Reading
Georgi, Chapter 7: Longitudinal Oscillations and Sound.

11.1 Displacement and pressure in a column of air

Consider air in a pipe of cross-section A along the x axis. Two fields describe the wave:

Definition — Longitudinal wave variables

(1) $\psi(x,t)$ is the *displacement*, along x, of the air whose equilibrium position is x.

(2) $\psi_P(x,t)$ is the *excess pressure*: the pressure at x minus the undisturbed room pressure P_0.

A wave in which the medium moves parallel to the direction of propagation is a *longitudinal wave*.

Look at the slab of air that sits between x and $x+\Delta x$ at equilibrium (Figure 27). Its equilibrium volume is $V_0 = A\,\Delta x$.

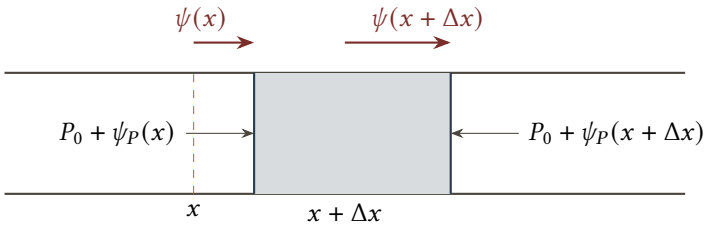


Figure 27 A slab of air (dashed at equilibrium, shaded when displaced). Its two faces move by different amounts, so its volume changes.

The two faces are displaced by different amounts, so the volume changes by

$$\Delta V = A\big[\psi(x+\Delta x,t) - \psi(x,t)\big] \approx A\,\frac{\partial \psi}{\partial x}\,\Delta x,$$

and the net pressure pushing the slab in the $+x$ direction is

$$\psi_P(x,t) - \psi_P(x+\Delta x,t) \approx -\frac{\partial \psi_P}{\partial x}\,\Delta x.$$

To connect these we need to know how the pressure of air responds to a change of volume.

History
Newton used the isothermal law and got a speed about 15% too low. Laplace supplied the adiabatic correction.

11.2 Isothermal or adiabatic?

The ideal-gas law $PV = nRT$ would give $P \propto 1/V$, but only at constant temperature. In a sound wave the compressions and rarefactions alternate so quickly that heat has no time to flow between them: the time scale of heat conduction is much longer than the period of the oscillation. The process is *adiabatic*.

Theorem — Adiabatic law

For an ideal gas undergoing a process with no heat exchange,

$$P\,V^{\gamma} = \text{constant}, \qquad \gamma = \frac{\alpha + 1}{\alpha},$$

where α is half the number of degrees of freedom per molecule.

Proof

The first law of thermodynamics is $\mathrm{d}U + \delta W = \delta Q$, with U the internal energy, $\delta W = P\,\mathrm{d}V$ the work done by the gas and δQ the heat supplied. For an adiabatic process $\delta Q = 0$. The internal energy of an ideal gas is $U = \alpha n R T = \alpha P V$, so

$$\alpha\,(V\,\mathrm{d}P + P\,\mathrm{d}V) = -P\,\mathrm{d}V,$$

$$\frac{\mathrm{d}P}{P} = -\frac{\alpha + 1}{\alpha}\,\frac{\mathrm{d}V}{V} = -\gamma\,\frac{\mathrm{d}V}{V}.$$

Integrating, $\ln P = -\gamma \ln V + \text{const}$. ∎

Gas	Degrees of freedom	α	γ
Monatomic (He)	3 translational	3/2	5/3
Diatomic (air)	3 translational + 2 rotational	5/2	7/5

Theorem — Pressure from displacement

For small disturbances, $|\psi_P| \ll P_0$ and $|\Delta V| \ll V_0$,

$$\psi_P = -\gamma P_0\,\frac{\partial \psi}{\partial x}.$$

Proof

Before and after the disturbance the slab satisfies the adiabatic law with the same constant:

$$P_0 V_0^{\gamma} = (P_0 + \psi_P)(V_0 + \Delta V)^{\gamma} = (P_0 + \psi_P)\,V_0^{\gamma}\left(1 + \frac{\Delta V}{V_0}\right)^{\gamma}.$$

Expand to first order, $(1 + \epsilon)^{\gamma} \approx 1 + \gamma\epsilon$, and drop the product of two small quantities:

$$P_0 \approx P_0 + \psi_P + \gamma P_0 \frac{\Delta V}{V_0},$$

so $\psi_P = -\gamma P_0\,\Delta V / V_0$. Insert $\Delta V / V_0 = \partial \psi / \partial x$. ∎

The sign makes sense: where $\partial \psi / \partial x < 0$ the air is squeezed together and the pressure rises.

11.3 The wave equation for sound

Theorem — Sound obeys the wave equation

$$\frac{\partial^2 \psi}{\partial t^2} = \frac{\gamma P_0}{\rho}\frac{\partial^2 \psi}{\partial x^2}, \qquad v_p = \sqrt{\frac{\gamma P_0}{\rho}},$$

where ρ is the equilibrium mass density of the gas.

Proof

The slab has mass $\Delta m = \rho A\,\Delta x$ and feels the net force $-A\,(\partial \psi_P/\partial x)\,\Delta x$. Newton's law gives

$$\rho A\,\Delta x\,\frac{\partial^2 \psi}{\partial t^2} = -A\,\Delta x\,\frac{\partial \psi_P}{\partial x},$$

and substituting $\psi_P = -\gamma P_0\,\partial\psi/\partial x$ gives the result. ∎

Example — The speed of sound in air

At sea level $P_0 \approx 1.0 \times 10^5$ Pa, $\rho \approx 1.2$ kg/m^3 and $\gamma = 7/5$:

$$v_p = \sqrt{\frac{1.4 \times 1.0 \times 10^5}{1.2}} \approx 342 \text{ m/s}.$$

The measured value at 70°F is 343 m/s. With the isothermal law ($\gamma \to 1$) the prediction would be only 289 m/s.

A lighter gas, or one with larger γ, has a higher sound speed. The resonant wavelengths of a cavity are set by its size, so a higher v_p means higher resonant frequencies $\omega = v_p k$: this is why a voice sounds higher after breathing helium.

Tip

Helium is both lighter and monatomic. Its much lower density is the main reason sound travels almost three times faster in it.

11.4 Standing sound waves in a pipe

Because sound obeys the wave equation, everything from **§9** carries over: $\omega = v_p k$, and the general motion in a pipe of length L is

$$\psi(x,t) = \sum_{m=1}^{\infty} A_m \sin(k_m x + \alpha_m)\sin(\omega_m t + \beta_m),$$

with k_m, α_m from the boundary conditions and A_m, β_m from the initial conditions.

Definition — Boundary conditions for sound

(1) **Closed end:** the air cannot move, so $\psi = 0$.

(2) **Open end:** the pressure must equal room pressure, so $\psi_P = 0$, that is $\partial\psi/\partial x = 0$.

Theorem — Modes of the three pipe configurations

(1) **Closed–closed:** $\alpha_m = 0$, $k_m = m\pi/L$, $\omega_m = m\pi v_p/L$.

(2) **Closed–open:** $\alpha_m = 0$, $k_m = (m - \tfrac{1}{2})\pi/L$, $\omega_m = (m - \tfrac{1}{2})\pi v_p/L$.

(3) **Open–open:** $\alpha_m = \pi/2$, $k_m = m\pi/L$, $\omega_m = m\pi v_p/L$.

In each case $m = 1, 2, 3, \ldots$

Proof

(1) $\psi(0) = 0$ gives $\sin \alpha_m = 0$; $\psi(L) = 0$ gives $\sin k_m L = 0$. (2) Again $\alpha_m = 0$; the open end at L needs $\cos k_m L = 0$. (3) $\partial\psi/\partial x = 0$ at $x = 0$ gives $\cos \alpha_m = 0$, so the shape is $\cos k_m x$; at $x = L$ its derivative vanishes when $\sin k_m L = 0$. ■

Example — Identifying a pipe from its spectrum

An audio analyser records peaks at frequencies in the ratio $1 : 3 : 5 : \ldots$ Only the closed–open pipe has $\omega_m \propto (2m - 1)$, so that is the configuration. Peaks at $1 : 2 : 3 : \ldots$ would indicate a pipe with two ends of the same kind.
For any motion of the closed–open pipe, every mode frequency is an odd multiple of ω_1, so the whole pattern repeats after the period of the lowest mode, $2\pi/\omega_1 = 4L/v_p$.

Watch Out
Pressure and displacement patterns are shifted by a quarter wavelength relative to each other.

! Displacement nodes are pressure antinodes

Since $\psi_P = -\gamma P_0 \, \partial\psi/\partial x$, a mode with displacement $\sin k_m x$ has pressure variation proportional to $\cos k_m x$. At a closed end the air does not move but the pressure swings the most; at an open end the air moves the most and the pressure does not change.

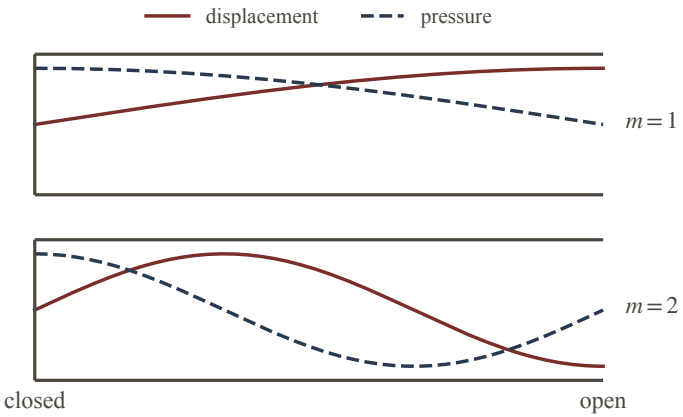


Figure 28 First two modes of a closed–open pipe: displacement amplitude (solid red) and pressure amplitude (dashed navy).

11.5 Driving a pipe

Example — Closed at one end, driven at the other

Let $\psi(0, t) = 0$ and $\psi(L, t) = D \cos \omega_d t$, for instance a loudspeaker cone at $x = L$. In the steady state the air oscillates at ω_d, and the dispersion relation fixes the wave number, $k_d = \omega_d / v_p$. Try

$$\psi(x, t) = A_d \sin(k_d x + \alpha) \cos \omega_d t.$$

The closed end gives $\alpha = 0$, and the driven end gives $A_d \sin k_d L = D$. Hence

$$\psi(x,t) = \frac{D}{\sin k_d L} \sin(k_d x) \cos \omega_d t.$$

The amplitude becomes very large when $\sin k_d L \to 0$, that is when $k_d = m\pi/L$: the wave numbers of the closed–closed pipe. A displacement drive of small amplitude looks, to the resonating air, like a closed end.

Summary of Lecture 11

(1) Sound is a longitudinal wave with $\psi_P = -\gamma P_0 \, \partial\psi/\partial x$.

(2) It obeys the wave equation with $v_p = \sqrt{\gamma P_0/\rho}$, about 342 m/s in air.

(3) Closed end: $\psi = 0$. Open end: $\partial\psi/\partial x = 0$.

(4) A closed–open pipe has only odd harmonics; pipes with like ends have all harmonics.

Check yourself — Lecture 11

(1) A pipe 0.5 m long is closed at one end and open at the other, with $v_p = 340$ m/s. Find its two lowest resonant frequencies.

(2) Is an open end a node or an antinode of pressure? Of displacement?

(3) Why is the adiabatic law used for sound, and not the isothermal one?

Answers are in Appendix B.

12 Lecture 12: Maxwell's Equations and Electromagnetic Waves

Reading

Georgi, Chapter 8 (§8.3 to end) and Chapter 9 (through §9.2).

Part II — Electromagnetic Waves. Every wave so far needed a medium: a string, a spring, air. This lecture derives a wave equation directly from Maxwell's equations in empty space. The wave speed that comes out is the speed of light, and what oscillates is not matter at all but the electric and magnetic fields themselves.

12.1 Three systems, one equation

All the continuous systems met in Part I obey $\partial^2\psi/\partial t^2 = v_p^2\,\partial^2\psi/\partial x^2$, differing only in the wave speed:

System	Parameters	v_p
Stretched string	tension T, mass per length ρ_L	$\sqrt{T/\rho_L}$
Massive spring	constant κ, length ℓ, mass per length ρ_L	$\sqrt{\kappa\ell/\rho_L}$
Organ pipe	pressure P_0, density ρ, index γ	$\sqrt{\gamma P_0/\rho}$

12.2 Vector calculus in brief

Definition — Fields, divergence and curl

A *scalar field* assigns a number to every point of space (temperature, for example). A *vector field* assigns a vector, $\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$. With $\vec{\nabla} = (\partial_x, \partial_y, \partial_z)$:

$$\vec{\nabla}\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

is the *divergence*, a measure of how much $\vec{A}$ spreads out from a point, and

$$\vec{\nabla}\times\vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ A_x & A_y & A_z \end{vmatrix}$$

is the *curl*, a measure of how much $\vec{A}$ circulates around a point. The *Laplacian* is $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$.

Theorem — Gauss's and Stokes's theorems

For a volume V bounded by the closed surface S,

$$\iiint_V (\vec{\nabla}\cdot\vec{A})\,d\tau = \oiint_S \vec{A}\cdot d\vec{a}.$$

For a surface S bounded by the closed path P,

$$\iint_S (\vec{\nabla}\times\vec{A})\cdot d\vec{a} = \oint_P \vec{A}\cdot d\vec{l}.$$

The first turns the total divergence inside a volume into the flux through its surface; the second turns the total curl through a surface into the circulation around its edge. They connect the differential form of Maxwell's equations used below to the integral form of an introductory course.

12.3 Maxwell's equations

Definition — Maxwell's equations

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0} \qquad \text{(Gauss's law)}$$

$$\vec{\nabla}\cdot\vec{B} = 0 \qquad \text{(Gauss's law for magnetism)}$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad \text{(Faraday's law)}$$

$$\vec{\nabla}\times\vec{B} = \mu_0\left(\vec{J} + \epsilon_0\frac{\partial\vec{E}}{\partial t}\right) \qquad \text{(Ampère–Maxwell law)}$$

In vacuum the charge density ρ and current density $\vec{J}$ vanish.

In vacuum the last two equations say that a changing magnetic field creates an electric field and a changing electric field creates a magnetic field. Each can sustain the other, with no charges present. The term $\mu_0\epsilon_0\,\partial\vec{E}/\partial t$ is the one Maxwell added, and without it there is no wave.

Theorem — The electromagnetic wave equation

In vacuum,

$$\nabla^2\vec{E} = \mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2}, \qquad \nabla^2\vec{B} = \mu_0\epsilon_0\frac{\partial^2\vec{B}}{\partial t^2}.$$

Each component of $\vec{E}$ and $\vec{B}$ obeys a three-dimensional wave equation with speed

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \approx 3.0\times10^8 \text{ m/s}.$$

Identity
$\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \nabla^2\vec{A}$ *for any vector field.*

Proof

Take the curl of Faraday's law. On the left use the identity, with $\vec{\nabla}\cdot\vec{E} = 0$ in vacuum:

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\nabla^2\vec{E}.$$

On the right exchange the order of the derivatives and use the Ampère–Maxwell law:

$$-\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B}) = -\mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2}.$$

Equating the two gives the equation for $\vec{E}$. Taking the curl of the Ampère–Maxwell law and using Faraday's law gives the same equation for $\vec{B}$. ∎

The constants μ_0 and ϵ_0 come from laboratory measurements of forces between currents and between charges. That their combination equals the measured speed of light (Rømer, 1676: about 2.2×10^8 m/s; Bradley, 1729: 3.01×10^8 m/s) was Maxwell's evidence that light *is* an electromagnetic wave.

12.4 The plane-wave solution

Theorem — Plane electromagnetic wave

The fields

$$\vec{E} = E_0 \cos(kz - \omega t)\,\hat{x}, \qquad \vec{B} = \frac{E_0}{c}\cos(kz - \omega t)\,\hat{y}$$

satisfy Maxwell's equations in vacuum provided $\omega = ck$.

Proof

$\vec{E}$ has only an x component, depending only on z and t, so the wave equation reduces to $\partial^2 E_x/\partial z^2 = \mu_0\epsilon_0\,\partial^2 E_x/\partial t^2$, that is

$$-k^2 E_0 \cos(kz - \omega t) = -\mu_0\epsilon_0\,\omega^2 E_0 \cos(kz - \omega t),$$

which holds when $\omega/k = 1/\sqrt{\mu_0\epsilon_0} = c$. For the magnetic field use Faraday's law:

$$\vec{\nabla}\times\vec{E} = \frac{\partial E_x}{\partial z}\,\hat{y} = -kE_0\sin(kz-\omega t)\,\hat{y} = -\frac{\partial \vec{B}}{\partial t}.$$

Integrating in time, $\vec{B} = (k/\omega)E_0\cos(kz-\omega t)\,\hat{y}$, and $k/\omega = 1/c$. ∎

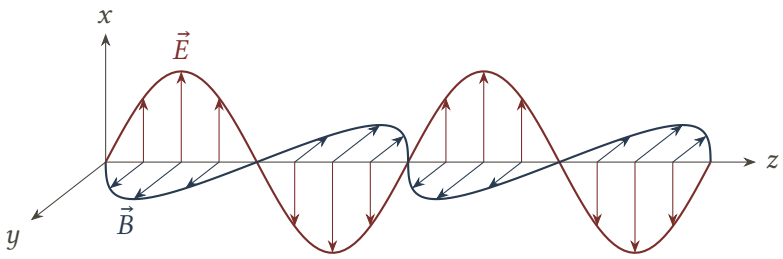


Figure 29 A plane electromagnetic wave travelling along z: $\vec{E}$ along x and $\vec{B}$ along y, in phase with each other.

Theorem — Properties of electromagnetic waves in vacuum

(1) $\vec{E}$ never comes without $\vec{B}$. The two are perpendicular to each other and to the direction of propagation $\hat{k}$, they are in phase, and

$$\vec{B} = \frac{1}{c}\,\hat{k}\times\vec{E}, \qquad |\vec{B}| = \frac{|\vec{E}|}{c}.$$

(2) The wave is non-dispersive: $\omega/k = c$ for every k.

(3) The wave travels in the direction of $\vec{E}\times\vec{B}$.

Definition — General plane wave

A plane wave travelling in an arbitrary direction is

$$\vec{E}(\vec{r},t) = \mathrm{Re}\left[\vec{E}_0\, e^{i(\vec{k}\cdot\vec{r}-\omega t+\phi)}\right], \qquad \omega = c\,|\vec{k}|,$$

with $\vec{k}$ the *wave vector* and $\vec{E}_0 \perp \vec{k}$. Its magnetic field is $\vec{B} = \frac{1}{c}\,\hat{k} \times \vec{E}$.

Example — A wave travelling diagonally

Let

$$\vec{k} = \frac{2\pi}{\lambda}\,\frac{\hat{x}+\hat{y}}{\sqrt{2}}, \qquad \vec{E}_0 = E_0\,\frac{-\hat{x}+\hat{y}}{\sqrt{2}}.$$

Check: $\vec{k}\cdot\vec{E}_0 \propto -1+1 = 0$, so the wave is transverse. Then $\vec{k}\cdot\vec{r} = \dfrac{2\pi}{\sqrt{2}\,\lambda}(x+y)$ and

$$\vec{E} = E_0\,\frac{-\hat{x}+\hat{y}}{\sqrt{2}}\,\cos\!\left(\frac{\sqrt{2}\,\pi}{\lambda}(x+y) - \omega t\right).$$

For the magnetic field,

$$\hat{k}\times\hat{E}_0 = \tfrac{1}{2}(\hat{x}+\hat{y})\times(-\hat{x}+\hat{y}) = \tfrac{1}{2}(\hat{z}+\hat{z}) = \hat{z},$$

so

$$\vec{B} = \frac{E_0}{c}\,\hat{z}\,\cos\!\left(\frac{\sqrt{2}\,\pi}{\lambda}(x+y) - \omega t\right).$$

12.5 Reflection from a perfect conductor

In empty space such a wave travels forever. Now put a perfect conductor in its way, filling the region $z > 0$. The charges in a perfect conductor move freely and without dissipation, and rearrange themselves so that the electric field parallel to the surface vanishes.

Theorem — Standing electromagnetic wave

An incident wave $\vec{E}_I = \frac{1}{2}E_0\cos(kz-\omega t)\,\hat{x}$ arriving at a perfect conductor at $z = 0$ produces a reflected wave $\vec{E}_R = -\frac{1}{2}E_0\cos(-kz-\omega t)\,\hat{x}$, and the total fields are

$$\vec{E} = E_0 \sin\omega t\,\sin kz\,\hat{x}, \qquad \vec{B} = \frac{E_0}{c}\cos\omega t\,\cos kz\,\hat{y}.$$

Proof

The boundary condition $\vec{E} = 0$ at $z = 0$ for all t forces a reflected wave of equal amplitude, opposite sign, travelling in $-z$. Its magnetic field follows from $\vec{B} = \frac{1}{c}\hat{k}\times\vec{E}$ with $\hat{k} = -\hat{z}$: $(-\hat{z})\times(-\hat{x}) = +\hat{y}$. So

$$\vec{B}_I = \frac{E_0}{2c}\cos(kz-\omega t)\,\hat{y}, \qquad \vec{B}_R = \frac{E_0}{2c}\cos(-kz-\omega t)\,\hat{y}.$$

Adding, with $\cos a - \cos b = -2\sin\frac{a+b}{2}\sin\frac{a-b}{2}$ and $\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2}$, where $a = kz-\omega t$ and $b = -kz-\omega t$, gives the stated products. ∎

The electric field has nodes at the conductor and every half wavelength in front of it; the magnetic field is largest exactly where the electric field vanishes.

Watch Out

In a traveling wave $\vec{E}$ and $\vec{B}$ are in phase. In a standing wave they are a quarter cycle apart in both time and space.

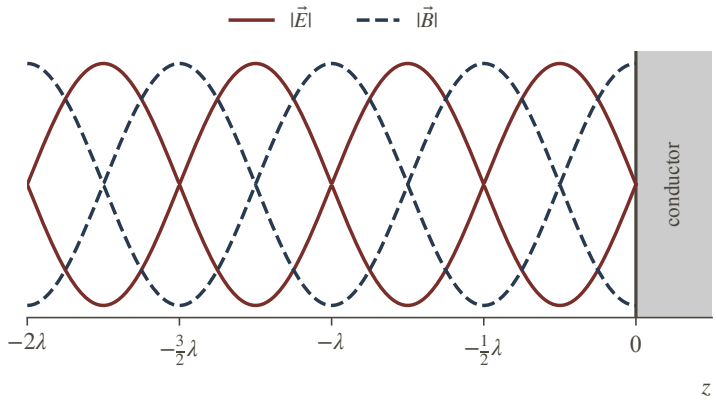


Figure 30 Envelopes of the standing wave in front of a perfect conductor. $\vec{E}$ (solid) vanishes at the surface and every half wavelength from it; $\vec{B}$ (dashed) is largest there.

12.6 Energy and the Poynting vector

Definition — Energy density and Poynting vector

The energy stored per unit volume in the fields is

$$u_E = \frac{\epsilon_0}{2} E^2, \qquad u_B = \frac{1}{2\mu_0} B^2.$$

The *Poynting vector*

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

is the energy flux: the energy crossing unit area per unit time, in the direction of $\vec{S}$.

Example — Energy in the standing wave

Using $1/\mu_0 c^2 = \epsilon_0$,

$$u_E = \frac{\epsilon_0}{2} E_0^2 \sin^2 \omega t \, \sin^2 kz, \qquad u_B = \frac{\epsilon_0}{2} E_0^2 \cos^2 \omega t \, \cos^2 kz.$$

The Poynting vector is

$$\begin{aligned} \vec{S} &= \frac{1}{\mu_0} E_x B_y \, \hat{z} = \frac{E_0^2}{\mu_0 c} \sin \omega t \cos \omega t \, \sin kz \cos kz \, \hat{z} \\ &= \frac{E_0^2}{4\mu_0 c} \sin 2\omega t \, \sin 2kz \, \hat{z}. \end{aligned}$$

Its time average is zero: no net energy is transported. The energy moves back and forth between electric form, concentrated near the antinodes of $\vec{E}$, and magnetic form, a quarter wavelength away.

This is how a microwave oven works. The walls are good conductors, the waves bounce between them and form standing waves, and the oscillating electric field shakes the molecules in the food. The fixed pattern of nodes and antinodes is why the food is rotated on a turntable.

Summary of Lecture 12

(1) Maxwell's equations in vacuum imply $\nabla^2 \vec{E} = \mu_0 \epsilon_0 \, \partial_t^2 \vec{E}$, and the same for $\vec{B}$.

(2) The wave speed is $c = 1/\sqrt{\mu_0 \epsilon_0}$; light is an electromagnetic wave.

(3) In a plane wave $\vec{B} = \frac{1}{c} \hat{k} \times \vec{E}$: transverse, perpendicular, in phase.

(4) A perfect conductor reflects the wave and produces a standing wave with zero average energy flux.

Check yourself — Lecture 12

(1) A plane wave in vacuum has $E_0 = 300$ V/m. Find B_0.

(2) At some instant $\vec{E}$ points along $\hat{y}$ and $\vec{B}$ along $\hat{z}$. Which way is the wave going?

(3) What is the time-averaged Poynting vector in the standing wave in front of a perfect conductor?

Answers are in Appendix B.

13 Lecture 13: Dispersion, Phase Velocity and Group Velocity

A pure sine wave that has been running forever carries no message. To send information one must send something that starts and stops: a pulse. This lecture asks what happens to a pulse as it travels, and finds that the answer depends entirely on the dispersion relation $\omega(k)$ of the medium.

Reading

Georgi, Chapter 8 (§8.3 to end) and Chapter 9: The Boundary at Infinity (through §9.2).

13.1 Why pulses, and when they keep their shape

Maxwell's equations in vacuum,

$$\nabla^2 \vec{E} = \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2}, \qquad \nabla^2 \vec{B} = \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2},$$

have not only harmonic plane-wave solutions but also solutions of arbitrary shape, such as $\vec{E} = E_0\,\hat{y}\,f(z-ct)$ with the matching $\vec{B}$. A localised f is a pulse: a chunk of energy localised in space and time.

Definition — Dispersive and non-dispersive media

A medium is *non-dispersive* if ω/k is the same constant v for all k, that is if $\omega = vk$. Otherwise it is *dispersive*: *dispersion* is the variation of wave speed with wavelength.

Theorem — Shape preservation

A pulse of arbitrary shape travels as $f(x-vt)$, without changing shape, if and only if the medium is non-dispersive.

Proof (idea)

Any pulse is a superposition of sinusoids $\cos(kx-\omega(k)t)$ with different k. Each can be written $\cos\big(k[x-v_p(k)t]\big)$ with $v_p(k)=\omega(k)/k$. If v_p is the same for all k, every component is shifted by the same distance vt and the sum is the original shape moved rigidly. If v_p depends on k, the components move by different distances and no longer add up to the original shape. ∎

The ideal string, with $\omega = vk$ and $v=\sqrt{T/\rho_L}$, is non-dispersive. A square pulse on it stays square.

13.2 A dispersive example: the stiff string

A real piano wire resists bending even when it carries no tension. The tension is then not the only restoring force.

Definition — Wave equation of a stiff string

Stiffness is modelled by a fourth-derivative term,

$$\frac{\partial^2 \psi}{\partial t^2} = v^2\left[\frac{\partial^2 \psi}{\partial x^2} - \alpha\,\frac{\partial^4 \psi}{\partial x^4}\right],$$

with $v^2 = T/\rho_L$ and $\alpha > 0$ a constant measuring the stiffness.

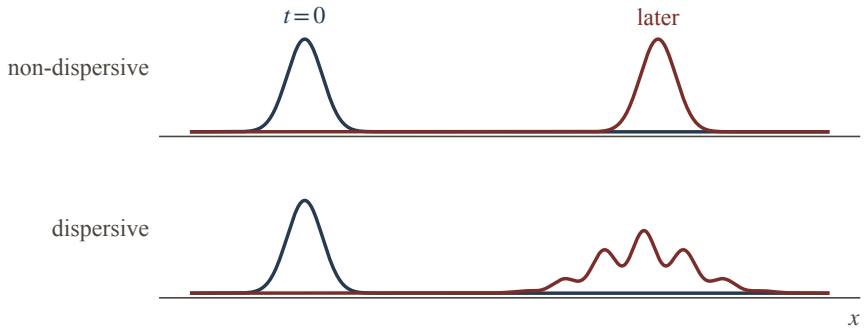


Figure 31 A pulse at two times (schematic). In a non-dispersive medium it arrives unchanged. In a dispersive medium its components travel at different speeds, so it spreads and develops ripples.

Theorem — Dispersion relation of the stiff string

$$\omega^2 = v^2\left(k^2 + \alpha k^4\right), \qquad v_p = \frac{\omega}{k} = v\sqrt{1 + \alpha k^2}.$$

Proof

Insert the test function $\psi = A\cos(kx - \omega t)$. Two time derivatives give $-\omega^2\psi$, two space derivatives $-k^2\psi$, and four space derivatives $+k^4\psi$. So $-\omega^2 = v^2(-k^2 - \alpha k^4)$. ∎

The phase velocity now grows with k. Short-wavelength components (large $k = 2\pi/\lambda$) run ahead of long-wavelength ones, and a pulse spreads out and loses its shape as it travels (Figure 32).

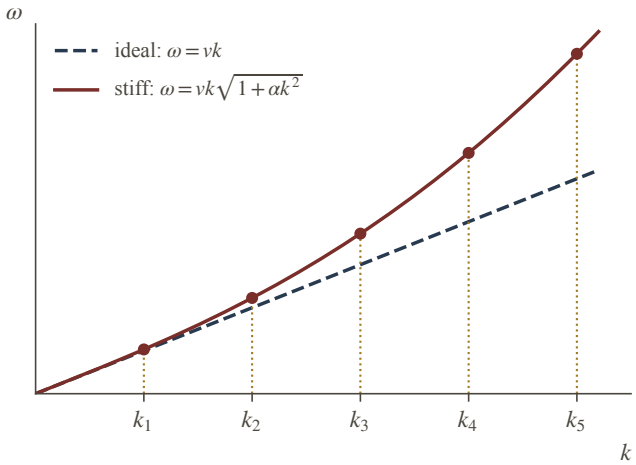


Figure 32 Dispersion relations of an ideal string (dashed) and a stiff string (solid). The dots mark the modes of a string fixed at both ends: equally spaced in k, but not in ω.

13.3 Two waves: phase and group velocity

The simplest signal with any structure is a sum of two sinusoids of equal amplitude and slightly different wave numbers:

Identity
$\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$.

$$\psi_1 = A\sin(k_1 x - \omega_1 t), \qquad \psi_2 = A\sin(k_2 x - \omega_2 t),$$

each moving with its own phase velocity ω_1/k_1, ω_2/k_2.

Theorem — Sum of two progressing waves

$$\psi_1 + \psi_2 = 2A\,\sin\!\left(\frac{k_1+k_2}{2}x - \frac{\omega_1+\omega_2}{2}t\right)\cos\!\left(\frac{k_1-k_2}{2}x - \frac{\omega_1-\omega_2}{2}t\right).$$

This is the travelling version of the beats of **§5**. When $k_1 \approx k_2 \approx k$ and $\omega_1 \approx \omega_2 \approx \omega$, the first factor is a rapidly varying *carrier* and the second a slowly varying *envelope*: the carrier's amplitude is modulated. The two factors move at different speeds.

Definition — Phase velocity and group velocity

The *phase velocity* is the speed of the carrier's crests,

$$v_p = \frac{\omega}{k}.$$

The *group velocity* is the speed of the envelope,

$$v_g = \frac{\omega_1-\omega_2}{k_1-k_2} \longrightarrow \frac{\mathrm{d}\omega}{\mathrm{d}k}.$$

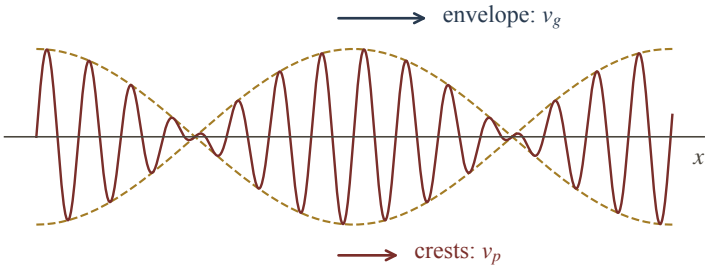


Figure 33 Two superposed waves of nearby wave number. The crests of the carrier move at v_p; the envelope moves at v_g.

Example — Velocities on the stiff string

From $\omega = vk\sqrt{1+\alpha k^2}$,

$$v_p = v\sqrt{1+\alpha k^2},$$

$$v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = v\,\frac{1+2\alpha k^2}{\sqrt{1+\alpha k^2}}.$$

For $\alpha = 0$ both equal v. For $\alpha > 0$, $v_g/v_p = (1+2\alpha k^2)/(1+\alpha k^2) > 1$: the envelope outruns the crests, and crests appear to move backwards through it.

! Which velocity carries the signal

In a non-dispersive medium $v_g = v_p$ and the distinction disappears. In a dispersive medium a modulation, and therefore the message, moves at the group velocity. A phase velocity larger than c can occur for electromagnetic waves in some media without contradiction, because nothing is transported at v_p.

13.4 A bounded dispersive system

Fix the stiff string at both ends, $\psi(0,t) = 0$ and $\psi(L,t) = 0$. Its motion is again a sum of normal modes,

$$\psi(x,t) = \sum_m A_m \sin(k_m x + \alpha_m) \sin(\omega_m t + \beta_m).$$

Theorem — Modes of a bounded dispersive string

(1) The boundary conditions give $\alpha_m = 0$ and $k_m = m\pi/L$, exactly as for the ideal string. The allowed wave numbers do *not* depend on the dispersion relation.

(2) Each mode oscillates at $\omega_m = \omega(k_m)$. The frequencies *do* depend on the dispersion relation, and in general are not equally spaced.

Proof

The mode shapes $\sin(k_m x + \alpha_m)$ solve the spatial part of the equation for any k_m, and the conditions at $x = 0$ and $x = L$ involve only the shape: $\sin\alpha_m = 0$ and $\sin k_m L = 0$. The time dependence of each shape is then fixed by substituting it into the wave equation, which is what the dispersion relation records. ■

Example — When does the shape come back?

Take a motion made of the first two modes, $\psi = \psi_1 + \psi_2$.
On an ideal string $\omega_2 = 2\omega_1$, so after the time $2\pi/\omega_1$ both modes have completed whole numbers of cycles and the string has returned to its starting shape.
On the stiff string

$$\frac{\omega_2}{\omega_1} = 2\sqrt{\frac{1 + 4\alpha\pi^2/L^2}{1 + \alpha\pi^2/L^2}} \neq 2.$$

The shape recurs only after a common multiple of $2\pi/\omega_1$ and $2\pi/\omega_2$, which may be very long, and if the ratio is irrational it never recurs exactly.

This is audible. The overtones of a stiff piano string are slightly sharper than exact multiples of the fundamental, and tuners stretch the octaves to compensate.

Summary of Lecture 13

(1) Information needs pulses; a pulse keeps its shape only if $\omega = vk$.

(2) A stiff string has $\omega^2 = v^2(k^2 + \alpha k^4)$ and is dispersive.

(3) Phase velocity $v_p = \omega/k$; group velocity $v_g = \mathrm{d}\omega/\mathrm{d}k$, the speed of the envelope.

(4) Boundary conditions fix k_m independently of dispersion; the dispersion relation then fixes $\omega_m = \omega(k_m)$.

Check yourself — Lecture 13

(1) A medium has $\omega = a\,k^2$. Find v_p and v_g.

(2) In what kind of medium are the two velocities equal?

(3) Do the allowed wave numbers of a string fixed at both ends depend on its stiffness?

Answers are in Appendix B.

14 Lecture 14: Fourier Transform and AM Radio

Lecture 13 showed that a dispersive medium destroys the shape of a pulse. Most real media are dispersive, yet radio carries speech and music across continents. This lecture develops the tool needed to follow an arbitrary signal through a dispersive medium, the Fourier transform, and then explains the trick that makes long-distance transmission work: amplitude modulation.

Reading

Georgi, Chapter 10: Signals and Fourier Analysis.

14.1 Dispersion, restated

In many media the wave speed depends on the wavelength; light in glass is the everyday example. The two velocities of **§13** are

$$v_p = \frac{\omega}{k}, \qquad v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k}.$$

In a non-dispersive medium $\omega = vk$, so $v_g = v_p = v$ and pulses keep their shape. In a dispersive medium a square pulse is built from many sinusoidal components travelling at different speeds, and it spreads out.

Example — A medium with a forbidden band

Suppose the dispersion relation has the form

$$\omega^2 = \omega_b^2 + c^2k^2,$$

which is a simple model of an electromagnetic wave in a medium with a lowest allowed frequency ω_b. Then

$$v_p = \frac{\omega}{k} = c\sqrt{1 + \frac{\omega_b^2}{c^2k^2}}, \qquad v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{c^2k}{\omega} = \frac{c}{\sqrt{1 + \omega_b^2/c^2k^2}}.$$

As functions of frequency, $v_p = c/\sqrt{1 - \omega_b^2/\omega^2}$ and $v_g = c\sqrt{1 - \omega_b^2/\omega^2}$, so that $v_pv_g = c^2$: the phase velocity exceeds c and the group velocity is below it. For $\omega < \omega_b$ there is no real k at all. A wave of such a frequency cannot propagate in the medium, and if the medium does not absorb energy either, the wave is *totally reflected.* (The lecture's example was an ionic crystal, shown only as a plotted curve; this formula is a stand-in with the same qualitative features.)

14.2 Sending a signal down a long string

Shake the end $x = 0$ of a very long string so that $\psi(0, t) = f(t)$. In a non-dispersive medium the result is simply

$$\psi(x, t) = f\left(t - \frac{x}{v}\right):$$

whatever happens at the end happens at x a time x/v later.

In a dispersive medium each frequency travels at its own speed. The strategy is to split $f(t)$ into components of definite frequency, propagate each one separately using the dispersion relation, and add them up again.

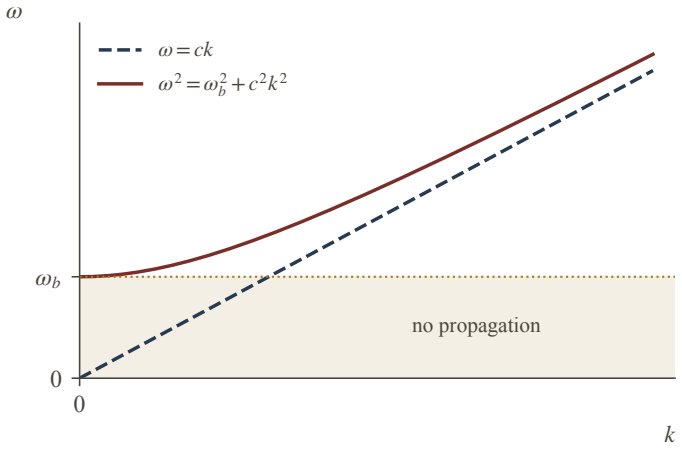


Figure 34 A dispersion relation with a lowest frequency ω_b. Below it there is no travelling-wave solution.

Definition — Fourier transform

A function $f(t)$ is written as a continuous superposition of harmonic oscillations,

$$f(t) = \int_{-\infty}^{\infty} \mathrm{d}\omega \; C(\omega)\, e^{-i\omega t},$$

where $C(\omega)$ is the amplitude of the component at frequency ω. The amplitude is recovered by

$$C(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}t \; f(t)\, e^{i\omega t}.$$

$C(\omega)$ is the *Fourier transform* of $f(t)$.

This is the continuous version of the Fourier series of **§9**: the sum over discrete modes has become an integral over all frequencies, and the orthogonality integral has become a delta function.

Definition — Dirac delta function

$\delta(x)$ is zero for $x \neq 0$, infinite at $x = 0$, and normalised so that

$$\int_{-\infty}^{\infty} \delta(x)\, \mathrm{d}x = 1, \qquad \int_{-\infty}^{\infty} \delta(x-\alpha)\, f(\alpha)\, \mathrm{d}\alpha = f(x).$$

The complex exponentials are orthogonal in the sense

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(\omega-\omega')t}\, \mathrm{d}t = \delta(\omega - \omega').$$

Proof of the inversion formula

Insert the expansion of $f(t)$ and exchange the order of integration:

$$\begin{aligned} \frac{1}{2\pi} \int \mathrm{d}t\, f(t)\, e^{i\omega t} &= \frac{1}{2\pi} \int \mathrm{d}\omega'\, C(\omega') \int \mathrm{d}t\; e^{i(\omega-\omega')t} \\ &= \int \mathrm{d}\omega'\, C(\omega')\, \delta(\omega - \omega') = C(\omega). \end{aligned}$$

■

Theorem — Propagating an arbitrary signal

If $\psi(0,t) = f(t)$ and the medium has dispersion relation $k = k(\omega)$, then for a wave travelling in the $+x$ direction

$$\psi(x,t) = \int_{-\infty}^{\infty} \mathrm{d}\omega\; C(\omega)\, e^{-i\omega t + ik(\omega)x}.$$

Proof

Each component $e^{-i\omega t}$ at $x = 0$ extends to the travelling wave $e^{i(k(\omega)x - \omega t)}$, which solves the wave equation of the medium. By linearity the superposition with weights $C(\omega)$ is a solution, and at $x = 0$ it reduces to $f(t)$. ∎

As a check, for a non-dispersive medium $k(\omega) = \omega/v$ and

$$\psi(x,t) = \int \mathrm{d}\omega\; C(\omega)\, e^{-i\omega(t - x/v)} = f\!\left(t - \frac{x}{v}\right),$$

as expected.

14.3 The idea of AM radio

We want to transmit a signal $f_s(t)$, for instance music, whose typical frequencies are around 1 kHz. Sending it directly fails twice over: it spreads out by dispersion, and such low frequencies are impractical to radiate. The solution is to use the signal to control the amplitude of a much faster oscillation.

Definition — Amplitude modulation

The transmitted wave is

$$f(t) = f_s(t)\cos\omega_0 t = \mathrm{Re}\left[f_s(t)\, e^{-i\omega_0 t}\right],$$

where $\cos\omega_0 t$ is the *carrier*, with ω_0 far above every frequency present in f_s, and $f_s(t)$ is the slowly varying *envelope*. AM carriers lie between about 0.3 and 30 MHz.

Multiplying by the carrier moves the signal's frequency content up to a narrow band around ω_0. For a single tone,

$$\cos\omega_s t\;\cos\omega_0 t = \tfrac{1}{2}\left[\cos(\omega_0 + \omega_s)t + \cos(\omega_0 - \omega_s)t\right],$$

so the Fourier amplitude of $f(t)$ is non-zero only for $\omega_0 - \omega_s \lesssim \omega \lesssim \omega_0 + \omega_s$, with $\omega_s \ll \omega_0$.

Why narrow

A signal tone at ω_s puts the transmitted energy at $\omega_0 \pm \omega_s$ only.

Over so narrow a band, any smooth dispersion relation is well approximated by its tangent line. Let $\omega_0 = \omega(k_0)$. Then

$$\omega \approx \omega_0 + (k - k_0)\, v_g, \qquad v_g = \left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k=k_0},$$

or, solved for k,

$$k \approx \frac{\omega}{v_g} + b, \qquad b = k_0 - \frac{\omega_0}{v_g}.$$

Theorem — A narrow-band signal travels undistorted

If $C(\omega)$ is negligible outside a band $|\omega - \omega_0| < \Delta\omega$ with $\Delta\omega \ll \omega_0$, small enough that the linear approximation above holds, then

$$\psi(x,t) = \mathrm{Re}\left[f_s\!\left(t - \frac{x}{v_g}\right) e^{-i(\omega_0 t - k_0 x)} \right].$$

Proof

Write $f_s(t) = \int \mathrm{d}\omega\, C(\omega) e^{-i\omega t}$, with $C(\omega)$ concentrated near $\omega = 0$. Then

$$f_s(t)\, e^{-i\omega_0 t} = \int \mathrm{d}\omega\; C(\omega)\, e^{-i(\omega+\omega_0)t} = \int \mathrm{d}\omega\; C(\omega-\omega_0)\, e^{-i\omega t}.$$

Propagate each component with $k \approx \omega/v_g + b$:

$$\int \mathrm{d}\omega\; C(\omega-\omega_0)\, e^{-i\omega t}\, e^{i(\omega/v_g + b)x} = e^{ibx} \int \mathrm{d}\omega\; C(\omega-\omega_0)\, e^{-i\omega(t - x/v_g)}.$$

Shift the integration variable back, $\omega \to \omega + \omega_0$:

$$= e^{ibx}\, e^{-i\omega_0(t-x/v_g)} \int \mathrm{d}\omega\; C(\omega)\, e^{-i\omega(t-x/v_g)}.$$

The integral is $f_s(t - x/v_g)$, and the prefactor is $e^{-i\omega_0 t} e^{i(\omega_0/v_g + b)x} = e^{-i(\omega_0 t - k_0 x)}$. ■

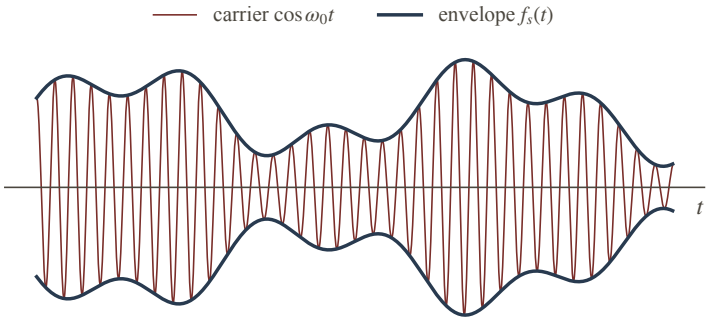


Figure 35 An amplitude-modulated wave: a fast carrier whose amplitude follows the slow signal.

The envelope arrives with exactly its original shape, delayed by x/v_g, while the carrier underneath runs at the phase velocity $v_p = \omega_0/k_0$. The dispersion of the medium has been made harmless, not by removing it, but by confining the signal to a band so narrow that only the local slope of $\omega(k)$ matters. This is what allows voice and music to be sent thousands of miles.

Summary of Lecture 14

(1) Fourier transform: $f(t) = \int \mathrm{d}\omega\, C(\omega) e^{-i\omega t}$, $C(\omega) = \frac{1}{2\pi}\int \mathrm{d}t\, f(t) e^{i\omega t}$.

(2) A signal propagates as $\psi = \int \mathrm{d}\omega\, C(\omega) e^{-i\omega t + ik(\omega)x}$.

(3) AM: transmit $f_s(t)\cos\omega_0 t$, which occupies a narrow band around ω_0.

(4) A narrow-band signal's envelope travels undistorted at v_g; the carrier travels at v_p.

Check yourself — Lecture 14

(1) A 1 MHz carrier is amplitude-modulated by a 1 kHz tone. Which frequencies are transmitted?

(2) At what speed does the envelope of a narrow-band signal travel? The carrier?

(3) If $f(t)$ has Fourier transform $C(\omega)$, what is the transform of the delayed signal $f(t - t_0)$?

Answers are in Appendix B.

15 Lecture 15: Uncertainty Principle and 2D Waves

Reading

Georgi, Chapter 10: Signals and Fourier Analysis.

A short pulse needs many frequencies; a pure tone lasts forever. This lecture makes that trade-off precise. The result, $\Delta t\,\Delta\omega \geq \frac{1}{2}$, is a theorem about waves of any kind, and it becomes Heisenberg's uncertainty principle once particles are described by waves. The lecture ends by starting on waves in two dimensions.

15.1 Recap: narrow-band transmission

For a signal $f(t) = f_s(t)\cos\omega_0 t$ of small bandwidth, **§14** gave

$$\psi(x,t) \approx \mathrm{Re}\left[f_s\left(t - \frac{x}{v_g}\right) e^{-i(\omega_0 t - k_0 x)}\right],$$

with the envelope travelling at $v_g = \mathrm{d}\omega/\mathrm{d}k|_{k_0}$ and the carrier at $v_p = \omega_0/k_0$. For AM radio ω_0 corresponds to 0.3–30 MHz and the bandwidth to about 5 kHz, so $\Delta\omega \ll \omega_0$ indeed.

15.2 A pulse and its spectrum

Example — Fourier transform of a two-sided exponential

Let $f(t) = e^{-\Gamma|t|}$ with $\Gamma > 0$. Then

$$\begin{aligned} C(\omega) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}t\; e^{-\Gamma|t|} e^{i\omega t} \\ &= \frac{1}{2\pi}\left[\int_{-\infty}^{0} \mathrm{d}t\; e^{(\Gamma+i\omega)t} + \int_{0}^{\infty} \mathrm{d}t\; e^{-(\Gamma-i\omega)t}\right] \\ &= \frac{1}{2\pi}\left[\frac{1}{\Gamma+i\omega} + \frac{1}{\Gamma-i\omega}\right] = \frac{\Gamma}{\pi(\Gamma^2+\omega^2)}. \end{aligned}$$

The pulse has duration of order $1/\Gamma$, and its spectrum is a peak of width of order Γ.

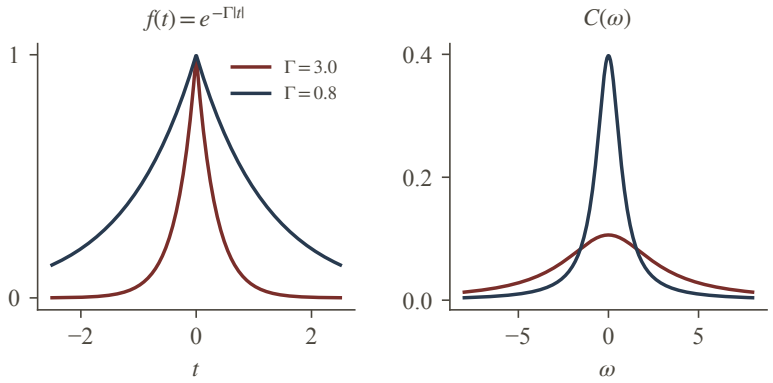


Figure 36 Large Γ (red): a narrow pulse with a wide spectrum. Small Γ (navy): a wide pulse with a narrow spectrum.

Making the pulse shorter always makes its spectrum wider, and vice versa. To turn this into a theorem we need precise definitions of "width".

15.3 Defining the spread

Definition — Averages and spreads of a signal

The *intensity* of a signal is proportional to $|f(t)|^2$. The average of any quantity $g(t)$, weighted by the intensity, is

$$\langle g \rangle = \frac{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}t\; g(t)\,|f(t)|^2}{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}t\; |f(t)|^2}.$$

The *spread in time* is the root-mean-square deviation from the average time,

$$\Delta t^2 \equiv \left\langle (t - \langle t \rangle)^2 \right\rangle.$$

The *spread in frequency* $\Delta\omega$ is defined the same way with weight $|C(\omega)|^2$: $\Delta\omega^2 \equiv \langle(\omega - \langle\omega\rangle)^2\rangle$.

Theorem — Frequency as a derivative

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\; \omega\, C(\omega)\, e^{-i\omega t} = i\frac{\partial}{\partial t} f(t).$$

Consequently the frequency averages can be computed from $f(t)$ directly:

$$\langle \omega \rangle = \frac{\int \mathrm{d}t\; f^*\big(i\,\partial_t f\big)}{\int \mathrm{d}t\; |f|^2}, \qquad \Delta\omega^2 = \frac{\int \mathrm{d}t\; \big|\big(i\,\partial_t - \langle\omega\rangle\big) f\big|^2}{\int \mathrm{d}t\; |f|^2}.$$

Proof

Differentiating $f(t) = \int \mathrm{d}\omega\, C(\omega) e^{-i\omega t}$ under the integral brings down a factor $-i\omega$, and multiplying by i gives the first statement. The second follows because the total intensity is the same whether it is added up over time or over frequency ($\int |f|^2\,\mathrm{d}t = 2\pi \int |C|^2\,\mathrm{d}\omega$), and the same holds for $i\partial_t f$, whose transform is $\omega C(\omega)$. ∎

15.4 The uncertainty principle

Theorem — Time–frequency uncertainty

For any signal $f(t)$ that vanishes as $t \to \pm\infty$,

$$\Delta t\;\Delta\omega \geq \frac{1}{2}.$$

Proof

Abbreviate $T \equiv t - \langle t\rangle$ and $\Omega \equiv i\,\partial_t - \langle\omega\rangle$. For a real parameter κ define

$$\gamma(\kappa, t) \equiv (T - i\kappa\,\Omega)\, f(t), \qquad R(\kappa) \equiv \frac{\int |\gamma|^2\,\mathrm{d}t}{\int |f|^2\,\mathrm{d}t} \geq 0.$$

Expand the numerator:

$$|\gamma|^2 = |Tf|^2 + \kappa^2 |\Omega f|^2 + i\kappa\big[Tf\,(\Omega f)^* - (\Omega f)\,Tf^*\big].$$

In the bracket the $\langle\omega\rangle$ terms cancel, and with $(\Omega f)^* = (-i\,\partial_t - \langle\omega\rangle)f^*$ the cross term is

$$i\kappa\,T\big[-i\,f\,\partial_t f^* - i\,f^*\partial_t f\big] = \kappa\,T\,\frac{\partial}{\partial t}\big(ff^*\big).$$

Integrate by parts. The boundary term vanishes because $f \to 0$ at $\pm\infty$, and $\partial T/\partial t = 1$:

$$\int \mathrm{d}t\;\kappa\,T\,\partial_t|f|^2 = -\kappa\int \mathrm{d}t\;|f|^2.$$

Dividing by $\int |f|^2\,\mathrm{d}t$ and recognising the definitions of the spreads,

$$R(\kappa) = \Delta t^2 + \kappa^2\Delta\omega^2 - \kappa \geq 0 \quad \text{for every } \kappa.$$

The quadratic is smallest at $\kappa = 1/(2\Delta\omega^2)$, where

$$R_{\min} = \Delta t^2 - \frac{1}{4\,\Delta\omega^2} \geq 0.$$

Rearranging gives $\Delta t^2\Delta\omega^2 \geq \tfrac{1}{4}$. ∎

Tip

Equality holds only for a Gaussian pulse, which is the problem-set example mentioned in the lecture.

Example — Time resolution of an AM broadcast

An AM station has bandwidth $\Delta\omega = 2\pi\Delta\nu \approx 3\times10^4\ \mathrm{s}^{-1}$. Then

$$\Delta t \geq \frac{1}{2\Delta\omega} \approx 1.7\times10^{-5}\ \mathrm{s}.$$

Features of the signal closer together than a few times 10^{-5} s cannot be told apart. To send faster variations, a wider band is required.

Theorem — Position–wave-number and Heisenberg's relation

For a pulse moving at speed v, $\Delta x = v\,\Delta t$ and $\Delta k = \Delta\omega/v$, so

$$\Delta x\;\Delta k \geq \frac{1}{2}.$$

In quantum mechanics a particle of momentum p is described by a wave with $p = \hbar k$, and therefore

$$\Delta x\;\Delta p \geq \frac{\hbar}{2}, \qquad \hbar = \frac{h}{2\pi} \approx 1.05\times10^{-34}\ \mathrm{J\,s} \approx 6.6\times10^{-16}\ \mathrm{eV\,s}.$$

Watch Out

The inequality is not about clumsy instruments. A wave simply does not have a sharp position and a sharp wave number at once.

! What the uncertainty principle says

The position and momentum of a particle cannot both be specified with arbitrary precision. This is the mathematical statement proved above, applied to matter waves; it has nothing to do with the quality of the measuring apparatus.

15.5 First look at two dimensions

Consider a rectangular mesh of beads of mass m, connected by strings with tension T_H and spacing a_H in the horizontal (x) direction and T_V, a_V in the vertical (y) direction. The beads move only perpendicular to the mesh. Label them by a pair of integers (j_x, j_y), with equilibrium position $x = j_x a_H$, $y = j_y a_V$.

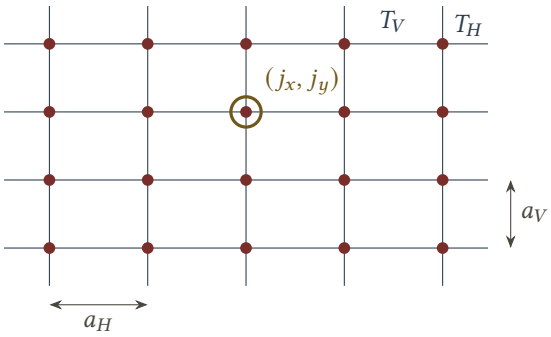


Figure 37 Part of the mesh of beads, seen from above. Each bead is tied to four neighbours and moves perpendicular to the page.

Remove the walls for the moment, so the mesh is infinite. It is then unchanged by a shift of one spacing in x, and separately by a shift of one spacing in y. By the argument of **§7**, the normal modes are eigenvectors of both translations.

Theorem — Normal modes of the infinite mesh

$$\psi(x,y,t) = \mathrm{Re}\left[A\, e^{i\vec{k}\cdot\vec{r}}\, e^{-i\omega t}\right], \qquad \vec{k} = k_x\hat{x} + k_y\hat{y}, \quad \vec{r} = x\hat{x} + y\hat{y},$$

with dispersion relation

$$\omega^2 = \frac{4T_H}{ma_H}\sin^2\frac{k_x a_H}{2} + \frac{4T_V}{ma_V}\sin^2\frac{k_y a_V}{2}.$$

Proof (sketch)

Translation in x gives the factor $e^{ik_x x}$ and translation in y the factor $e^{ik_y y}$; their product is $e^{i\vec{k}\cdot\vec{r}}$. The force on a bead is the sum of the one-dimensional beaded-string force from its two horizontal neighbours and that from its two vertical neighbours, so the one-dimensional result $\omega^2 = (4T/ma)\sin^2(ka/2)$ of **§8** applies to each direction and the two contributions add. ∎

A qualitative difference from one dimension appears at once. In one dimension a given ω had just two modes, $e^{\pm ikx}$. In two dimensions a given ω has infinitely many: lowering k_x a little can always be compensated by raising k_y. The next lecture puts the walls back and sees which of these survive.

Summary of Lecture 15

(1) $e^{-\Gamma|t|}$ has spectrum $\Gamma/\pi(\Gamma^2+\omega^2)$: short pulse, wide spectrum.

(2) Frequency acts on $f(t)$ as $i\,\partial/\partial t$.

(3) $\Delta t\,\Delta\omega \geq \frac{1}{2}$, hence $\Delta x\,\Delta k \geq \frac{1}{2}$ and $\Delta x\,\Delta p \geq \hbar/2$.

(4) In two dimensions the modes are $e^{i\vec{k}\cdot\vec{r}}$, and one frequency corresponds to infinitely many $\vec{k}$.

Check yourself — Lecture 15

(1) A pulse lasts about $\Delta t = 1$ ns. What is the smallest possible spread in angular frequency?

(2) Which pulse shape reaches the limit $\Delta t\,\Delta\omega = \frac{1}{2}$?

(3) An electron is confined to $\Delta x = 0.1$ nm. Estimate the minimum spread of its momentum.

Answers are in Appendix B.

16 Lecture 16: 2D and 3D Waves; Snell's Law

This lecture completes the move from one dimension to two and three: standing waves on a rectangular membrane and in a box, then plane waves travelling in an arbitrary direction. When such a wave meets the boundary between two media, matching the wave on the two sides yields the law of reflection and Snell's law of refraction. Geometrical optics turns out to be a consequence of wave physics.

Reading

Georgi, Chapter 11: Two and Three Dimensions, and Chapter 12 (through §12.3).

16.1 The mesh with walls

Recall the one-dimensional beaded string of **§8**: beads of mass m, spacing a, tension T, with modes $e^{\pm ijka}$ and

$$\omega(k) = 2\sqrt{\frac{T}{ma}}\left|\sin\frac{ka}{2}\right|.$$

For the two-dimensional mesh of **§15** the modes of the infinite system are

$$A\,e^{ik_x x}e^{ik_y y} = A\,e^{i\vec{k}\cdot\vec{r}},$$

with

$$\omega^2 = \frac{4T_H}{ma_H}\sin^2\frac{k_x a_H}{2} + \frac{4T_V}{ma_V}\sin^2\frac{k_y a_V}{2}.$$

Since $\omega/|\vec{k}|$ is not constant, the mesh is a dispersive medium.

In one dimension the two modes $e^{\pm ikx}$ of equal frequency combine into $\cos kx$ and $\sin kx$, and a boundary condition picks one of them. In two dimensions infinitely many (k_x, k_y) share each frequency, until walls are imposed.

Theorem — Normal modes of the bounded mesh

Let the mesh be fixed on all four sides of a rectangle $0 \le x \le L_H$, $0 \le y \le L_V$:

$$\psi(0,y,t) = \psi(L_H,y,t) = \psi(x,0,t) = \psi(x,L_V,t) = 0.$$

The normal modes are

$$\psi_{(n_x,n_y)} = A_{(n_x,n_y)}\,\sin\!\left(\frac{n_x\pi x}{L_H}\right)\sin\!\left(\frac{n_y\pi y}{L_V}\right), \qquad k_x = \frac{n_x\pi}{L_H}, \quad k_y = \frac{n_y\pi}{L_V}.$$

Proof

For fixed $|k_x|$ and $|k_y|$ the four modes $e^{\pm ik_x x}e^{\pm ik_y y}$ have the same frequency, so any combination of them is a mode. The only combination vanishing at $x = 0$ and at $y = 0$ is $\sin k_x x\,\sin k_y y$. Vanishing at $x = L_H$ and $y = L_V$ then requires $\sin k_x L_H = 0$ and $\sin k_y L_V = 0$. ∎

Example — Counting modes

In the lecture's mesh $L_H = 5a_H$ and $L_V = 4a_V$, so there are $4 \times 3 = 12$ moving beads. The allowed values are $n_x = 1,\dots,4$ and $n_y = 1,2,3$, giving 12 modes, one per degree of freedom. The general motion is

$$\psi(x,y,t) = \sum_{n_x,n_y}\psi_{(n_x,n_y)}(x,y)\,\sin\!\big(\omega_{(n_x,n_y)}t + \phi_{(n_x,n_y)}\big),$$

with $\psi_{(n_x,n_y)}$ the mode shapes above.

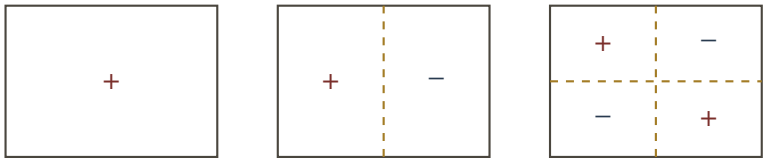


Figure 38 Three modes of a rectangular membrane. Dashed lines are nodal lines; the signs show regions moving in opposite directions. Sand sprinkled on a vibrating plate collects on such lines (Chladni figures).

16.2 The continuum limit: wave equation in two and three dimensions

Take $T_H = T_V = T$ and $a_H = a_V = a \to 0$. For $ka \ll 1$, $\sin^2(ka/2) \approx k^2a^2/4$, so

$$\omega^2 \approx \frac{Ta}{m}\left(k_x^2 + k_y^2\right).$$

Definition — Membrane parameters

The *surface mass density* is $\rho_s = m/a^2$ and the *surface tension* is $T_s = T/a$ (force per unit length of edge). In terms of them

$$\omega^2 = \frac{T_s}{\rho_s}\,|\vec{k}|^2, \qquad v = \sqrt{\frac{T_s}{\rho_s}}.$$

Theorem — Wave equation in higher dimensions

A continuous membrane obeys

$$\frac{\partial^2\psi}{\partial t^2} = v^2\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\psi = v^2\,\nabla^2\psi,$$

and in three dimensions, for example for sound,

$$\frac{\partial^2\psi}{\partial t^2} = v^2\,\nabla^2\psi(x, y, z, t).$$

The continuous medium is non-dispersive: $\omega = v\,|\vec{k}|$.

Proof (as in one dimension)

Acting on $e^{i(\vec{k}\cdot\vec{r}-\omega t)}$, each ∂_t gives $-i\omega$ and ∇^2 gives $-|\vec{k}|^2$. The equation $\partial_t^2\psi = v^2\nabla^2\psi$ therefore reproduces $\omega^2 = v^2|\vec{k}|^2$ for every mode, and by superposition holds for every motion built from them. ∎

Example — Sound in a rectangular box

A box with sides a, b, c and boundary conditions like those of the membrane has modes

$$\psi \propto \sin(k_x x)\,\sin(k_y y)\,\sin(k_z z)\,\sin(\omega t + \phi),$$

$$k_x = \frac{n_x\pi}{a}, \qquad k_y = \frac{n_y\pi}{b}, \qquad k_z = \frac{n_z\pi}{c}.$$

Substituting into the wave equation,

$$\omega^2 = v^2\left[\left(\frac{n_x\pi}{a}\right)^2 + \left(\frac{n_y\pi}{b}\right)^2 + \left(\frac{n_z\pi}{c}\right)^2\right],$$

with n_x, n_y, n_z integers. These are the resonant frequencies of a room.

16.3 Plane waves

Definition — Plane wave

The progressing-wave solution

$$\psi(\vec{r}, t) = A\, e^{i(\vec{k}\cdot\vec{r} - \omega t)}$$

is a *plane wave*. Its phase is constant on planes (in two dimensions, lines) perpendicular to $\vec{k}$, and these *wavefronts* move in the direction of $\vec{k}$ at speed $\omega/|\vec{k}|$.

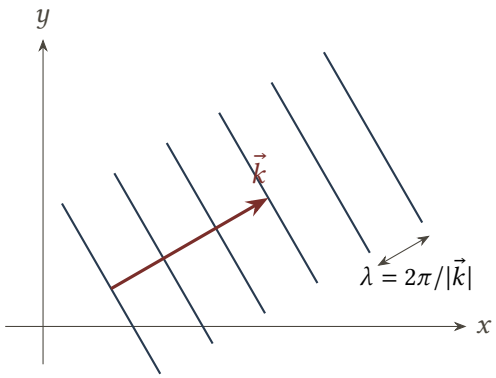


Figure 39 A plane wave in two dimensions. The lines are wavefronts, one wavelength apart and perpendicular to $\vec{k}$.

The same form describes electromagnetic waves, sound, and waves on membranes. In a uniform medium a plane wave goes on forever. Something interesting happens only where the medium changes.

16.4 Reflection and refraction at a boundary

Two membranes are joined along the line $x = 0$. The one at $x < 0$ has wave speed v, the one at $x > 0$ has wave speed v'. A plane wave with wave vector $\vec{k} = (k_x, k_y)$ and frequency ω arrives from the left (Figure 40).

In general one must allow reflected and transmitted waves in every direction consistent with the frequency:

$$\psi_L = A\, e^{i(\vec{k}\cdot\vec{r} - \omega t)} + \sum_\alpha R_\alpha A\, e^{i(\vec{k}_\alpha\cdot\vec{r} - \omega t)} \qquad (x \le 0),$$

$$\psi_R = \sum_\beta T_\beta A\, e^{i(\vec{k}_\beta\cdot\vec{r} - \omega t)} \qquad (x \ge 0),$$

with $|\vec{k}_\alpha|^2 = \omega^2/v^2$ and $|\vec{k}_\beta|^2 = \omega^2/v'^2$, as the dispersion relation of each membrane demands.

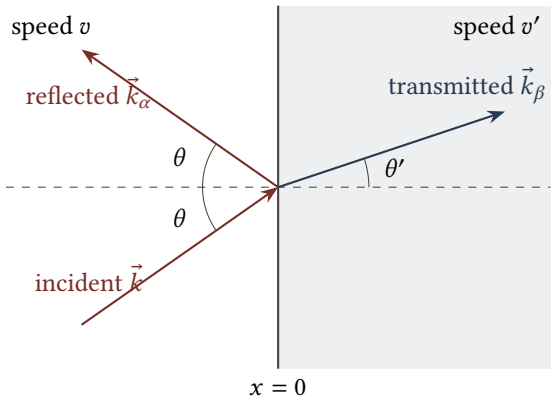


Figure 40 A plane wave meeting the boundary between two media. Angles are measured from the normal to the boundary.

Theorem — Law of reflection and Snell's law

(1) The component of the wave vector along the boundary is the same for the incident, reflected and transmitted waves: $k_{\alpha y} = k_{\beta y} = k_y$.

(2) The angle of reflection equals the angle of incidence.

(3) The angle of refraction θ' satisfies

$$n \sin\theta = n' \sin\theta', \qquad n \equiv \frac{c}{v},$$

where n is the *refractive index* of a medium.

Proof

The membrane does not tear, so $\psi_L = \psi_R$ at $x = 0$ for all y and t:

$$e^{i(k_y y - \omega t)} + \sum_\alpha R_\alpha \, e^{i(k_{\alpha y} y - \omega t)} = \sum_\beta T_\beta \, e^{i(k_{\beta y} y - \omega t)}.$$

Exponentials with different k_y are independent functions of y, so this can hold for all y only if every term has the same k_y. That proves (1), and leaves a single reflected and a single transmitted wave, with

$$k_{\alpha x} = -\sqrt{\frac{\omega^2}{v^2} - k_y^2} = -k_x, \qquad k_{\beta x} = +\sqrt{\frac{\omega^2}{v'^2} - k_y^2}.$$

The reflected wave vector is the mirror image of the incident one, which is (2). For (3), $k_y = |\vec{k}| \sin\theta = |\vec{k}'| \sin\theta'$, and $|\vec{k}| = \omega/v = n\,\omega/c$, $|\vec{k}'| = n'\omega/c$. ∎

Generality

Only continuity at the boundary was used, so the result holds for water waves, sound and light alike.

These are the two basic laws of geometrical optics, obtained here from nothing more than the requirement that the wave be continuous along the boundary. The amplitudes R and T need a second boundary condition and are found for light in Lecture 19.

Theorem — Total internal reflection

If the wave goes from a slower to a faster medium ($n > n'$) and

$$\sin\theta > \frac{n'}{n},$$

there is no transmitted travelling wave.

Proof

Snell's law would require $\sin\theta' = (n/n')\sin\theta > 1$, which no real angle satisfies. Equivalently $k_{\beta x}^2 = \omega^2/v'^2 - k_y^2 < 0$: the transmitted wave number is imaginary, and the disturbance decays exponentially with distance into the second medium instead of propagating. ■

The limiting angle, $\sin\theta_c = n'/n$, is the *critical angle*. For glass ($n \approx 1.5$) to air it is about 42°. Light striking the inside of a glass fibre at a more grazing angle is reflected with no loss, which is what guides it along an optical fibre.

Summary of Lecture 16

(1) A rectangular membrane has modes $\sin(n_x\pi x/L_H)\sin(n_y\pi y/L_V)$.

(2) In the continuum limit $\partial_t^2\psi = v^2\nabla^2\psi$ with $\omega = v|\vec{k}|$.

(3) Plane waves: $A\,e^{i(\vec{k}\cdot\vec{r}-\omega t)}$.

(4) Continuity along a boundary conserves k_y, which gives the law of reflection, Snell's law $n\sin\theta = n'\sin\theta'$, and total internal reflection.

Check yourself — Lecture 16

(1) For a square membrane of side L, compare the frequencies of the modes $(1,2)$ and $(2,1)$.

(2) Find the critical angle for light going from water ($n = 1.33$) to air.

(3) Which two quantities are the same for the incident, reflected and transmitted waves?

Answers are in Appendix B.

17 Lecture 17: Polarization and Polarizers

Reading
Georgi, Chapter 12: Polarization (through §12.3).

Part III — Optics. An electromagnetic wave is transverse, so its electric field has two independent components perpendicular to the direction of travel. How those two components are related, in size and in phase, is the *polarization* of the wave. This lecture classifies the possibilities, introduces a compact two-component notation for them, and describes the simplest device that acts on polarization: the polarizer.

17.1 Describing polarization

Recall Snell's law from **§16**, $n_1 \sin\theta_1 = n_2 \sin\theta_2$; it will be needed again in Lecture 19. For now consider a single plane wave, and choose the z axis along its direction of travel.

Definition — Polarization vector

A plane wave travelling along z is

$$\vec{E}(z,t) = \mathrm{Re}\left[\vec{\psi}_0\, e^{i(kz-\omega t)}\right], \qquad \vec{\psi}_0 = \psi_1 \hat{x} + \psi_2 \hat{y},$$

with complex amplitudes $\psi_1 = A_1 e^{i\phi_1}$ and $\psi_2 = A_2 e^{i\phi_2}$. It is a superposition of two waves, one polarized along x and one along y. The column vector

$$Z = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

specifies the polarization completely.

Only the ratio A_2/A_1 and the phase difference $\phi_2 - \phi_1$ matter for the shape traced out by $\vec{E}$; an overall phase only shifts the origin of time.

17.2 Linear polarization

Example — Two waves in phase

Add

$$\vec{E}_1 = E_0 \cos(kz-\omega t)\,\hat{x}, \qquad \vec{E}_2 = E_0 \cos(kz-\omega t)\,\hat{y}.$$

The sum is $\vec{E} = \mathrm{Re}\left[(E_0\hat{x} + E_0\hat{y})e^{i(kz-\omega t)}\right]$, that is

$$Z = E_0 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

At every instant $E_x = E_y$, so the tip of $\vec{E}$ moves back and forth along a fixed line at 45° to the x axis.

Definition — Linearly polarized wave

A wave is *linearly polarized* if the two components have no phase difference (or a phase difference of π). Then $\vec{E}$ oscillates along a fixed direction. For polarization at angle θ to the x axis,

$$Z = E_0 \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix},$$

with the special cases $Z = E_0(1, 0)$ along x and $Z = E_0(0, 1)$ along y.

17.3 Circular polarization

Example — Equal amplitudes, phase difference $\pi/2$

Add

$$\vec{E}_1 = E_0 \cos(kz - \omega t)\,\hat{x}, \qquad \vec{E}_2 = E_0 \sin(kz - \omega t)\,\hat{y} = E_0 \cos\!\left(kz - \omega t - \tfrac{\pi}{2}\right)\hat{y}.$$

Since $e^{-i\pi/2} = -i$,

$$\vec{E} = \mathrm{Re}\left[(E_0\hat{x} - iE_0\hat{y})\, e^{i(kz-\omega t)}\right], \qquad Z = E_0 \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

At a fixed position, say $z = 0$, the field is $(E_x, E_y) = E_0(\cos\omega t, -\sin\omega t)$: a vector of constant length E_0 rotating clockwise in the x–y plane.

Definition — Circularly polarized wave

A wave is *circularly polarized* if the two components have equal amplitudes and a phase difference of $\pm\pi/2$. The field vector has constant magnitude and rotates at angular frequency ω:

$$Z = E_0 \begin{pmatrix} 1 \\ -i \end{pmatrix} \text{ (clockwise)}, \qquad Z = E_0 \begin{pmatrix} 1 \\ i \end{pmatrix} \text{ (counter-clockwise)}.$$

The sense of rotation is that seen at a fixed point, looking at the x–y plane with the z axis pointing towards the viewer. Books assign the names "right-handed" and "left-handed" in opposite ways, so it is safer to state the sense of rotation and the viewpoint.

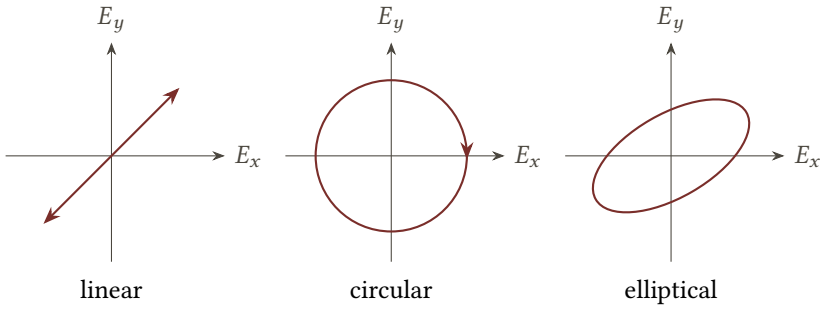


Figure 41 The path traced by the tip of $\vec{E}$ at a fixed point in space for the three kinds of polarization.

17.4 Elliptical polarization

Definition — Elliptically polarized wave

In every other case the tip of $\vec{E}$ traces an ellipse, and the wave is *elliptically polarized*. There are two ways to obtain it:

(1) a phase difference of $\pi/2$ with *unequal* amplitudes, for example

$$\vec{E}_1 = \tfrac{1}{2}E_0 \cos(kz - \omega t)\,\hat{x}, \qquad \vec{E}_2 = E_0 \sin(kz - \omega t)\,\hat{y},$$

which is $Z = E_0\,(\tfrac{1}{2},\, -i)$;

(2) a phase difference $\Delta\phi$ that is neither a multiple of π (linear) nor, for equal amplitudes, an odd multiple of $\pi/2$ (circular), for example

$$\vec{E}_1 = E_0 \cos(kz - \omega t)\,\hat{x}, \qquad \vec{E}_2 = E_0 \cos(kz - \omega t + \Delta\phi)\,\hat{y}.$$

In case (1) the axes of the ellipse lie along x and y; in case (2) they are tilted. Linear and circular polarization are the two limiting shapes of an ellipse: one with zero width, the other with equal axes.

Theorem — General polarization state

Every polarization vector can be written

$$Z = e^{i\phi}\begin{pmatrix} A\cos\theta - iB\sin\theta \\ A\sin\theta + iB\cos\theta \end{pmatrix}, \qquad A \geq |B|,$$

which describes an ellipse with semi-axes A and $|B|$ whose major axis makes the angle θ with the x axis. $B = 0$ is linear polarization, $|B| = A$ circular, and the sign of B gives the sense of rotation.

Proof (sketch)

For $\theta = 0$ the vector is $(A,\ iB)$, i.e. $E_x = A\cos(kz - \omega t)$, $E_y = -B\sin(kz - \omega t)$, an ellipse along the axes. Multiplying by the rotation matrix through angle θ gives the stated form, and the factor $e^{i\phi}$ is the freedom to choose the origin of time. This uses four real parameters, the same number as two complex amplitudes, so nothing is left out. ∎

17.5 Unpolarized light

Definition — Unpolarized light

Unpolarized light is produced by a very large number of independent, uncorrelated emitters, each radiating a short wave train at a slightly different time and frequency. Its polarization changes randomly and far too quickly to follow, and on average no direction is preferred.

Watch Out

Unpolarized is not a fixed sum of all polarizations at once. Adding equal fields in every direction would give zero.

Light from the Sun or from a hot filament is unpolarized. It cannot be represented by a single vector Z; it must be treated as a rapidly changing mixture.

17.6 Polarizers

A simple polarizer for microwaves is a grid of parallel metal wires.

(1) A field *along* the wires drives the electrons up and down the wires. As with the metal plate of **§12**, the wave is reflected (and partly absorbed) instead of passing through.

(2) A field *across* the wires cannot move the electrons any appreciable distance, so the wave passes almost unaffected.

Definition — Polarizer and easy axis

A *polarizer* transmits the component of $\vec{E}$ along one direction, its *easy axis*, and removes the perpendicular component. For the wire grid the easy axis is perpendicular to the wires. Acting on Z, a polarizer with easy axis at angle θ to the x axis is the matrix

$$P_\theta = \begin{pmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{pmatrix}.$$

Special cases:

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{\pi/2} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad P_{\pi/4} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Why P_θ has this form

Let $\hat{e} = (\cos\theta, \sin\theta)$ be the unit vector along the easy axis. The transmitted field is the projection of Z on $\hat{e}$, namely $\hat{e}\,(\hat{e}\cdot Z)$. Writing this out, the matrix has entries $(P_\theta)_{ij} = e_i e_j$. Note that $P_\theta^2 = P_\theta$: a second identical polarizer changes nothing. ∎

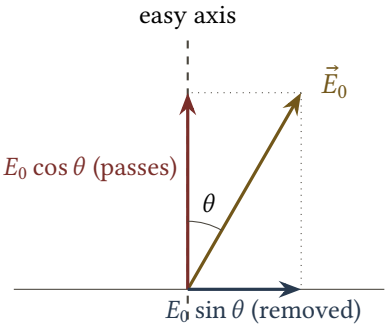


Figure 42 A polarizer keeps the component of $\vec{E}_0$ along its easy axis and removes the rest.

Theorem — Malus's law

Linearly polarized light of intensity I_0, whose polarization makes the angle θ with the easy axis of an ideal polarizer, emerges with

$$I = I_0\cos^2\theta,$$

polarized along the easy axis.

Proof

The component of the field along the easy axis is $|\vec{E}_f| = |\vec{E}_0| \cos\theta$; the perpendicular component is removed. The intensity is proportional to $\langle \vec{E}^{\,2} \rangle$, hence to $\cos^2\theta$. ■

Example — Three polarizers

Unpolarized light of intensity I_0 meets a polarizer with easy axis along x. Averaging $\cos^2\theta$ over all directions gives $\frac{1}{2}$, so $I_1 = I_0/2$, polarized along x. A second polarizer along y then gives $I_2 = I_1 \cos^2 90^\circ = 0$: crossed polarizers are dark.
Now insert a third polarizer at 45° *between* them. After it, $I = I_1 \cos^2 45^\circ = I_0/4$, polarized at 45°; after the last one, $I = (I_0/4)\cos^2 45^\circ = I_0/8$. Adding an absorbing element has let light through, because the middle polarizer rotates the polarization. In matrices, $P_{\pi/2}P_0 = 0$ but $P_{\pi/2}P_{\pi/4}P_0 \neq 0$.

Summary of Lecture 17

(1) Polarization is described by $Z = (\psi_1, \psi_2)$, the complex amplitudes of the x and y components.

(2) No phase difference: linear. Equal amplitudes and phase difference $\pm\pi/2$: circular. Otherwise: elliptical.

(3) Unpolarized light is a rapidly fluctuating mixture from uncorrelated emitters.

(4) A polarizer projects onto its easy axis: $I = I_0 \cos^2\theta$, and $I_0/2$ for unpolarized light.

Check yourself — Lecture 17

(1) What polarization does $Z = E_0\,(1,\, i)$ describe?

(2) Linearly polarized light meets a polarizer whose easy axis is at 30° to the polarization. What fraction of the intensity passes?

(3) Unpolarized light of intensity I_0 passes two polarizers whose easy axes differ by 60°. Find the final intensity.

Answers are in Appendix B.

18 Lecture 18: Wave Plates and Radiation

A polarizer changes polarization by throwing part of the wave away. A wave plate changes it without loss, by delaying one component relative to the other. After that, the lecture turns to a question so far avoided: where do electromagnetic waves come from? The answer is that they are produced by accelerated charges.

Reading
Georgi, Chapter 12, §12.4 to end.

18.1 Wave plates

Some transparent crystals have a refractive index that depends on the direction of the electric field: n_x for light polarized along x and n_y for light polarized along y. The two components then have different wavelengths inside the crystal, $\lambda_x = \lambda/n_x$ and $\lambda_y = \lambda/n_y$.

Definition — Wave plate, fast and slow axes

A *wave plate* is a slab of such a material of thickness l. On passing through it the two components acquire a relative phase

$$\Delta\phi = \frac{2\pi l}{\lambda_x} - \frac{2\pi l}{\lambda_y} = \frac{n_x - n_y}{c}\,\omega\, l.$$

The axis with the smaller index (smaller accumulated phase, larger speed) is the *fast axis*; the other is the *slow axis*. A *quarter-wave plate* has $|\Delta\phi| = \pi/2$, and a *half-wave plate* has $|\Delta\phi| = \pi$.

Theorem — Matrix of a quarter-wave plate

A quarter-wave plate with fast axis along x acts on the polarization vector as

$$Q_0 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix},$$

up to an overall phase. With the fast axis at angle θ to the x axis,

$$Q_\theta = \begin{pmatrix} \cos^2\theta + i\sin^2\theta & (1-i)\sin\theta\cos\theta \\ (1-i)\sin\theta\cos\theta & \sin^2\theta + i\cos^2\theta \end{pmatrix}.$$

Proof

The slow component is delayed by a quarter cycle relative to the fast one, which multiplies it by $e^{i\pi/2} = i$. For a tilted plate, the fast and slow directions are $\hat{e}_f = (\cos\theta, \sin\theta)$ and $\hat{e}_s = (-\sin\theta, \cos\theta)$, and the plate leaves the projection on $\hat{e}_f$ alone while multiplying the projection on $\hat{e}_s$ by i:

$$Q_\theta = P_\theta + i\,P_{\theta+\pi/2},$$

with P_θ the projection matrix of **§17**. Writing out the entries gives the result. ∎

Example — Linear to circular

Light linearly polarized at 45°, $Z = E_0(1, 1)$, passes through Q_0:

$$Q_0 Z = E_0 \begin{pmatrix} 1 \\ i \end{pmatrix},$$

which is circularly polarized. Light polarized along x or along y is unchanged, because it has only one component. Two quarter-wave plates in succession give $Q_0^2 = \mathrm{diag}(1, -1)$, a half-wave plate, which turns polarization at +45° into polarization at −45°. No light is absorbed in any of these steps.

18.2 Why radiation must fall off as $1/r$

In vacuum an electromagnetic wave neither gains nor loses energy. The energy flux is the Poynting vector $\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$. Surround a source by two spheres of radii $r_1 < r_2$. The same power must cross both:

$$\langle S \rangle_1 \cdot 4\pi r_1^2 = \langle S \rangle_2 \cdot 4\pi r_2^2.$$

Theorem — Radiation fields

For energy to be carried to infinity,

$$\langle S \rangle \propto \frac{1}{r^2}, \qquad |\vec{E}| \propto \frac{1}{r}, \quad |\vec{B}| \propto \frac{1}{r}.$$

This is a demanding condition. The familiar fields of charges fall off faster.

Source	$\lvert\vec{E}\rvert$	$\lvert\vec{B}\rvert$	Radiation?
Charge at rest	$\propto 1/r^2$	0	No: $\vec{S} = 0$.
Charge at constant velocity $\vec{u}$	$\propto 1/r^2$	$\propto 1/r^2$	No: $S \propto 1/r^4$.
Accelerated charge	$\propto 1/r$	$\propto 1/r$	Yes: $S \propto 1/r^2$.

For a charge moving at constant velocity $\vec{u}$, with $\beta = u/c$,

$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \frac{1 - \beta^2}{(1 - \beta^2 \sin^2\theta)^{3/2}} \hat{r}, \qquad \vec{B} = \frac{\vec{u} \times \vec{E}}{c^2}.$$

Shortcut

Go to the frame in which the uniformly moving charge is at rest. There is no radiation there, so there is none in any frame.

Both fall as $1/r^2$, so $S \propto 1/r^4$ and the power through a large sphere tends to zero. To radiate, a charge must be accelerated.

18.3 The kink argument

A charge q sits at rest, is given a small acceleration a for a short time Δt, and then moves with the small constant velocity $u = a\,\Delta t \ll c$. Look at the field a time t later (Figure 43).

(1) News travels at the speed of light. Outside a sphere of radius $c\,t$ the field is still that of a charge at rest at the original position.

(2) Inside a sphere of radius $c(t - \Delta t)$ the field is that of a charge moving uniformly, pointing radially away from the charge's present position.

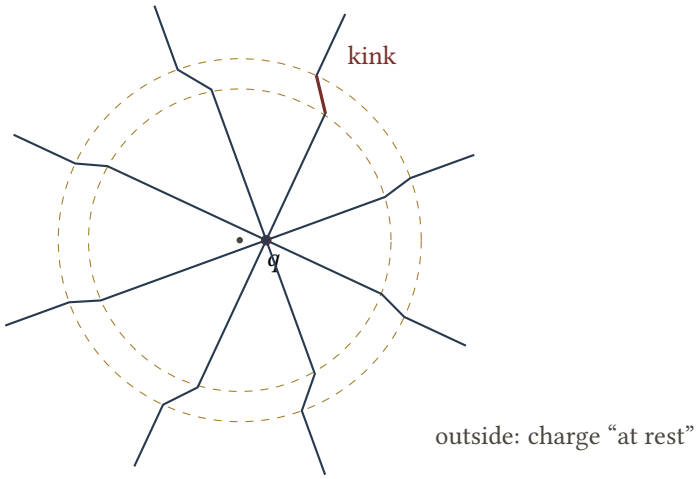


Figure 43 Field lines of a charge that was briefly accelerated to the right. The transverse kink in the shell between the two dashed spheres is the radiation field, moving outwards at speed c.

(3) In the thin shell between them there is no charge, so the field lines must be continuous. They have to join the inner lines to the displaced outer lines, and so they acquire a *kink*: a component perpendicular to $\hat{r}$.

Theorem — Radiation field of an accelerated charge

At a distance r large compared with the size of the charge's motion,

$$\vec{E}_{\text{rad}}(\vec{r},t) = -\frac{q\,\vec{a}_\perp(t-r/c)}{4\pi\epsilon_0 c^2\, r},$$

$$\vec{B}_{\text{rad}} = \frac{1}{c}\,\hat{r}\times\vec{E}_{\text{rad}}, \qquad \vec{S}_{\text{rad}} = \frac{1}{\mu_0}\vec{E}_{\text{rad}}\times\vec{B}_{\text{rad}},$$

where $\vec{a}_\perp = \vec{a} - (\vec{a}\cdot\hat{r})\,\hat{r}$ is the part of the acceleration perpendicular to the line of sight, evaluated at the *retarded time* $t' = t - r/c$.

Proof (geometrical sketch)

In time t the charge has moved a distance $u\,t$, so the inner and outer field lines are offset sideways by $u_\perp t$, with $u_\perp = a_\perp \Delta t$. The shell has radial thickness $c\,\Delta t$. A field line crossing the shell therefore has

$$\frac{E_\perp}{E_\parallel} = \frac{u_\perp t}{c\,\Delta t} = \frac{a_\perp t}{c} = \frac{a_\perp r}{c^2},$$

using $r = ct$, and it points opposite to $\vec{a}_\perp$. The radial component is fixed by Gauss's law to be the Coulomb field, $E_\parallel = q/4\pi\epsilon_0 r^2$. Multiplying, $E_\perp = q\,a_\perp/4\pi\epsilon_0 c^2 r$. The full derivation is in Georgi, pp. 355–360. ∎

One power of r has been cancelled: the transverse field falls only as $1/r$, so $S_{\text{rad}} \propto 1/r^2$ and energy really is carried off to arbitrarily large distances.

Recipe — Computing radiation

(1) Find the acceleration $\vec{a}(t)$ of the charge.

(2) Choose the observer's position $\vec{r}$, and form $\vec{a}_\perp = \vec{a} - (\vec{a}\cdot\hat{r})\hat{r}$.

(3) $\vec{E}_{\text{rad}} = -q\,\vec{a}_\perp(t - r/c)/4\pi\epsilon_0 c^2 r$.

(4) $\vec{B}_{\text{rad}} = \frac{1}{c}\hat{r}\times\vec{E}_{\text{rad}}$.

(5) $\vec{S}_{\text{rad}} = \frac{1}{\mu_0}\vec{E}_{\text{rad}}\times\vec{B}_{\text{rad}}$.

(6) Integrate $\vec{S}_{\text{rad}}$ over a sphere for the total power.

Theorem — Total radiated power

Integrating $\vec{S}_{\text{rad}}$ over a sphere of radius r gives

$$P(t) = \frac{q^2\,a^2(t - r/c)}{6\pi\epsilon_0 c^3}.$$

Proof

If $\vec{a}$ makes the angle θ with $\hat{r}$ then $a_\perp = a\sin\theta$ and $S = E_{\text{rad}}^2/\mu_0 c = q^2a^2\sin^2\theta/(16\pi^2\epsilon_0 c^3 r^2)$. With $\mathrm{d}A = r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi$ and $\int_0^\pi \sin^3\theta\,\mathrm{d}\theta = \frac{4}{3}$,

$$P = \frac{q^2a^2}{16\pi^2\epsilon_0 c^3}\cdot 2\pi\cdot\frac{4}{3} = \frac{q^2a^2}{6\pi\epsilon_0 c^3}.$$

■

This is the *Larmor formula.* The power grows as the square of the acceleration and does not depend on r.

18.4 Worked examples

Example — A harmonically oscillating charge

A charge moves as $\vec{x}(t) = \hat{z}\,d\cos\omega t$, observed at distance $R \gg d$. Its acceleration is $\vec{a} = -\hat{z}\,d\,\omega^2\cos\omega t$.

(1) Observer on the z axis. $\vec{a}\parallel\hat{r}$, so $\vec{a}_\perp = 0$: no radiation along the line of oscillation.

(2) Observer at $R\,\hat{y}$. $\vec{a}\perp\hat{r}$, so $\vec{a}_\perp = \vec{a}$, and

$$\vec{E}_{\text{rad}} = \frac{q\,d\,\omega^2}{4\pi\epsilon_0 c^2 R}\cos\big(\omega(t - R/c)\big)\,\hat{z},$$

$$\vec{B}_{\text{rad}} = \frac{1}{c}\,\hat{y}\times\vec{E}_{\text{rad}} = \frac{q\,d\,\omega^2}{4\pi\epsilon_0 c^3 R}\cos\big(\omega(t - R/c)\big)\,\hat{x}.$$

This is a harmonic wave, linearly polarized along z, with amplitude falling as $1/R$.

(3) Observer at 30° from the z axis, with $\hat{r} = \frac{1}{2}\hat{y} + \frac{\sqrt{3}}{2}\hat{z}$. Then $\vec{a}\cdot\hat{r} =$

$-\frac{\sqrt{3}}{2}\, d\,\omega^2 \cos\omega t$ and

$$\vec{a}_\perp = -\omega^2 d \cos\omega t \left(\tfrac{1}{4}\hat{z} - \tfrac{\sqrt{3}}{4}\hat{y}\right),$$

$$\vec{E}_{\text{rad}} = \frac{q\,\omega^2 d}{8\pi\epsilon_0 c^2 R} \cos\big(\omega(t - R/c)\big)\left(\tfrac{1}{2}\hat{z} - \tfrac{\sqrt{3}}{2}\hat{y}\right).$$

The bracket is a unit vector perpendicular to $\hat{r}$, and the amplitude is half that of case (2), in agreement with $a_\perp = a \sin 30°$.

This is the pattern of a dipole antenna: it radiates most strongly sideways and not at all along its own axis, with intensity proportional to $\sin^2\theta$.

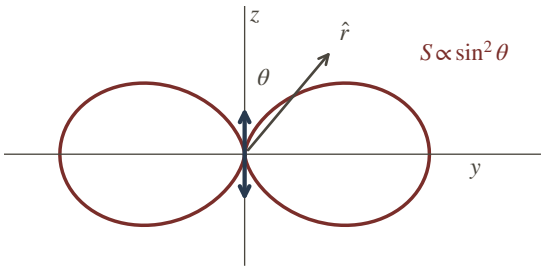


Figure 44 Angular pattern of the power radiated by a charge oscillating along z (double arrow). The distance from the origin to the curve is proportional to the intensity in that direction.

Example — Polarization from an elliptical orbit

A charge q moves on the ellipse

$$x(t) = \sqrt{2}\,A\cos\omega t, \qquad y(t) = A\sin\omega t,$$

so $\vec{a} = -\sqrt{2}A\omega^2\cos\omega t\,\hat{x} - A\omega^2\sin\omega t\,\hat{y}$. What polarization do distant observers on the three axes see?

On the x axis: only the y part of $\vec{a}$ is transverse,

$$\vec{E}_{\text{rad}} = \frac{q\,\omega^2 A}{4\pi\epsilon_0 c^2 R} \sin\big(\omega(t - R/c)\big)\,\hat{y},$$

linearly polarized along y.

On the y axis: only the x part is transverse; the wave is linearly polarized along x, with amplitude larger by $\sqrt{2}$.

On the z axis: all of $\vec{a}$ is transverse. The two components have unequal amplitudes and a phase difference of $\pi/2$, so the wave is elliptically polarized, with axis ratio $\sqrt{2}$.

The observer sees the charge's motion projected onto the plane perpendicular to the line of sight, and the polarization of the radiation reproduces that projected motion.

Summary of Lecture 18

(1) A wave plate adds a relative phase $(n_x - n_y)\omega l/c$; a quarter-wave plate turns 45° linear light into circular.

(2) Radiation requires fields falling as $1/r$; static and uniformly moving charges do not radiate.

(3) $\vec{E}_{\text{rad}} = -q\,\vec{a}_\perp(t - r/c)/4\pi\epsilon_0 c^2 r$, with $\vec{B}_{\text{rad}} = \frac{1}{c}\hat{r} \times \vec{E}_{\text{rad}}$.

(4) Total power $P = q^2a^2/6\pi\epsilon_0 c^3$; none is emitted along the direction of acceleration.

Check yourself — Lecture 18

(1) A crystal has $n_x - n_y = 0.01$. How thick must a quarter-wave plate be for $\lambda = 600$ nm?

(2) In which direction does an accelerated charge radiate nothing?

(3) A charge oscillates with fixed amplitude. By what factor does the radiated power change if the frequency is doubled?

Answers are in Appendix B.

19 Lecture 19: Waves in Media and Brewster's Angle

Light slows down in glass and water, and part of it is reflected at every surface. This lecture derives both facts from Maxwell's equations in matter. It then computes how much light is reflected at a boundary, and finds an angle of incidence at which, for one polarization, nothing is reflected at all.

Reading
Georgi, Chapter 12: Polarization (§12.4 to end).

19.1 Mathematical and physical preliminaries

Two integral theorems from **§12** convert Maxwell's equations into boundary conditions:

$$\int_V (\vec{\nabla}\cdot\vec{A})\,d\tau = \oint_S \vec{A}\cdot d\vec{a}, \qquad \int_S (\vec{\nabla}\times\vec{A})\cdot d\vec{a} = \oint_L \vec{A}\cdot d\vec{l}.$$

Matter responds to fields in two ways. Charges bound in atoms shift slightly, giving each small region an electric dipole moment; bound currents circulate, giving it a magnetic dipole moment.

Definition — Polarization and magnetization

The *polarization* $\vec{P}$ is the electric dipole moment per unit volume of a material, and the *magnetization* $\vec{M}$ is its magnetic dipole moment per unit volume. They produce a bound charge density and a bound current density,

$$\rho_b = -\vec{\nabla}\cdot\vec{P}, \qquad \vec{J}_b = \vec{\nabla}\times\vec{M},$$

and a changing polarization is itself a current, $\vec{J}_p = \partial\vec{P}/\partial t$.

Material	Behaviour of its charges
Perfect conductor	An unlimited supply of charges that move freely at no cost; the field inside is zero.
Dielectric	Every charge is attached to a particular atom or molecule and can shift only slightly within it.

Watch Out
*"Polarization" here means the dipole density $\vec{P}$ of matter. It is unrelated to the polarization of a wave in **§17**.*

19.2 Maxwell's equations in matter

Split the charge and current into free and bound parts, $\rho = \rho_f + \rho_b$ and $\vec{J} = \vec{J}_f + \vec{J}_b + \vec{J}_p$.

Definition — The fields $\vec{D}$ and $\vec{H}$

$$\vec{D} \equiv \epsilon_0\vec{E} + \vec{P}, \qquad \vec{H} \equiv \frac{\vec{B}}{\mu_0} - \vec{M}.$$

$\vec{D}$ is the *electric displacement* and $\vec{H}$ the *magnetic field strength*. They are constructed so that only free charges and free currents act as their sources.

Theorem — Maxwell's equations in matter

$$\vec{\nabla}\cdot\vec{D} = \rho_f, \qquad \vec{\nabla}\cdot\vec{B} = 0,$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}, \qquad \vec{\nabla}\times\vec{H} = \vec{J}_f + \frac{\partial\vec{D}}{\partial t}.$$

Proof

Gauss's law reads $\epsilon_0 \vec{\nabla}\cdot\vec{E} = \rho_f + \rho_b = \rho_f - \vec{\nabla}\cdot\vec{P}$, so $\vec{\nabla}\cdot(\epsilon_0\vec{E}+\vec{P}) = \rho_f$. The Ampère–Maxwell law reads

$$\frac{1}{\mu_0}\vec{\nabla}\times\vec{B} = \vec{J}_f + \vec{\nabla}\times\vec{M} + \frac{\partial\vec{P}}{\partial t} + \epsilon_0\frac{\partial\vec{E}}{\partial t}.$$

Moving the $\vec{M}$ term to the left and combining the last two terms gives $\vec{\nabla}\times\vec{H} = \vec{J}_f + \partial\vec{D}/\partial t$. The other two equations contain no sources and are unchanged. ■

Definition — Linear medium

For weak fields in a homogeneous, isotropic material, $\vec{P}\propto\vec{E}$ and $\vec{M}\propto\vec{B}$, so that

$$\vec{D} = \epsilon\,\vec{E}, \qquad \vec{H} = \frac{\vec{B}}{\mu}.$$

$\epsilon = k_e\epsilon_0$ is the *permittivity* ($k_e \geq 1$ is the dielectric constant) and μ the *permeability*. For most transparent materials $\mu \approx \mu_0$.

Theorem — Speed of light in a medium

In a linear medium with no free charges or currents, Maxwell's equations have the vacuum form with $\epsilon_0 \to \epsilon$ and $\mu_0 \to \mu$. Waves therefore travel at

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}, \qquad n = \sqrt{\frac{\mu\,\epsilon}{\mu_0\epsilon_0}} \approx \sqrt{k_e}\,.$$

Proof

With $\rho_f = 0$, $\vec{J}_f = 0$, $\vec{D} = \epsilon\vec{E}$ and $\vec{H} = \vec{B}/\mu$ the equations become

$$\vec{\nabla}\cdot\vec{E} = 0, \quad \vec{\nabla}\cdot\vec{B} = 0, \quad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}, \quad \vec{\nabla}\times\vec{B} = \mu\epsilon\frac{\partial\vec{E}}{\partial t}.$$

The derivation of **§12** goes through unchanged and gives a wave equation with $1/v^2 = \mu\epsilon$. ■

Since $\epsilon > \epsilon_0$, we have $n > 1$: the phase velocity of light in matter is *smaller* than c. A plane wave $\vec{E} = \vec{E}_0\cos(kz-\omega t)$ in the medium has $\vec{E}_0 \perp \vec{k}$ as before, $\omega/k = c/n$, magnetic field $\vec{B} = \frac{1}{v}\hat{k}\times\vec{E}$, and Poynting vector $\vec{S} = \frac{1}{\mu}\vec{E}\times\vec{B} \approx \frac{1}{\mu_0}\vec{E}\times\vec{B}$.

The refractive index generally depends on the frequency. Matter is a dispersive medium for light.

Dispersion

n depends on frequency, $n = n(\omega)$. In glass it is larger for blue light than for red, which is why a prism spreads colours.

19.3 Reflection and transmission at an interface

A plane wave in medium 1 (index n_1, speed v_1) strikes the flat surface $z = 0$ of medium 2 (index n_2, speed v_2). There are three waves:

$$\vec{E}_I = \vec{E}_{0I}\cos(\vec{k}_I\cdot\vec{r}-\omega t), \qquad \vec{E}_R = \vec{E}_{0R}\cos(\vec{k}_R\cdot\vec{r}-\omega t),$$

$$\vec{E}_T = \vec{E}_{0T}\cos(\vec{k}_T\cdot\vec{r}-\omega t),$$

each with $\vec{B} = \frac{1}{v}\hat{k} \times \vec{E}$ for its own medium. The incident amplitude is given; $\vec{E}_{0R}$ and $\vec{E}_{0T}$ are unknown. The kinematics was settled in **§16**: on the boundary

(1) $\vec{k}_I \cdot \vec{r} = \vec{k}_R \cdot \vec{r} = \vec{k}_T \cdot \vec{r}$,

(2) $\theta_I = \theta_R$,

(3) $n_1 \sin\theta_I = n_2 \sin\theta_T$.

To find the amplitudes we need boundary conditions specific to electromagnetic fields.

Theorem — Boundary conditions on $\vec{E}$

At an interface with no free charge:

(1) the component of $\vec{D}$ *perpendicular* to the surface is continuous, $\epsilon_1 E_\perp^{(1)} = \epsilon_2 E_\perp^{(2)}$;

(2) the component of $\vec{E}$ *parallel* to the surface is continuous, $E_\parallel^{(1)} = E_\parallel^{(2)}$.

Proof

(1) Apply Gauss's theorem to $\vec{\nabla} \cdot \vec{D} = 0$ on a thin pillbox straddling the surface. The flux out of the two faces must cancel, so $D_\perp$ is the same on both sides. (2) Apply Stokes's theorem to $\vec{\nabla} \times \vec{E} = -\partial\vec{B}/\partial t$ on a thin rectangular loop straddling the surface. As the loop's width shrinks, the magnetic flux through it vanishes, so the line integrals of $\vec{E}$ along the two long sides cancel. ■

19.4 Polarization in the plane of incidence

Take the incident wave polarized *in* the plane of incidence, the x–z plane (Figure 45). In medium 1 the total field is $\vec{E}_{0I} + \vec{E}_{0R}$, and in medium 2 it is $\vec{E}_{0T}$.

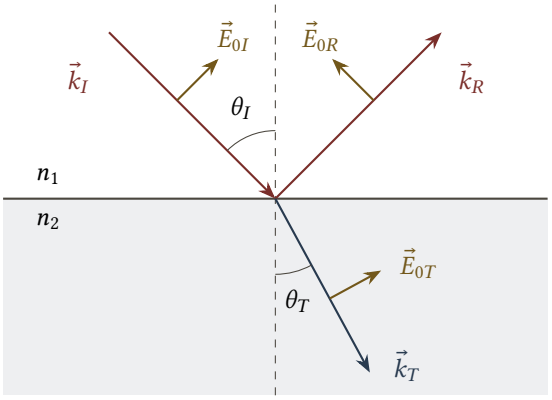


FIGURE 45 Reflection and refraction with the electric field in the plane of incidence. Each field is perpendicular to its own wave vector.

Theorem — Fresnel coefficients, in-plane polarization

Define

$$\alpha \equiv \frac{\cos\theta_T}{\cos\theta_I}, \qquad \beta \equiv \frac{\epsilon_2 n_1}{\epsilon_1 n_2} \approx \frac{n_2}{n_1} \;\; (\mu_1 \approx \mu_2 \approx \mu_0).$$

Then $E_{0R} = R\,E_{0I}$ and $E_{0T} = T\,E_{0I}$ with

$$R = \frac{\alpha - \beta}{\alpha + \beta}, \qquad T = \frac{2}{\alpha + \beta}.$$

Proof

Resolve each amplitude into components perpendicular (z) and parallel (x) to the surface. The perpendicular condition gives

$$\epsilon_1\left(-E_{0I}\sin\theta_I + E_{0R}\sin\theta_I\right) = -\epsilon_2 E_{0T}\sin\theta_T,$$

and the parallel condition gives

$$E_{0I}\cos\theta_I + E_{0R}\cos\theta_I = E_{0T}\cos\theta_T.$$

Using Snell's law, $\sin\theta_T/\sin\theta_I = n_1/n_2$, these become

$$E_{0I} - E_{0R} = \beta\,E_{0T}, \qquad E_{0I} + E_{0R} = \alpha\,E_{0T}.$$

Adding gives E_{0T}; subtracting gives E_{0R}. For $\mu \approx \mu_0$, $\epsilon \propto n^2$ and so $\beta = (n_2^2/n_1^2)(n_1/n_2) = n_2/n_1$. ∎

19.5 Three special cases

Example — Normal incidence

For $\theta_I = 0$, $\alpha = 1$ and $\beta = n_2/n_1$:

$$R = \frac{n_1 - n_2}{n_1 + n_2}, \qquad T = \frac{2n_1}{n_1 + n_2}.$$

These have the same form as the string results of **§10** with $v \propto 1/n$. From air into glass ($n_2 = 1.5$), $R = -0.2$: the reflected field is inverted, and the reflected intensity is $R^2 = 4\%$.

Example — Grazing incidence

As $\theta_I \to 90°$, $\cos\theta_I \to 0$ and $\alpha \to \infty$, so

$$R \to 1, \qquad T \to 0.$$

Any smooth surface becomes a good mirror when viewed at a glancing angle.

Handy form

Brewster's angle satisfies $\tan\theta_B = n_2/n_1$. *For air to glass it is about* $56°$.

Theorem — Brewster's angle

For light polarized in the plane of incidence there is an angle of incidence θ_B at which nothing is reflected. It is the angle at which the reflected and transmitted directions are perpendicular:

$$\theta_B + \theta_T = \frac{\pi}{2}, \qquad \text{equivalently} \qquad \tan\theta_B = \frac{n_2}{n_1}.$$

Proof

$R = 0$ requires $\alpha = \beta$:

$$\frac{\cos\theta_T}{\cos\theta_B} = \frac{n_2}{n_1} = \frac{\sin\theta_B}{\sin\theta_T},$$

the last step by Snell's law. Cross-multiplying, $\sin\theta_T\cos\theta_T = \sin\theta_B\cos\theta_B$, that is $\sin 2\theta_T = \sin 2\theta_B$. Either $\theta_T = \theta_B$, which would need $n_1 = n_2$ (no boundary at all), or $2\theta_B = \pi - 2\theta_T$. Then $\sin\theta_T = \cos\theta_B$, and Snell's law gives $n_1 \sin\theta_B = n_2 \cos\theta_B$. ■

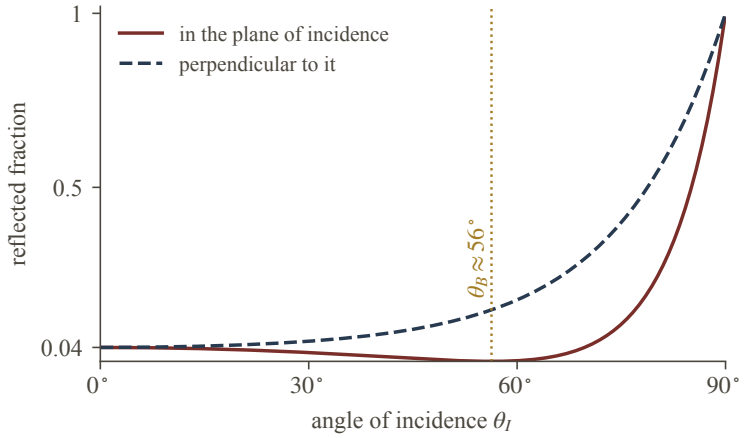


Figure 46 Fraction of the intensity reflected from air to glass ($n = 1.5$). Solid: R^2 for polarization in the plane of incidence, from the theorem above. Dashed: the other polarization, quoted from the standard Fresnel result for comparison (not derived in the lecture).

At Brewster's angle $T = 1/\beta = n_1/n_2$, and all of the incident energy enters the second medium.

Remark — Why the reflection vanishes

The reflected wave is radiated by the charges of medium 2, which oscillate along $\vec{E}_{0T}$. At Brewster's angle the reflected direction is perpendicular to the refracted ray, hence *parallel* to $\vec{E}_{0T}$. By **§18**, a charge does not radiate along its direction of acceleration, so no reflected wave is produced.

! Brewster's angle polarizes reflected light

Unpolarized light is an even mixture of both polarizations. The one with $\vec{E}$ perpendicular to the plane of incidence is reflected at every angle. At Brewster's angle only the component perpendicular to the plane of incidence is reflected, so the reflected light is completely linearly polarized, parallel to the surface. This is the origin of glare from roads and water, and the reason polarizing sunglasses with a vertical easy axis remove it.

Summary of Lecture 19

(1) In matter, $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ and $\vec{H} = \vec{B}/\mu_0 - \vec{M}$ have only free sources.

(2) Light travels at $v = c/n$ with $n = \sqrt{\mu\epsilon/\mu_0\epsilon_0}$.

(3) At a boundary $\epsilon E_\perp$ and $E_\parallel$ are continuous, giving $R = (\alpha - \beta)/(\alpha + \beta)$ and $T = 2/(\alpha + \beta)$.

(4) $R = 0$ at Brewster's angle, $\tan\theta_B = n_2/n_1$, for in-plane polarization.

Check yourself — Lecture 19

(1) A glass has dielectric constant $k_e = 2.25$ and $\mu \approx \mu_0$. Find n and the speed of light in it.

(2) Find Brewster's angle for light going from air into water ($n = 1.33$).

(3) What fraction of the intensity is reflected at normal incidence from air into water?

Answers are in Appendix B.

20 Lecture 20: Interference and the Soap Bubble

When two waves overlap, their fields add. Their intensities do not: the result can be brighter than the sum of the two, or completely dark. This lecture works out the rule for two superposed electromagnetic waves and uses it to explain a familiar sight, the colours of a soap bubble, and to estimate how thin the film must be.

Reading
Georgi, Chapter 13: Interference and Diffraction.

20.1 Intensity of two superposed waves

Consider two waves of the same frequency, travelling in the same direction with the same polarization but different amplitudes and phases:

$$\vec{E}_1 = A_1 \cos(\omega t - kz + \phi_1)\,\hat{x}, \qquad \vec{E}_2 = A_2 \cos(\omega t - kz + \phi_2)\,\hat{x},$$

and $\vec{E} = \vec{E}_1 + \vec{E}_2$.

Definition — Intensity

The *intensity* I is the power transported per unit area, the magnitude of the Poynting vector $\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$. In a medium of index n, where $\vec{B} = \frac{1}{v}\hat{k}\times\vec{E}$ and $v = c/n$,

$$I = |\vec{S}| = \frac{n}{\mu_0 c}\,|\vec{E}|^2 = c\,n\,\epsilon_0\,|\vec{E}|^2.$$

What a detector or the eye registers is the time average $\langle I\rangle = \frac{1}{T}\int_0^T I\,dt$.

Theorem — Two-wave interference

$$\langle I\rangle = c\,n\,\epsilon_0\left[\frac{A_1^2}{2} + \frac{A_2^2}{2} + A_1 A_2\cos\delta\right], \qquad \delta \equiv \phi_1 - \phi_2.$$

Proof

Write $u = \omega t - kz$. Then

$$|\vec{E}|^2 = A_1^2\cos^2(u+\phi_1) + A_2^2\cos^2(u+\phi_2) + 2A_1A_2\cos(u+\phi_1)\cos(u+\phi_2).$$

Each $\cos^2$ averages to $\frac{1}{2}$, since $\cos^2 x = \frac{1}{2}(1+\cos 2x)$. For the cross term,

$$\cos(u+\phi_1)\cos(u+\phi_2) = \tfrac{1}{2}\big[\cos(2u+\phi_1+\phi_2) + \cos(\phi_1-\phi_2)\big],$$

and the first piece averages to zero while the second is constant. ∎

The first two terms are the intensities of the separate waves. The third is the *interference term*, and it can have either sign.

Definition — Constructive and destructive interference

(1) *Constructive*: $\delta = 2N\pi$. The amplitudes add, $\langle I\rangle \propto \frac{1}{2}(A_1+A_2)^2$.

(2) *Destructive*: $\delta = (2N+1)\pi$. The amplitudes subtract, $\langle I\rangle \propto \frac{1}{2}(A_1-A_2)^2$.

Here N is any integer. If $A_1 = A_2$, destructive interference gives complete darkness.

Tip
Energy is not destroyed in the dark regions. It is redistributed to the bright ones.

The cleanest way to see this is in the complex plane. Write

$$\vec{E}_1 = \mathrm{Re}\left[A_1 e^{i\phi_1} e^{i(\omega t - kz)}\right]\hat{x}, \qquad \vec{E}_2 = \mathrm{Re}\left[A_2 e^{i\phi_2} e^{i(\omega t - kz)}\right]\hat{x}.$$

The two complex amplitudes $A_1 e^{i\phi_1}$ and $A_2 e^{i\phi_2}$ are arrows in the complex plane, and the total amplitude is their vector sum (Figure 47). Its squared length is $A_1^2 + A_2^2 + 2A_1A_2\cos\delta$ by the law of cosines. Arrows pointing the same way add; arrows pointing opposite ways cancel.

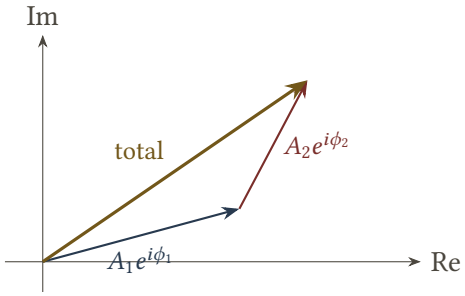


Figure 47 Adding two waves of the same frequency is adding two arrows in the complex plane.

20.2 *Reflection at a dielectric surface, revisited*

At normal incidence, **§19** gave

$$R = \frac{n_1 - n_2}{n_1 + n_2}, \qquad T = \frac{2n_1}{n_1 + n_2}.$$

Theorem — Phase change on reflection

(1) If $R > 0$ (from higher to lower index, $n_1 > n_2$) the reflected amplitude is not inverted.

(2) If $R < 0$ (from lower to higher index, $n_1 < n_2$) the reflected amplitude is inverted, which is a phase shift of π.

The transmitted wave is never inverted, since $T > 0$.

Example — Air to soap solution

With $n_1 = 1$ and $n_2 = 1.5$,

$$R = \frac{1 - 1.5}{1 + 1.5} = -0.2, \qquad T = \frac{2}{2.5} = 0.8.$$

Using $I = \frac{1}{2} c\, n\, \epsilon_0 E^2$ for the time-averaged intensity,

$$I_R = R^2 I_0 = 0.04\, I_0, \qquad I_T = \frac{n_2}{n_1} T^2 I_0 = 1.5 \times 0.64\, I_0 = 0.96\, I_0.$$

The two add up to I_0, as energy conservation demands. Only 4% of the light is reflected at each surface.

20.3 The soap film

A soap bubble is a thin layer of soapy water, of thickness d and index $n_2 \approx 1.5$, with air on both sides. Take light of vacuum wavelength λ arriving at normal incidence (Figure 48). Two reflected waves reach the eye:

(1) wave 1, reflected at the front surface (air to soap);

(2) wave 2, transmitted into the film, reflected at the back surface (soap to air) and transmitted out again.

Both have amplitude of roughly $0.2E_0$, so they can cancel almost completely.

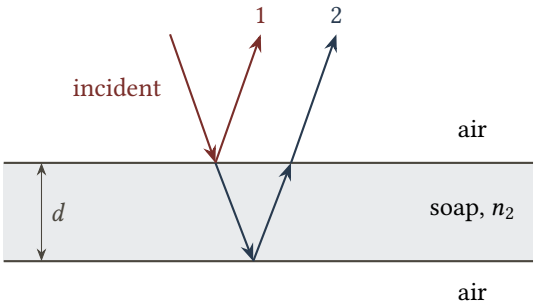


Figure 48 The two reflections from a thin film, drawn at a slight angle for clarity. Wave 1 is inverted on reflection; wave 2 is not, but travels the extra distance $2d$ inside the film.

Theorem — Phase difference for a thin film in air

$$\delta = 2\pi \frac{2d}{\lambda/n_2} + \pi.$$

Proof

Wave 2 travels an extra distance $2d$ inside the film, where the wavelength is λ/n_2; each wavelength of path contributes 2π. Wave 1 is reflected from a medium of higher index and is inverted, a phase shift of π. Wave 2 is reflected from a medium of lower index and is not inverted, and its two transmissions cause no shift. ■

Theorem — Bright and dark conditions

With $N = 1, 2, 3, \ldots$:

(1) **Constructive** ($\delta = 2N\pi$): $d = \dfrac{(2N-1)\lambda}{4n_2}$.

(2) **Destructive** ($\delta = (2N+1)\pi$): $d = \dfrac{N\lambda}{2n_2}$, and also $d \to 0$.

(3) For a fixed thickness, the strongly reflected wavelengths are $\lambda_{\max} = \dfrac{4\,d\,n_2}{2N-1}$.

Watch Out

For a film with different media on its two sides, count the inversions again. Two inversions or none remove the extra π.

! A very thin film is dark, not bright

As $d \to 0$ the path difference vanishes, and one might expect the two reflections to reinforce. But the inversion of wave 1 remains, so $\delta \to \pi$ and the reflections cancel. Just before a soap film breaks, its thinnest part at the top turns black.

20.4 Why bubbles are coloured

Example — A thin film, $d \approx 100$ nm

$$\lambda_{\max} = \frac{4 \times 100\ \text{nm} \times 1.5}{2N - 1} = \frac{600\ \text{nm}}{2N - 1} = 600,\ 200,\ 120\ \text{nm},\ \ldots$$

Only one of these, 600 nm (orange), lies in the visible range of roughly 400–700 nm. The film reflects that colour strongly and others weakly, so it looks coloured. A slightly different thickness selects a different colour, which is why the colours form bands and swirl as the film drains.

Example — A thick film, $d \approx 100\ \mu$m

Now $4dn_2 = 600\ \mu$m and

$$\lambda_{\max} = \frac{600\ \mu\text{m}}{2N - 1}.$$

$N = 1$ gives 600 μm, far in the infrared. The visible range is reached only near $N \approx 500$:

$$N = 500:\ 600.6\ \text{nm}, \qquad N = 501:\ 599.4\ \text{nm}, \qquad N = 502:\ 598.2\ \text{nm}, \ \ldots$$

Hundreds of closely spaced wavelengths across the whole visible spectrum are reflected strongly. The eye cannot separate them, and the film looks white.

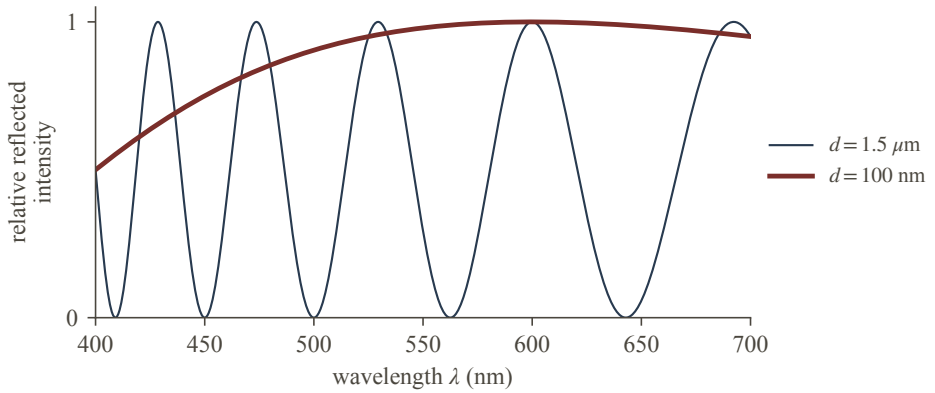


Figure 49 Relative reflected intensity across the visible spectrum for soap films ($n_2 = 1.5$) of two thicknesses. The thin film favours one broad band of colour. The thicker one has several narrow peaks, and a much thicker film has hundreds, which the eye averages to white.

Remark — An estimate from colour alone

Seeing colours in a soap film tells us, with no instrument, that its thickness is of the order of the wavelength of light divided by a few: about 100 nm, a few hundred molecules thick.

Summary of Lecture 20

(1) $\langle I \rangle \propto \frac{1}{2}A_1^2 + \frac{1}{2}A_2^2 + A_1 A_2 \cos\delta$.

(2) Reflection from a higher-index medium inverts the wave (phase π); from a lower-index medium it does not.

(3) Thin film in air: $\delta = 4\pi n_2 d/\lambda + \pi$; bright when $d = (2N-1)\lambda/4n_2$.

(4) $d \approx 100$ nm gives colours; thick films look white; $d \longrightarrow 0$ looks black.

Check yourself — Lecture 20

(1) A soap film ($n_2 = 1.33$) is 120 nm thick. Which visible wavelength does it reflect most strongly at normal incidence?

(2) Why does a soap film look black just before it bursts?

(3) Two beams, each of intensity I, are superposed. What is the intensity if they are in phase? Exactly out of phase?

Answers are in Appendix B.

21 Lecture 21: Phased Radar and Single-Electron Interference

Reading

Georgi, Chapter 13: Interference and Diffraction (§13.4.1, the single slit, is the key section).

Lecture 20 treated two waves travelling along the same line. Here the sources are separated in space, so the phase difference depends on the direction of observation and an interference *pattern* appears. Going from two sources to many explains how a radar beam is made narrow and steered without moving parts. The same pattern, built up one electron at a time, is the central experiment of quantum mechanics.

21.1 Huygens' principle

Definition — Huygens' principle

Every point on a wavefront acts as a source of spherical waves. The wave at a later time is the superposition of all these secondary waves.

The principle holds in an odd number of space dimensions and can be derived from Maxwell's equations. Its practical use is this: when a plane wave falls on a screen with narrow slits, each slit behaves as a point source, and all the slits oscillate in phase.

21.2 Two slits

Two slits A and B a distance d apart are illuminated by a plane wave. The pattern is observed in a direction θ from the forward direction, at a distance $L \gg d$, so that the two paths to the observation point are nearly parallel (Figure 50).

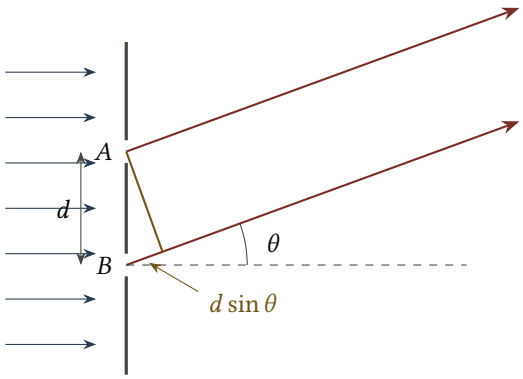


Figure 50 Two slits illuminated from the left. In the direction θ the wave from B travels $d \sin\theta$ farther than the wave from A.

Theorem — Two-slit intensity pattern

The path difference and the resulting phase difference are

$$r_B - r_A = d\sin\theta, \qquad \delta = \frac{2\pi}{\lambda}\, d\sin\theta = k\, d\sin\theta,$$

and the time-averaged intensity is

$$\langle I\rangle = I_0 \cos^2\frac{\delta}{2},$$

where I_0 is the intensity in the forward direction, $\delta = 0$.

Proof

In complex notation, with equal amplitudes E_0 from the two slits,

$$E = E_0\, e^{i(\omega t - kr_A)} + E_0\, e^{i(\omega t - kr_B)} = E_0\, e^{i(\omega t - kr_A)}\left[1 + e^{-i\delta}\right]$$
$$= E_0\, e^{i(\omega t - kr_A)}\, e^{-i\delta/2}\left[e^{i\delta/2} + e^{-i\delta/2}\right] = 2E_0 \cos\frac{\delta}{2}\, e^{i(\omega t - kr_A - \delta/2)}.$$

The intensity is proportional to $|E|^2 = EE^* = 4E_0^2 \cos^2(\delta/2)$. ∎

Bright fringes lie at $\delta = 2N\pi$, that is $d \sin\theta = N\lambda$, and dark ones halfway between. A given δ corresponds to the direction

$$\sin\theta = \frac{\lambda}{2\pi d}\,\delta.$$

21.3 From two slits to N

Add more equally spaced slits, each a distance d from the next. The phase difference between neighbours is the same, $\delta = k\,d\sin\theta$. The waves are N arrows of equal length in the complex plane, each rotated by δ from the one before. They cancel when the arrows close into a regular polygon.

Number of slits	Smallest δ for complete cancellation
2	π
3	$2\pi/3$
4	$2\pi/4 = \pi/2$
5	$2\pi/5$
N	$2\pi/N$



$\delta = 0$: sum $= 3E_0$ $\quad$ $N = 3,\ \delta = 2\pi/3$ $\quad$ $N = 4,\ \delta = \pi/2$

Figure 51 Adding the waves from N slits as arrows in the complex plane, each turned by δ from the previous one. When the arrows close into a polygon the total is zero.

The first dark direction moves closer to the central maximum as slits are added: the bright peak becomes *narrower*.

Theorem — N-source interference

For N equally spaced sources of equal amplitude, oscillating in phase,

$$\langle I\rangle = I_0\left[\frac{\sin(N\delta/2)}{\sin(\delta/2)}\right]^2, \qquad \delta = k\,d\sin\theta,$$

where I_0 is the intensity from a single source.

Proof

Let R be the distance from the first source. Summing the geometric series,

$$E = E_0\, e^{i(\omega t - kR)} \sum_{m=0}^{N-1} \left(e^{-i\delta}\right)^m = E_0\, e^{i(\omega t - kR)}\, \frac{1 - e^{-iN\delta}}{1 - e^{-i\delta}}.$$

Factor half the phase out of numerator and denominator:

$$\frac{1 - e^{-iN\delta}}{1 - e^{-i\delta}} = \frac{e^{-iN\delta/2}\left(e^{iN\delta/2} - e^{-iN\delta/2}\right)}{e^{-i\delta/2}\left(e^{i\delta/2} - e^{-i\delta/2}\right)} = e^{-i(N-1)\delta/2}\, \frac{\sin(N\delta/2)}{\sin(\delta/2)}.$$

The phase factors have modulus 1 and drop out of $|E|^2$. ∎

Theorem — Features of the N-source pattern

(1) **Principal maxima** at $\delta = 0, \pm 2\pi, \ldots$, where all N arrows line up: $\langle I \rangle = N^2 I_0$.

(2) **Zeros** at $\delta = 2\pi m/N$ for $m = 1, \ldots, N-1$.

(3) **Secondary maxima:** $N - 2$ small ones between neighbouring principal maxima.

(4) **Width** of a principal maximum, from zero to zero: $4\pi/N$ in δ. It shrinks as $1/N$.

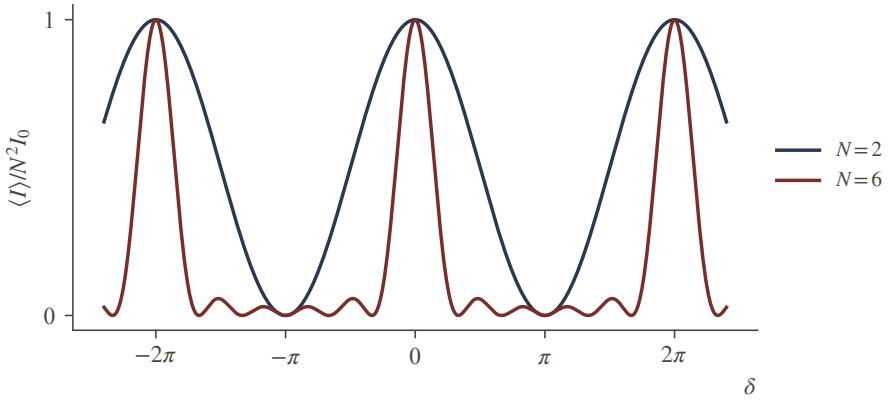


Figure 52 Interference patterns for 2 and 6 sources, each normalised to its own peak. More sources give narrower principal maxima, with $N - 2$ weak secondary maxima between them.

21.4 Resolution and beam steering

The half-width of the central beam is the direction of its first zero, $\delta = 2\pi/N$:

$$\frac{2\pi}{\lambda}\, d \sin\theta = \frac{2\pi}{N} \qquad \Longrightarrow \qquad \sin\theta = \frac{\lambda}{Nd}.$$

Remark — How to make a narrow beam

The beam is narrow when λ/Nd is small: use a short wavelength λ, a large spacing d, or many sources N. What matters is the total width Nd of the array measured in wavelengths.

A narrow beam is useful only if it can be pointed. Instead of rotating the antenna, give each source an extra phase lag relative to its neighbour.

Theorem — Phased array

If neighbouring sources are driven with an additional phase difference $\Delta\varphi$, then

$$\delta = \frac{2\pi}{\lambda}\, d \sin\theta - \Delta\varphi,$$

and the principal maximum, $\delta = 0$, points in the direction

$$\sin\theta = \frac{\lambda\,\Delta\varphi}{2\pi d}.$$

Making the offset grow steadily in time, $\Delta\varphi = \varphi\, t$, sweeps the beam across the sky: $\sin\theta(t) = \lambda\varphi\, t/2\pi d$.

Proof

The total phase difference between neighbours is the geometric one from the path difference plus the one imposed electronically. The N-source formula depends only on that total, so its maximum moves to wherever the two contributions cancel. ■

This is a *phased-array radar* (radio waves, roughly 3 kHz to 300 GHz). The beam direction is controlled entirely by the relative timing of the antenna elements, and can be changed far faster than any dish could turn. The same interference appears with light, water ripples and sound.

21.5 Single-electron interference

Replace the light by a beam of electrons aimed at two slits, and turn the source down until only one electron is in the apparatus at a time. Each electron is detected at a single point on the screen, like a particle. Yet as the points accumulate, they build up the two-slit pattern $\cos^2(\delta/2)$: many arrivals at the bright fringes and none at the dark ones.

! One electron interferes with itself

There is never a second electron present for the first to interfere with. The pattern therefore cannot come from electrons influencing each other. A single electron must be described by a wave that passes through both slits; the squared magnitude of that wave gives the probability of finding the electron at each point. This is taken up in Lecture 23.

Watch Out

No electron "splits". Each is detected whole, at one point. It is the probability of arrival that shows the pattern.

21.6 Towards diffraction: a slit of finite width

Interference was built up from two sources to N. A real slit of width D is, by Huygens' principle, a *continuous* row of infinitely many sources. The interference of infinitely many sources is called *diffraction*.

Remark — Interference or diffraction?

Feynman observed that no one has ever defined the difference between interference and diffraction satisfactorily; it is a question of usage. One usually says "interference" for a few sources and "diffraction" for many.

Theorem — Minima of a single slit

A slit of width D gives zero intensity in the directions

$$\sin\theta = \frac{m\lambda}{D}, \qquad m = \pm 1, \pm 2, \ldots$$

Proof (pairing argument)

Divide the slit into two halves. Each source in the upper half has a partner in the lower half a distance $D/2$ away, with phase difference

$$\delta = \frac{D}{2}\,\frac{2\pi}{\lambda}\sin\theta.$$

When $\delta = \pi$ every pair cancels, so the whole slit gives zero: $\sin\theta = \lambda/D$. Dividing the slit into three equal parts, the three contributions cancel when neighbouring parts differ by $2\pi/3$ or $4\pi/3$, giving $\sin\theta = \lambda/D$ and $2\lambda/D$. Dividing into N parts gives $\sin\theta = \lambda/D,\ 2\lambda/D, \ldots, (N-1)\lambda/D$. ∎

The full intensity pattern of the single slit is derived in the next lecture.

Summary of Lecture 21

(1) Two slits: $\delta = k\,d\sin\theta$ and $\langle I\rangle = I_0\cos^2(\delta/2)$.

(2) N sources: $\langle I\rangle = I_0\sin^2(N\delta/2)/\sin^2(\delta/2)$, with beam half-width $\sin\theta = \lambda/Nd$.

(3) An added phase offset between sources steers the beam: phased radar.

(4) Single electrons build up the same pattern; a single slit has minima at $\sin\theta = m\lambda/D$.

Check yourself — Lecture 21

(1) Two slits 0.1 mm apart are lit with $\lambda = 500$ nm. What is the fringe spacing on a screen 1 m away?

(2) For ten slits, how bright is a principal maximum compared with one slit alone, and how many secondary maxima lie between two principal maxima?

(3) An array has spacing $d = \lambda/2$. What phase offset between neighbours steers the beam to $30°$?

Answers are in Appendix B.

22 Lecture 22: Diffraction and Resolution

Reading

Georgi, Chapter 13: Interference and Diffraction (§13.4.1; §13.4.2 and §13.4.4 are not covered).

A slit of finite width is a continuous line of Huygens sources. This lecture adds them all up, in two different ways, to obtain the single-slit diffraction pattern. The pattern sets a fundamental limit on how sharply any instrument with a finite opening can form an image, including a telescope, the human eye and a laser pointer.

22.1 Method I: the limit of N slits

Recall the N-source result of **§21**:

$$\langle I \rangle \propto \left[\frac{\sin(N\delta/2)}{\sin(\delta/2)}\right]^2, \qquad \delta = \frac{2\pi}{\lambda}\, d \sin\theta,$$

with δ the phase difference between neighbouring slits. To describe one slit of width D, fill it with sources ever more densely:

$$d \to 0, \qquad N \to \infty, \qquad Nd = D \text{ fixed}.$$

Theorem — Single-slit diffraction pattern

A slit of width D illuminated at normal incidence by light of wavelength λ gives

$$\langle I \rangle = I_0 \left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta \equiv \frac{\pi D \sin\theta}{\lambda},$$

where I_0 is the intensity in the forward direction.

Proof

In the limit, $\delta \to 0$ while

$$\frac{N\delta}{2} = \frac{\pi N d \sin\theta}{\lambda} = \frac{\pi D \sin\theta}{\lambda} = \beta$$

stays fixed. The numerator is $\sin\beta$. In the denominator $\sin(\delta/2) \approx \delta/2 = \beta/N$. Hence

$$\left[\frac{\sin(N\delta/2)}{\sin(\delta/2)}\right]^2 \longrightarrow N^2 \left(\frac{\sin\beta}{\beta}\right)^2.$$

The factor N^2 is absorbed by letting each of the N sources carry an intensity proportional to N^{-2} (amplitude $\propto 1/N$), so that the total amplitude through the slit stays finite. Since $\sin\beta/\beta \to 1$ as $\beta \to 0$, the prefactor is the forward intensity I_0. ∎

22.2 Method II: a Fourier transform

Add the contributions of all the point sources in the opening directly. If $f(x, y)$ describes the opening (1 where light passes, 0 where it is blocked), the total field in the direction with transverse wave vector (k_x, k_y) is proportional to

Key idea

The far-field diffraction pattern is the Fourier transform of the aperture.

$$C(k_x, k_y) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy\, f(x, y)\, e^{-i\vec{k}\cdot\vec{r}}.$$

Derivation for a slit

For a slit, $f = 1$ for $|x| \le D/2$ and 0 otherwise, for all y:

$$C(k_x, k_y) = \frac{1}{4\pi^2} \int_{-D/2}^{D/2} e^{-ik_x x}\, dx \int_{-\infty}^{\infty} e^{-ik_y y}\, dy$$
$$= \frac{1}{2\pi}\, \delta(k_y) \left[\frac{e^{-ik_x x}}{-ik_x} \right]_{-D/2}^{D/2} = \frac{1}{2\pi}\, \delta(k_y)\, \frac{2\sin(k_x D/2)}{k_x}.$$

The delta function says the light is not deflected along the slit. In the x direction $k_x = k \sin\theta = (2\pi/\lambda)\sin\theta$, so $k_x D/2 = \beta$ and

$$I \propto |C|^2 \propto \frac{\sin^2(k_x D/2)}{k_x^2} \propto \left(\frac{\sin\beta}{\beta} \right)^2,$$

the same result as Method I. ∎

This is the uncertainty principle of **§15** in another guise: confining the wave to a width $\Delta x \sim D$ forces a spread $\Delta k_x \sim 1/D$ in its transverse wave number, hence a spread in direction.

22.3 Reading the pattern

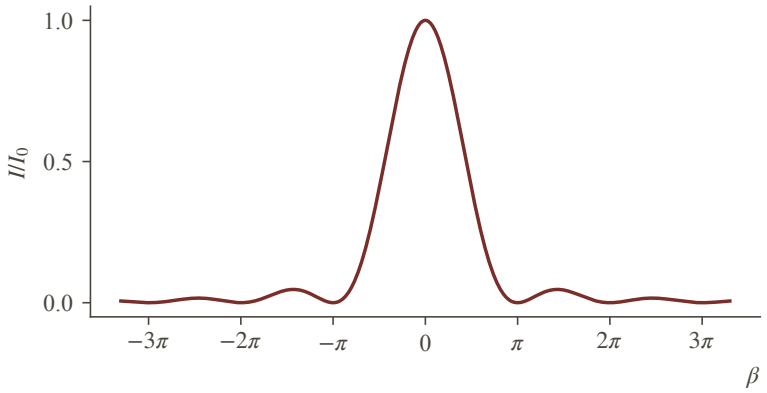


Figure 53 Single-slit diffraction pattern $(\sin\beta/\beta)^2$. The central maximum is twice as wide as the side maxima and contains most of the light.

Theorem — Properties of the single-slit pattern

(1) **Zeros** at $\beta = m\pi$, that is $\sin\theta = m\lambda/D$, $m = \pm 1, \pm 2, \ldots$, as found by the pairing argument in **§21**.

(2) **Wider slit, narrower pattern.** The half-width of the central maximum is $\sin\theta = \lambda/D$; increasing D shrinks it.

(3) **Longer wavelength, wider pattern.** The spacing of the minima is proportional to λ, so red light spreads more than blue.

(4) **Fall-off.** The side maxima decrease as $1/\beta^2$. For a wide slit this is a rapid decrease in angle; for a narrow slit it is slow.

Watch Out

Making a slit narrower does not make the beam narrower. Below a few wavelengths it does the opposite.

! Squeezing a beam makes it spread

Geometrical optics suggests that a narrower slit gives a narrower strip of light. That is true only while $D \gg \lambda$. Once the slit is comparable to the wavelength, narrowing it further *widens* the pattern, as λ/D.

22.4 Realistic multi-slit patterns

The slits of **§21** were treated as points. Real slits have a width D as well as a separation d.

Theorem — N slits of finite width

$$I = I_0 \underbrace{\left(\frac{\sin\beta}{\beta}\right)^2}_{\text{diffraction}} \underbrace{\left[\frac{\sin(N\delta/2)}{\sin(\delta/2)}\right]^2}_{\text{interference}},$$

$$\beta = \frac{\pi D \sin\theta}{\lambda}, \qquad \delta = k\,d\sin\theta = \frac{2\pi d\sin\theta}{\lambda}.$$

Proof

Every slit produces the same diffracted amplitude, $\propto \sin\beta/\beta$, in the direction θ. The N slits are then N identical sources with that common amplitude and relative phases $0, \delta, 2\delta, \ldots$, so the amplitude factorises into the single-slit factor times the N-source sum. ■

Because $d > D$, the interference factor varies faster. The pattern is a set of sharp interference fringes whose brightness is modulated by the broad single-slit envelope (Figure 54).

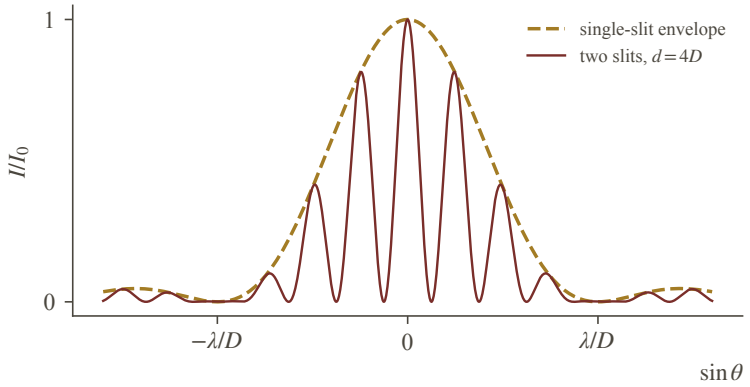


Figure 54 Two slits of finite width with $d = 4D$: two-slit fringes (solid) under the single-slit envelope (dashed).

22.5 Circular apertures and resolution

Most optical instruments have round openings. The same integral over a circular aperture of diameter D gives a pattern of concentric rings.

Theorem — Diffraction by a circular aperture

$$I(\theta) = I_0 \left[\frac{2J_1(\beta)}{\beta} \right]^2, \qquad \beta = \frac{\pi D \sin\theta}{\lambda},$$

where J_1 is the Bessel function of the first kind of order one. The first dark ring lies at $J_1(\beta) = 0$, $\beta \approx 3.83$, that is

$$\sin\theta \approx 1.22 \, \frac{\lambda}{D}.$$

Definition — Angular resolution

Two point sources can just be told apart when the central maximum of one falls on the first dark ring of the other. The smallest resolvable angular separation of an instrument with aperture D is therefore

$$\theta_{\min} \approx 1.22 \, \frac{\lambda}{D}.$$

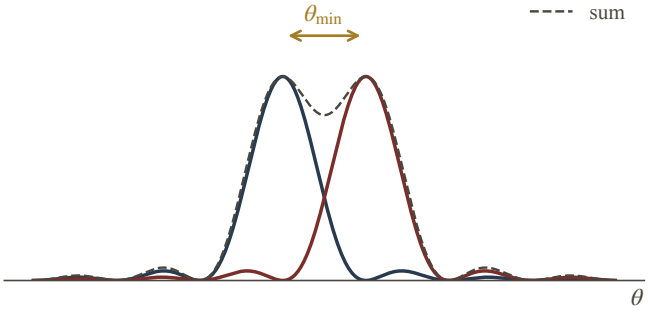


Figure 55 Two point sources at the limit of resolution (schematic, drawn with the slit pattern): the peak of one image sits on the first zero of the other. The dashed sum still shows a dip between two bumps.

Better resolution needs a larger aperture or a shorter wavelength. This is why telescopes are built large, and why it is a statement about waves, not about the quality of the glass.

Example — The human eye and a phone screen

The pupil is roughly 2–8 mm across; take $D \approx 5$ mm and visible light of $\lambda \approx 500$ nm:

$$\theta_{\min} \approx 1.22 \times \frac{5 \times 10^{-7}}{5 \times 10^{-3}} \approx 1.2 \times 10^{-4} \text{ rad}.$$

A screen with 401 pixels per inch has pixel spacing

$$\Delta x \approx \frac{2.54 \text{ cm}}{401} \approx 6.3 \times 10^{-3} \text{ cm}.$$

Held at 25 cm this subtends $\Delta\theta \approx 2.5 \times 10^{-4}$ rad, about twice $\theta_{\min}$: the eye can still just resolve the pixels. Packing about 2000 pixels into 6 cm gives $\Delta x \approx 3 \times 10^{-3}$ cm and $\Delta\theta \approx 1.2 \times 10^{-4}$ rad, which reaches the diffraction limit of the eye. A screen with 40 000 pixels per inch would show nothing more.

Example — A laser pointer aimed at the Moon

A laser beam leaves through an opening of $D = 1$ mm with $\lambda = 500$ nm. It is not a perfect pencil of light; it spreads with half-angle

$$\theta \approx 1.22\,\frac{\lambda}{D} = 1.22 \times \frac{5 \times 10^{-7}}{1 \times 10^{-3}} \approx 6 \times 10^{-4}\ \text{rad}.$$

At the distance of the Moon, $L \approx 4 \times 10^{8}$ m, the central spot has radius

$$L\,\theta \approx 4 \times 10^{8} \times 6 \times 10^{-4}\ \text{m} \approx 240\ \text{km}.$$

Summary of Lecture 22

(1) Single slit: $I = I_0(\sin\beta/\beta)^2$ with $\beta = \pi D \sin\theta/\lambda$; zeros at $\sin\theta = m\lambda/D$.

(2) The far-field pattern is the Fourier transform of the aperture.

(3) Finite-width multi-slit pattern = diffraction envelope × interference fringes.

(4) Circular aperture: resolution $\theta_{\text{min}} \approx 1.22\,\lambda/D$.

Check yourself — Lecture 22

(1) A slit 0.1 mm wide is lit with $\lambda = 500$ nm. Where is the first minimum?

(2) Estimate the angular resolution of a telescope with a 2.4 m mirror at $\lambda = 500$ nm.

(3) Two slits of width D are $d = 3D$ apart. Which interference maxima are missing?

Answers are in Appendix B.

23 Lecture 23: Quantum Waves and Gravitational Waves

The course has dealt with two kinds of waves: waves of matter (strings, springs, air) and waves of vector fields (light). This last lecture introduces two more. One is the probability wave of quantum mechanics; the other is the gravitational wave, a travelling distortion of space-time. Both are understood with the tools already developed.

23.1 Recap: diffraction

From **§22**, a single slit gives

$$I = I_0\left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta = \frac{\pi D \sin\theta}{\lambda},$$

and a laser beam leaving a 1 mm opening spreads with half-angle $\theta \approx 1.22\lambda/D \approx 6\times10^{-4}$, making a spot of radius about 240 km on the Moon. A laser pointer does not produce a perfect pencil beam.

Reading

No textbook reading. The gravitational-wave part follows the lecture video; it is not in the lecture notes.

23.2 Light as particles, electrons as waves

Everything so far described light as an electromagnetic wave, and that description accounted for interference and diffraction completely. In the twentieth century it was found to be incomplete.

Example — The photoelectric effect

Light can knock electrons out of a material. Hertz found that this is easier with high-frequency light than with low-frequency light. That is strange for a wave, whose intensity depends on the square of the field amplitude and not on its frequency. Einstein's explanation was that light arrives in discrete quanta, *photons*, each of energy

$$E = h\nu = \frac{hc}{\lambda}.$$

An electron that absorbs one photon leaves with kinetic energy at most

$$K_{\max} = h\nu - \phi,$$

where ϕ is the energy needed to free an electron from the material. If $h\nu < \phi$ no electron is emitted, however intense the light.

So light, which interferes like a wave, is absorbed like a particle. Electrons, which are elementary particles, turn out to interfere like waves.

	Bullets	Electrons
Arrival	One whole bullet at one point	One whole electron at one point
Slit 1 only open	Distribution P_1	Distribution $I_1 = \lvert\psi_1\rvert^2$
Slit 2 only open	Distribution P_2	Distribution $I_2 = \lvert\psi_2\rvert^2$
Both open	$P_1 + P_2$: no fringes	$\lvert\psi_1 + \psi_2\rvert^2$: fringes

An electron (1) arrives like a particle, producing a single hit in the detector, and (2) interferes with itself like a wave. In reality it is like neither.

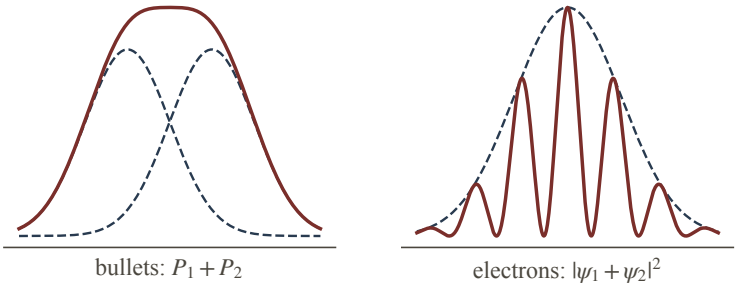


Figure 56 Distribution of arrivals on the screen with both slits open. Bullets simply add. Electrons show fringes, although each one arrives at a single point.

23.3 Watching the electrons

One can try to find out which slit each electron goes through by shining light on the slits and looking for a flash of scattered light near slit 1 or slit 2.

(1) **Bright light.** Every electron is seen, and we know its slit. The fringes disappear; the distribution is $P_1 + P_2$, as for bullets. Observing the electron has disturbed it.

(2) **Dimmer light.** Light consists of photons, so dimmer light means fewer photons, not gentler ones. Some electrons are seen and some are not. The pattern is a mixture: no fringes for the seen electrons, fringes for the unseen ones.

(3) **Redder light.** To disturb each electron less, lower the photon energy $E = hc/\lambda$ by increasing the wavelength. The fringes return once λ exceeds the separation of the slits. But by **§22**, light of that wavelength can no longer resolve which slit the flash came from.

Watch Out
No one has found a way round this. If someone does, quantum mechanics has to be discarded.

Theorem — Heisenberg's uncertainty principle

It is not possible to determine which slit the electron passed through without destroying the interference pattern. Quantitatively,

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}.$$

This is the relation proved for waves in general in **§15**. A single slit shows it directly. Passing through a slit of width D fixes the electron's transverse position to within D, and the amplitude for transverse wave number k_x is then

$$C(k_x) \propto \int_{-D/2}^{D/2} e^{-ik_x x} \, \mathrm{d}x.$$

A narrow slit gives a wide spread in k_x, hence in the transverse momentum $p_x = \hbar k_x$: we know where the electron was but not where it is heading. As $D \to \infty$, $C(k_x)$ approaches a delta function: the momentum is sharp, and the position is unknown.

23.4 The wave function

Definition — Wave function and probability

The state of an electron is described by a complex *wave function* ψ. The probability of finding the electron at a given place is

$$P \propto |\psi|^2.$$

! Quantum mechanics predicts only the odds

The theory gives the exact evolution of the wave function, and therefore the probability distribution of outcomes. It does not give the outcome of an individual measurement. It is now believed to be impossible to predict exactly what will happen in a given situation.

Could the electron have "already decided" which slit to use, by some hidden variable? That is not tenable here: whether fringes appear depends on what we choose to do at the slits, which the electron could not have anticipated. What travels through both slits is a new kind of wave, a *probability wave*.

23.5 A particle in a box

Example — Normal modes of a confined particle

A particle is confined to $0 < x < L$ by infinitely high potential walls. The wave function must vanish at the walls,

$$\psi(0) = 0, \qquad \psi(L) = 0,$$

which are exactly the boundary conditions of a string fixed at both ends. The normal modes are therefore

$$\psi_m(x,t) = A_m \sin(k_m x)\, e^{-i\omega_m t}, \qquad k_m = \frac{m\pi}{L}, \quad m = 1, 2, 3, \ldots$$

The probability density $|\psi_m|^2 \propto \sin^2(k_m x)$ has m humps and does not change in time.

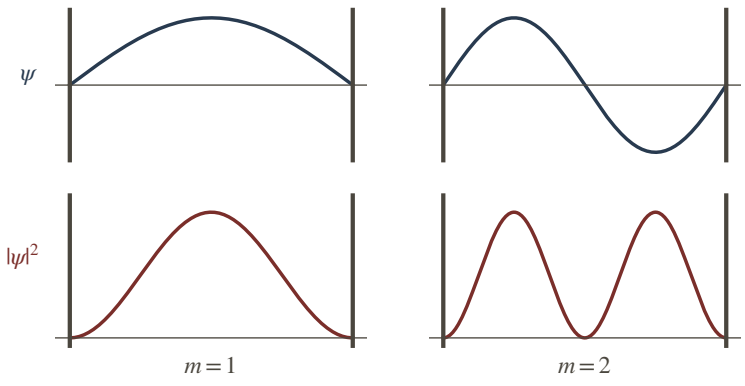


Figure 57 The first two modes of a particle in a box: wave function (top) and probability density (bottom).

The boundary conditions fix k_m. To find the frequencies we need what the string had and this problem so far lacks: a wave equation, and with it a dispersion relation.

Definition — Schrödinger's equation

The wave equation obeyed by the wave function of a particle of mass m in a potential V is

$$i\hbar\,\frac{\partial}{\partial t}\psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\psi(x,t).$$

Feynman

Quoted in the lecture: it is not possible to derive this equation from anything you know; it came out of the mind of Schrödinger.

Theorem — Dispersion relation and group velocity of matter waves

For a free particle ($V = 0$),

$$\omega = \frac{\hbar k^2}{2m}.$$

With de Broglie's relation $p = \hbar k$, the group velocity is

$$v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{\hbar k}{m} = \frac{p}{m},$$

the classical velocity of the particle.

Proof

Insert $\psi_m = A_m \sin(k_m x) e^{-i\omega_m t}$ with $V = 0$. The left side gives $\hbar\omega_m\psi_m$ and the right side $(\hbar^2 k_m^2/2m)\,\psi_m$. Differentiating $\omega = \hbar k^2/2m$ with respect to k gives v_g. ∎

This ties the two pictures together. A "particle" is a wave packet, a superposition of many waves of different wavelengths, and the packet moves at the group velocity of **§13**. That velocity is p/m, which is what Newtonian mechanics calls the velocity. The medium is dispersive, since $\omega \propto k^2$: the phase velocity $\omega/k = p/2m$ is only half of it, and a free packet spreads as it travels. The energies of the particle in a box follow too: $E = \hbar\omega = \hbar^2 k^2/2m$ with $k = k_m$, so they are proportional to the square of the mode number.

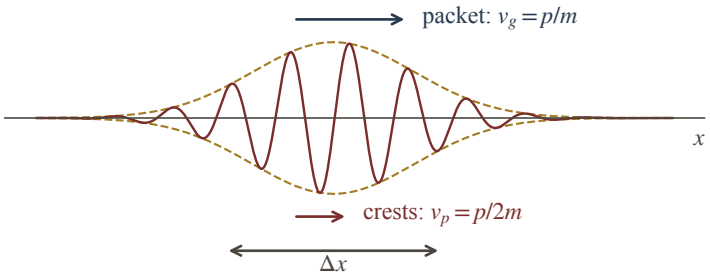


Figure 58 A particle as a wave packet. The packet as a whole moves at the group velocity, which is the classical velocity; its width Δx and its spread of wave numbers obey $\Delta x\,\Delta k \geq \frac{1}{2}$.

23.6 Gravitational waves

General relativity predicts a fourth kind of wave. Einstein showed that the distortion of space-time produced by massive objects can travel through space as a *gravitational wave*. The derivation is beyond this course; the lecture asks us to take the result on trust.

Any accelerating mass radiates gravitational waves, a person waving their arms included, but for objects of ordinary mass the distortion is far too small to detect. Einstein himself doubted it would ever be measured.

Definition — Effect of a gravitational wave

A gravitational wave passing perpendicular to a plane stretches space along one direction in that plane while squeezing it along the perpendicular direction, and half a period later does the opposite. A ring of free particles is deformed into an ellipse, alternately elongated along the two directions.

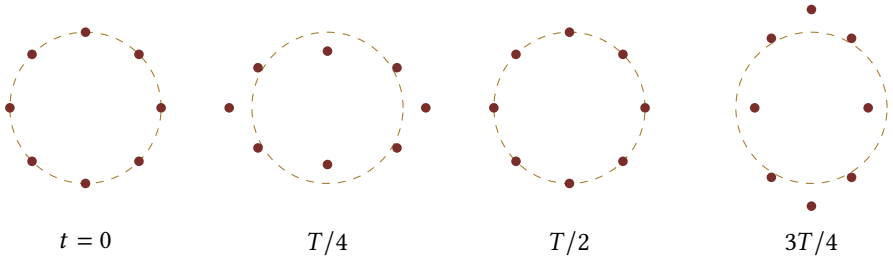


Figure 59 A ring of free particles as a gravitational wave passes through the page, shown at quarter-period intervals. The distortion is enormously exaggerated.

23.7 LIGO: detecting the distortion by interference

The Laser Interferometer Gravitational-wave Observatory measures this stretching and squeezing with the interference of **§20**.

(1) A laser beam is split into two beams travelling along two perpendicular arms, each 4 km long.

(2) A mirror at the end of each arm sends the beam back.

(3) The returning beams are recombined on a photodetector, which records their interference.

Theorem — Response of the interferometer

The two beams arrive at the detector with a phase difference set by the difference of the round-trip path lengths,

$$\delta = \frac{2\pi}{\lambda} \cdot 2\,(L_1 - L_2) + \delta_0,$$

where δ_0 is a fixed offset from the optics. The instrument is adjusted so that with no wave present the beams cancel: $\delta = \pi$ and the detector is dark. A change ΔL in the difference of the arm lengths changes δ by $4\pi\,\Delta L/\lambda$, and light reaches the detector.

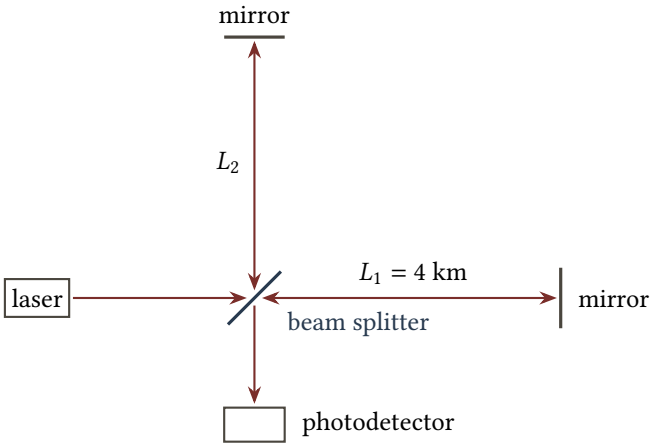


Figure 60 Layout of the LIGO interferometer.

Proof

This is two-wave interference, with $\langle I \rangle \propto 1 + \cos\delta$ for equal amplitudes. Each beam goes out and back, so a length difference $L_1 - L_2$ gives a path difference of twice that. At $\delta = \pi$ the intensity vanishes; any departure from π lets light through. ■

A gravitational wave lengthens one arm while shortening the other, which is exactly the change the interferometer is sensitive to. Long arms make the change in length as large as possible for a given distortion.

There are two detectors, at Hanford and at Livingston, about 3000 km apart. A disturbance local to one site does not appear at the other, so a genuine signal must show up at both, with a small time shift corresponding to the travel time between them.

Tip
Requiring the same signal at two distant sites removes local disturbances such as earthquakes or passing traffic.

Example — The first detection

On 14 September 2015 both detectors recorded the same signal, announced on 11 February 2016. It matched the calculated waveform for two black holes, of about 29 and 36 solar masses, orbiting each other faster and faster and then merging, after which the signal stopped. The source was about 1.3 billion light years away.

Remark — A new way to observe the universe

Until this measurement, nearly everything known about the distant universe came from electromagnetic waves. Gravitational waves are a second, independent channel, and they were detected with an instrument whose principle is the two-beam interference of this course.

23.8 Closing

The course began with one mass on one spring and ends with waves of probability and of space-time. The same ideas ran through all of it: linear equations of motion, normal modes, superposition, dispersion relations, boundary conditions, and interference. As the lecture puts it, this is not the end of vibrations and waves but only the beginning; quantum mechanics and relativity take up the story from here.

Summary of Lecture 23

(1) Light is absorbed in quanta of energy $h\nu$; electrons interfere like waves but arrive as particles.

(2) Finding out which slit an electron used destroys the fringes: $\Delta x\,\Delta p \geq \hbar/2$.

(3) The wave function gives probabilities, $P \propto |\psi|^2$, and obeys Schrödinger's equation; for a free particle $\omega = \hbar k^2/2m$ and $v_g = p/m$.

(4) Gravitational waves stretch and squeeze space; LIGO detects them as a change in the interference of two laser beams.

Check yourself — Lecture 23

(1) What is the energy of a photon of wavelength 500 nm? Can it eject an electron from a metal with $\phi = 2.3$ eV?

(2) For a particle in a box, what is the ratio of the energy of the second mode to that of the first?

(3) Why does LIGO need two detectors far apart?

Answers are in Appendix B.

Appendix A: Formula Sheet

The main results of the course, with the lecture in which each is derived. Use it for revision, not as a substitute for the derivations.

Part I — Mechanical vibrations and waves

Result	Formula	Lec.
Simple harmonic motion	$\ddot{x} + \omega_0^2 x = 0,\ x = A\cos(\omega_0 t + \phi)$	1
Small oscillations	$k = V''(0),\ \omega_0 = \sqrt{V''(0)/m}$	1
Energy	$E = \frac{1}{2}\mu\dot{x}^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$	2
Damped oscillator	$\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0$	2
Underdamped	$x = Ae^{-\Gamma t/2}\cos(\omega t + \phi),\ \omega^2 = \omega_0^2 - \Gamma^2/4$	2
Critical; overdamped	$(A + Bt)e^{-\Gamma t/2};\ A_+ e^{-\Gamma_+ t} + A_- e^{-\Gamma_- t}$	2
Driven amplitude	$A(\omega_d) = f_0 \big/ \sqrt{(\omega_0^2 - \omega_d^2)^2 + \omega_d^2\Gamma^2}$	3
Phase lag; quality factor	$\tan\delta = \Gamma\omega_d/(\omega_0^2 - \omega_d^2);\ Q = \omega_0/\Gamma$	3
Normal modes	$\det(M^{-1}K - \omega^2 I) = 0,\ (M^{-1}K - \omega^2 I)A = 0$	4
Beats	carrier $\frac{1}{2}(\omega_1 + \omega_2)$, modulation $\frac{1}{2}(\omega_2 - \omega_1)$	5
Driven coupled system	$(M^{-1}K - \omega_d^2 I)B = M^{-1}F$	6
Symmetry	$[S, M^{-1}K] = 0$; translation gives $A_j \propto e^{ijka}$	7
Chain dispersion	$\omega^2 = 4\omega_0^2 \sin^2(ka/2)$	7, 8
Fixed ends; free ends	$ka = n\pi/(N+1);\ ka = n\pi/N$	8
Wave equation	$\partial_t^2\psi = v_p^2\,\partial_x^2\psi,\ v_p = \sqrt{T/\rho_L}$	8
Standing waves	$\psi_m = A_m \sin(\omega_m t + \beta_m)\sin(k_m x + \alpha_m)$	9
Fourier coefficients	$A_m = \frac{2}{L}\int_0^L \psi(x,0)\sin(k_m x)\,dx$	9
Traveling waves	$\psi = f(x - v_p t) + g(x + v_p t)$	10
String energy densities	$\frac{1}{2}\rho_L(\partial_t\psi)^2,\ \frac{1}{2}T(\partial_x\psi)^2$	10
Junction	$R = \dfrac{v_2 - v_1}{v_2 + v_1},\ \mathcal{T} = \dfrac{2v_2}{v_1 + v_2}$	10
Sound	$v_p = \sqrt{\gamma P_0/\rho},\ \psi_P = -\gamma P_0\,\partial_x\psi$	11

Part II — Electromagnetic waves

Result	Formula	Lec.
EM wave equation	$\nabla^2\vec{E} = \mu_0\epsilon_0\,\partial_t^2\vec{E},\ c = 1/\sqrt{\mu_0\epsilon_0}$	12
Plane wave	$\vec{B} = \frac{1}{c}\,\hat{k}\times\vec{E},\ \omega = ck$	12
Energy; Poynting vector	$u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0}B^2,\ \vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$	12
Phase and group velocity	$v_p = \omega/k,\ v_g = \mathrm{d}\omega/\mathrm{d}k$	13
Stiff string	$\omega^2 = v^2(k^2 + \alpha k^4)$	13
Fourier transform	$f(t) = \int \mathrm{d}\omega\, C(\omega)e^{-i\omega t},\ C(\omega) = \frac{1}{2\pi}\int \mathrm{d}t\, f(t)e^{i\omega t}$	14
Narrow-band signal	$\psi = \mathrm{Re}\left[f_s(t - x/v_g)\, e^{-i(\omega_0 t - k_0 x)}\right]$	14
Uncertainty	$\Delta t\,\Delta\omega \geq \frac{1}{2},\ \Delta x\,\Delta k \geq \frac{1}{2}$	15
Membrane modes	$\sin(n_x\pi x/L_H)\sin(n_y\pi y/L_V),\ \omega = v\lvert\vec{k}\rvert$	16
Snell's law	$n_1\sin\theta_1 = n_2\sin\theta_2,\ n = c/v$	16
Critical angle	$\sin\theta_c = n_2/n_1$	16

Part III — Optics

Result	Formula	Lec.
Polarization vector	linear $(\cos\theta, \sin\theta)$; circular $(1, \pm i)$	17
Malus's law	$I = I_0\cos^2\theta$; unpolarized light: $I_0/2$	17
Wave plate	$\Delta\phi = (n_x - n_y)\,\omega l/c$	18
Radiation field	$\vec{E}_{\mathrm{rad}} = -\dfrac{q\,\vec{a}_\perp(t - r/c)}{4\pi\epsilon_0 c^2 r}$	18
Radiated power	$P = \dfrac{q^2a^2}{6\pi\epsilon_0 c^3}$	18
Light in matter	$v = c/n,\ n = \sqrt{\mu\epsilon/\mu_0\epsilon_0}$	19
Fresnel (in-plane)	$R = \dfrac{\alpha-\beta}{\alpha+\beta},\ T = \dfrac{2}{\alpha+\beta},\ \alpha = \dfrac{\cos\theta_T}{\cos\theta_I},\ \beta = \dfrac{n_2}{n_1}$	19
Brewster's angle	$\tan\theta_B = n_2/n_1$	19
Two-wave interference	$\langle I\rangle \propto \frac{1}{2}A_1^2 + \frac{1}{2}A_2^2 + A_1A_2\cos\delta$	20
Thin film in air	$\delta = 4\pi n_2 d/\lambda + \pi$	20
N sources	$I = I_0\,\dfrac{\sin^2(N\delta/2)}{\sin^2(\delta/2)},\ \delta = kd\sin\theta$	21
Single slit	$I = I_0\left(\dfrac{\sin\beta}{\beta}\right)^2,\ \beta = \dfrac{\pi D\sin\theta}{\lambda}$	22
Resolution	$\theta_{\min} \approx 1.22\,\lambda/D$	22
Photon; matter wave	$E = h\nu,\ p = \hbar k,\ \omega = \hbar k^2/2m$	23

Appendix B: Answers to the Check-Yourself Questions

Try each question before reading its answer. If an answer surprises you, go back to the lecture section it comes from.

Lecture 1

(1) $\omega = \sqrt{k/m} = 4$ rad/s and $T = 2\pi/\omega \approx 1.57$ s.

(2) $A = v_0/\omega$ and $\phi = -\pi/2$, so that $x = (v_0/\omega)\sin\omega t$.

(3) $V''(0) = V_0/a^2$, so $\omega = \sqrt{V_0/ma^2}$.

Lecture 2

(1) Compare $\Gamma^2/4$ with $\omega_0^2 = 100$: $\Gamma = 4$ is underdamped, $\Gamma = 20$ critically damped, $\Gamma = 30$ overdamped.

(2) The envelope is $e^{-\Gamma t/2}$, so $t = 2/\Gamma$.

(3) $\omega_0 = 1/\sqrt{LC} \approx 3.2 \times 10^4$ rad/s, about 5 kHz.

Lecture 3

(1) $Q = \omega_0/\Gamma = 50$, so the amplitude is $Q\Delta = 5$ cm.

(2) 0, $\pi/2$ and π.

(3) After several times $2/\Gamma$, when the transient $e^{-\Gamma t/2}$ has died away.

Lecture 4

(1) Five: one per coordinate.

(2) All blocks move together, no spring is stretched, so there is no restoring force: it is a uniform translation.

(3) The centre of mass must stay at rest: $2m \cdot B + m(-B) + m(-B) = 0$.

Lecture 5

(1) Carrier 10.5 s^{-1}, modulation 0.5 s^{-1}, and $2\pi/(\omega_2 - \omega_1) \approx 6.3$ s.

(2) Release from rest with $x_1 = -x_2$.

(3) $x_1 + x_2$ and $x_1 - x_2$; each obeys its own uncoupled oscillator equation.

Lecture 6

(1) $+1$: the response has the shape of the symmetric mode.

(2) $\omega_d^2 = g/\ell + k/m$, where the numerator of B_1 vanishes.

(3) The model has no damping. Any drag makes the peaks finite.

Lecture 7

(1) Reflecting twice is the identity, $S^2 = I$, so $\beta^2 = 1$.

(2) $2\omega_0$, at $ka = \pi$: neighbouring masses move exactly opposite to each other.

(3) Otherwise $A_j = \beta^j$ grows without limit towards one end of the chain.

Lecture 8

(1) $ka = n\pi/(N+1) = \pi/4,\ \pi/2,\ 3\pi/4$.

(2) $y_0 = y_1$: the last segment is horizontal, so the tension has no transverse component.

(3) $v_p = \sqrt{T/\rho_L} = 100$ m/s.

Lecture 9

(1) $4L$ and $2L$.

(2) k_m and α_m by the boundary conditions; A_m and β_m by the initial conditions.

(3) By orthogonality $A_3 = 1$ and every other $A_m = 0$: the string stays in the third mode.

Lecture 10

(1) In the $-x$ direction at speed 2.

(2) $R = \frac{1}{3}$ (not inverted) and $\mathcal{T} = \frac{4}{3}$.

(3) All of it is kinetic: the string is flat but moving.

Lecture 11

(1) $f_1 = v_p/4L = 170$ Hz and $f_2 = 3f_1 = 510$ Hz.

(2) A pressure node and a displacement antinode.

(3) The oscillation is too fast for heat to flow between compressed and rarefied regions.

Lecture 12

(1) $B_0 = E_0/c = 1 \times 10^{-6}$ T.

(2) Along $\vec{E} \times \vec{B}$, that is $+\hat{x}$.

(3) Zero: energy only moves back and forth over a quarter wavelength.

Lecture 13

(1) $v_p = ak$ and $v_g = 2ak$.

(2) A non-dispersive one, $\omega = vk$.

(3) No. $k_m = m\pi/L$ comes from the boundary conditions alone; stiffness changes the frequencies $\omega(k_m)$.

Lecture 14

(1) 999 kHz and 1001 kHz.

(2) The envelope at v_g, the carrier at v_p.

(3) $e^{i\omega t_0}\,C(\omega)$: a delay only changes the phases.

Lecture 15

(1) $\Delta\omega \geq 1/2\Delta t = 5\times 10^{8}\ \mathrm{s}^{-1}$.

(2) A Gaussian.

(3) $\Delta p \geq \hbar/2\Delta x \approx 5\times 10^{-25}\ \mathrm{kg\,m/s}$.

Lecture 16

(1) They are equal, $\omega = \sqrt{5}\,\pi v/L$.

(2) $\sin\theta_c = 1/1.33$, so $\theta_c \approx 49°$.

(3) The frequency ω and the component of $\vec{k}$ along the boundary.

Lecture 17

(1) Circular, rotating counter-clockwise.

(2) $\cos^2 30° = 75\%$.

(3) $\frac{1}{2} I_0 \cos^2 60° = I_0/8$.

Lecture 18

(1) $l = \lambda/4(n_x - n_y) = 15\ \mu\mathrm{m}$.

(2) Along its acceleration, where $\vec{a}_\perp = 0$.

(3) $a \propto \omega^2$ and $P \propto a^2$, so the power rises by a factor of 16.

Lecture 19

(1) $n = \sqrt{k_e} = 1.5$ and $v = c/n = 2\times 10^{8}\ \mathrm{m/s}$.

(2) $\tan\theta_B = 1.33$, so $\theta_B \approx 53°$.

(3) $R^2 = \left[(1-1.33)/(1+1.33)\right]^2 \approx 2\%$.

Lecture 20

(1) $\lambda_{\max} = 4dn_2 \approx 640$ nm for $N = 1$; the next, 213 nm, is ultraviolet.

(2) As $d \to 0$ only the inversion at the front surface remains, so the two reflections cancel.

(3) $4I$ and 0.

Lecture 21

(1) $\lambda L/d = 5$ mm.

(2) $N^2 = 100$ times brighter; $N - 2 = 8$ secondary maxima.

(3) $\Delta\varphi = (2\pi d/\lambda)\sin 30^\circ = \pi/2$.

Lecture 22

(1) $\sin\theta = \lambda/D = 5 \times 10^{-3}$.

(2) $\theta_{\text{min}} \approx 1.22\lambda/D \approx 2.5 \times 10^{-7}$ rad.

(3) Every third one, $m = \pm 3, \pm 6, \ldots$, where a fringe falls on a zero of the single-slit envelope.

Lecture 23

(1) $E = hc/\lambda \approx 2.48$ eV. Yes, with $K_{\text{max}} \approx 0.18$ eV.

(2) $E \propto k_m^2 \propto m^2$, so the ratio is 4.

(3) A real gravitational wave appears at both, slightly shifted in time; a local disturbance appears at only one.